



集成电路设计工程 2025秋

Lecture 4 集成电路设计电路模块与电路仿真介绍

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2025年10月14日

第一次作业情况



■目前网络学堂已经有9位同学按时提交作业

■考虑到软件setup等困难，一次作业我们给的延期，本周日(10月19日)晚上23: 59PM为最终deadline，请注意，此为最终Firm Deadline，在这个日期前提交不扣分。

■今晚（10月14日）23: 59PM提交的同学，我们会有额外的奖励机制。



仿真操作案例说明

-以反相器为例

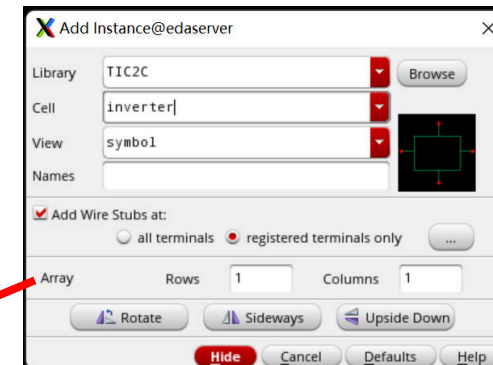
仿真测试用例 (Test Bench, TB)



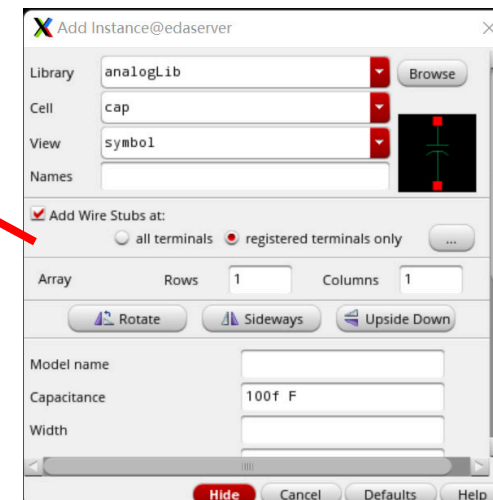
- TB是用于仿真待验电路模块的原理图
- 注意！在正式的模块设计中要将TB和电路原理图分开，即一定要单独建立TB和Symbol，不要以Cell内部原理图作为TB。Why?
- Symbol命名方式：今天设计一个反相器电路，可采用以下命名方式。
 - 电路原理图： **INV**
 - TB原理图： **TB_INV**



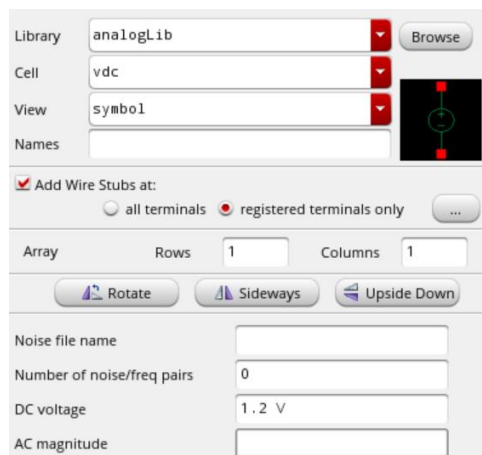
反相器



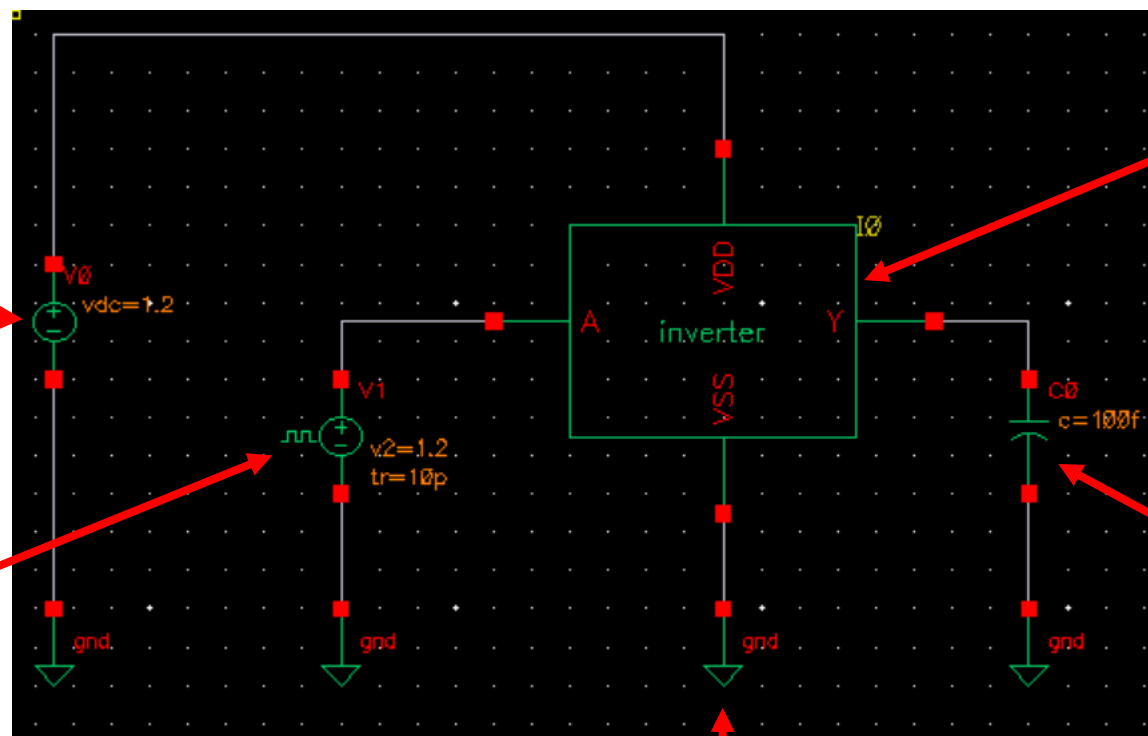
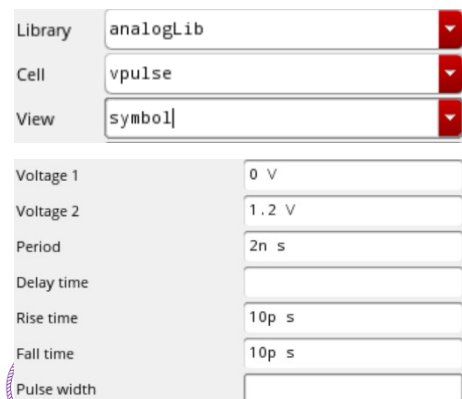
负载电容



直流电压源



方波电压信号源



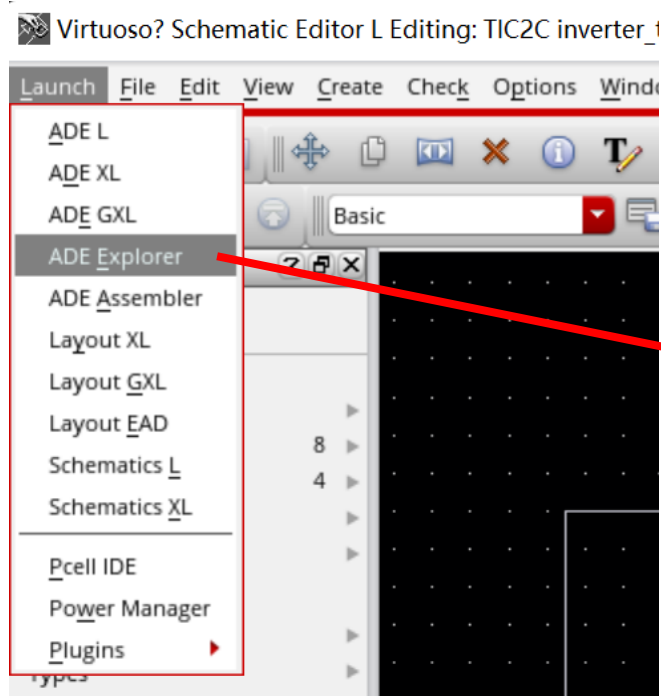
电路地



启动仿真环境

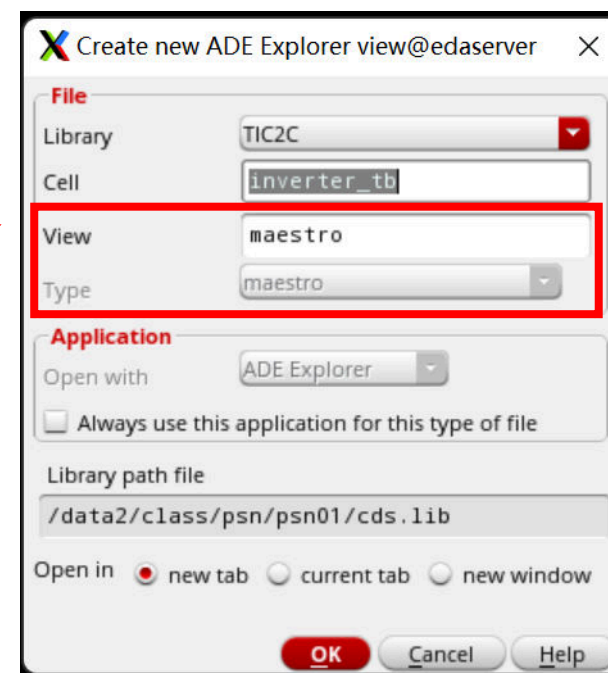


- 保存TB原理图后，从原理图编辑器菜单栏 – Launch – ADE Explorer打开仿真环境 (Analog Design Environment, ADE) 。



第一次启动

后继启动





仿真项目列表

原理图页面

ADE 页面

仿真输出项目列表

← 添加仿真项目

← 新增输出项目

← 运行仿真

← 显示观测项目

添加仿真项目



■从ADE右侧工具栏添加仿真项目，以瞬态仿真为例：

- 其他仿真项目用法参考手册或 “help”

设置完毕后，左侧列表中
将多出一项

仿真项目选择

仿真参数设置

添加仿真输出



■从ADE右侧工具栏可添加仿真输出项

- 最常见的为Signal类型，即节点电压或端口电流
- 也可添加Expression，填入自定义表达式作为输出

The screenshot illustrates the ADE interface for adding simulation outputs. On the left, the 'Signal' option is selected in the output type menu. The top table shows two 'signal' outputs, with the second one highlighted. The bottom schematic diagram shows an inverter circuit with nodes labeled A and Y, and a capacitor labeled Cb.

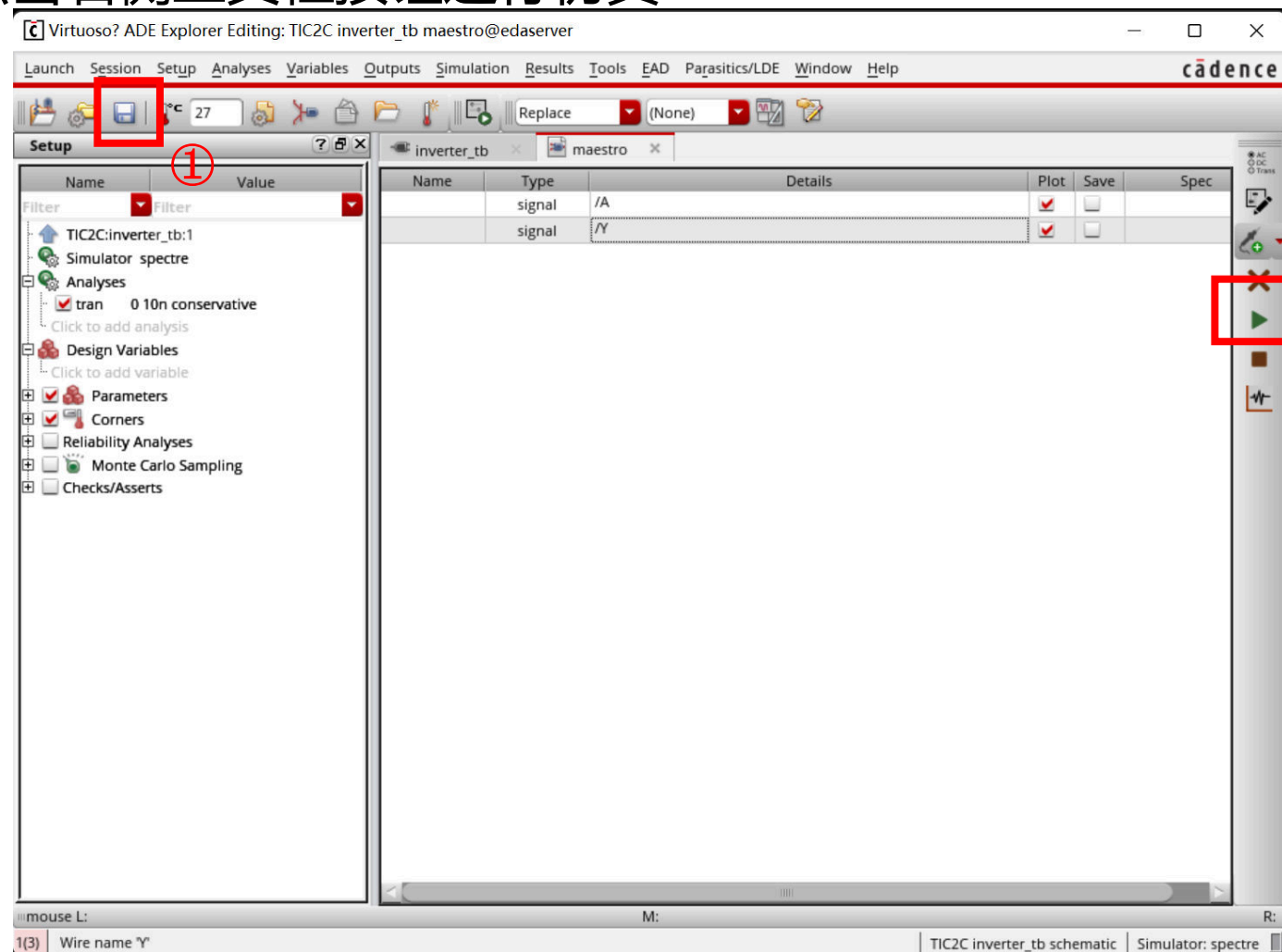
Name	Type	Details	Plot	Save	Spec
	signal	/A	☒	☐	
	signal		☒	☐	

点击“...”后，进入原理图选择要观察的节点

运行仿真



■保存ADE设置后，点击右侧工具栏按钮运行仿真

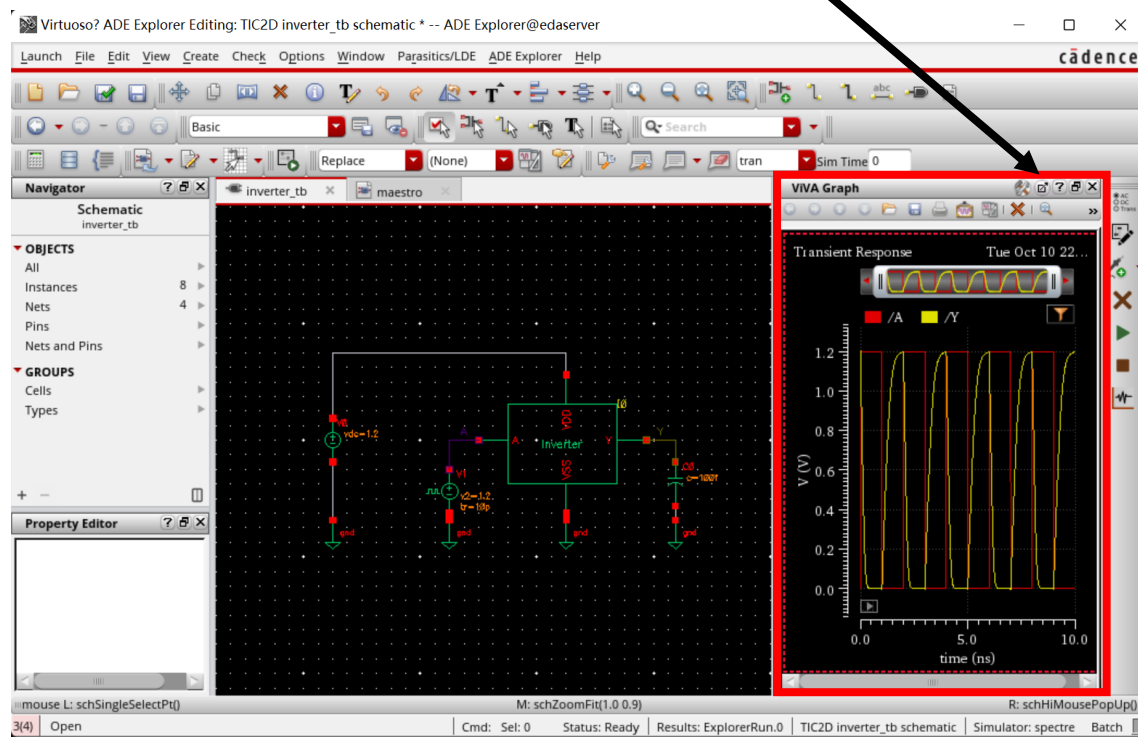


查看仿真输出



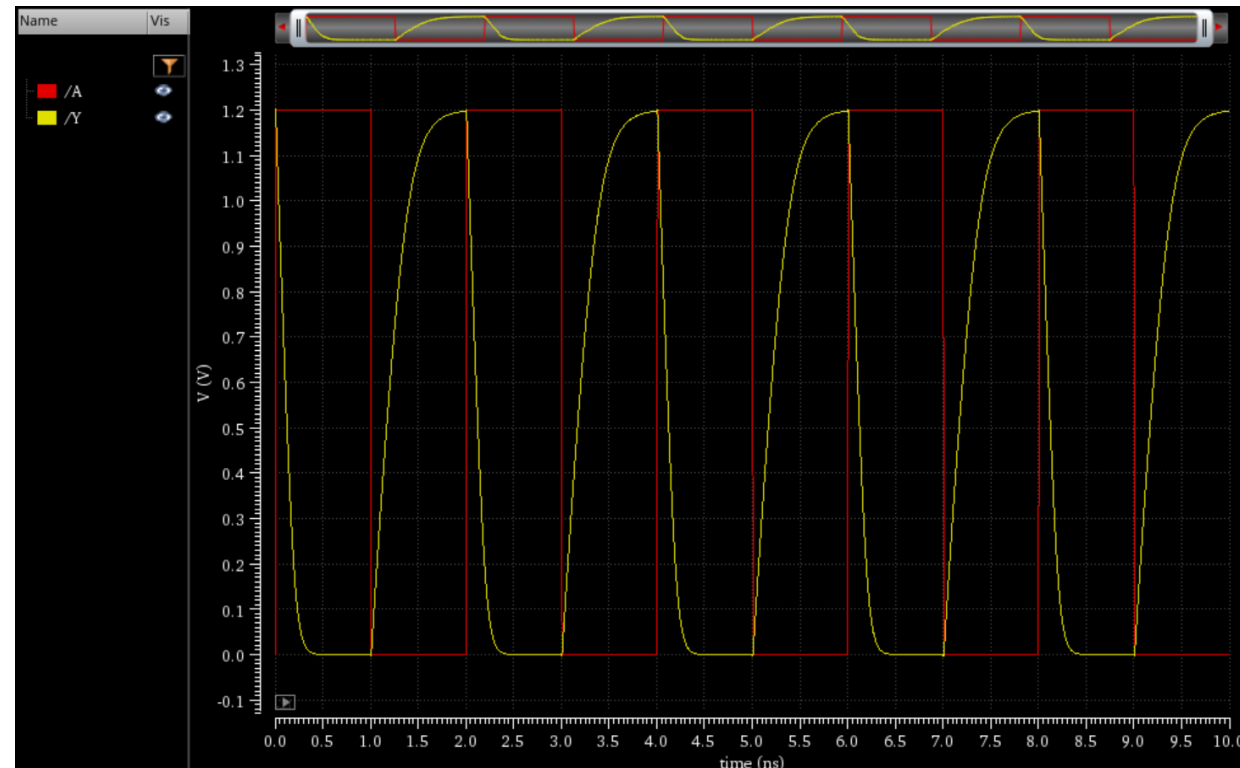
■默认设置下，仿真输出项将自动绘图显示

点击“”可将输出窗口独立



仿真输出窗口

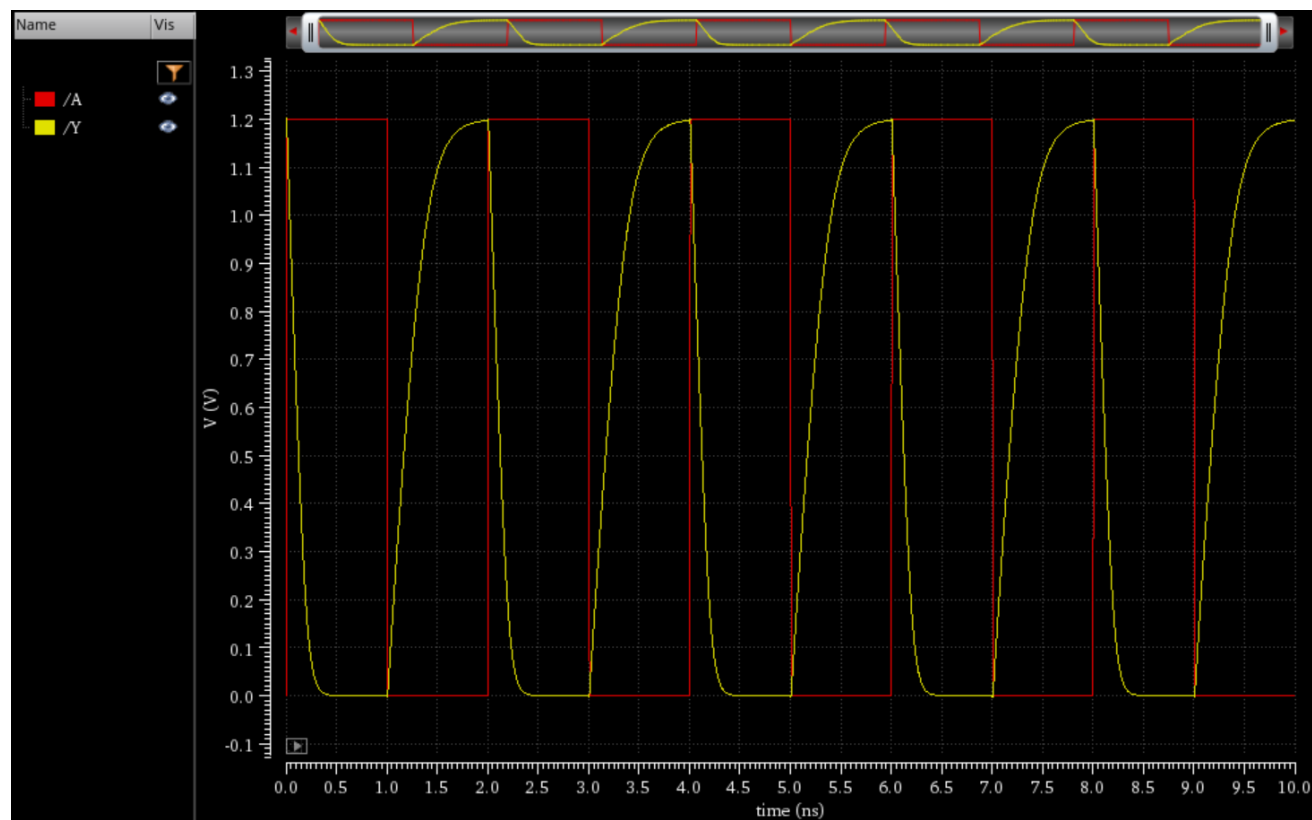
瞬态仿真输出波形



波形观察窗口操作



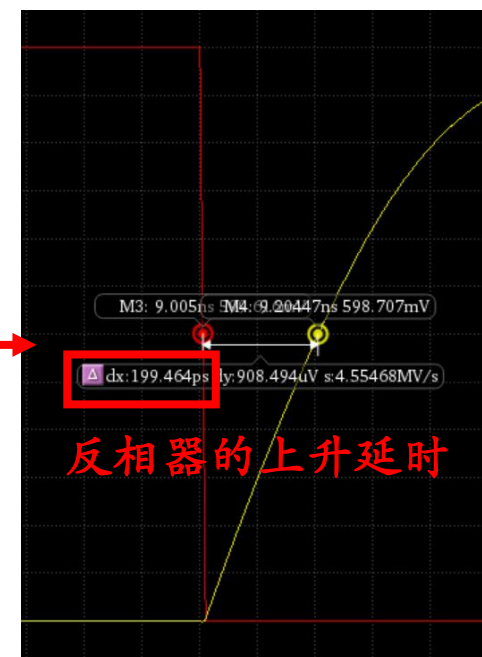
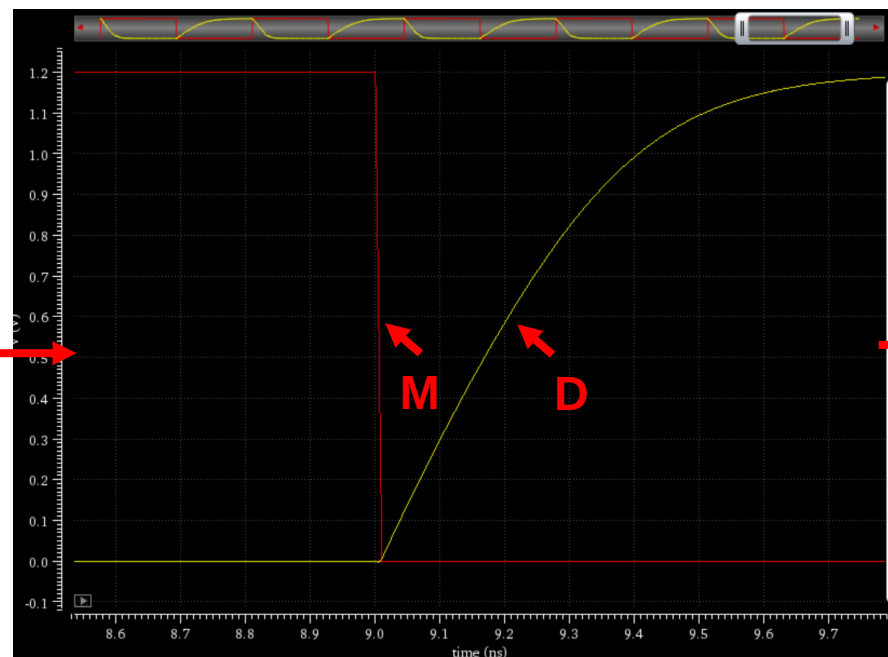
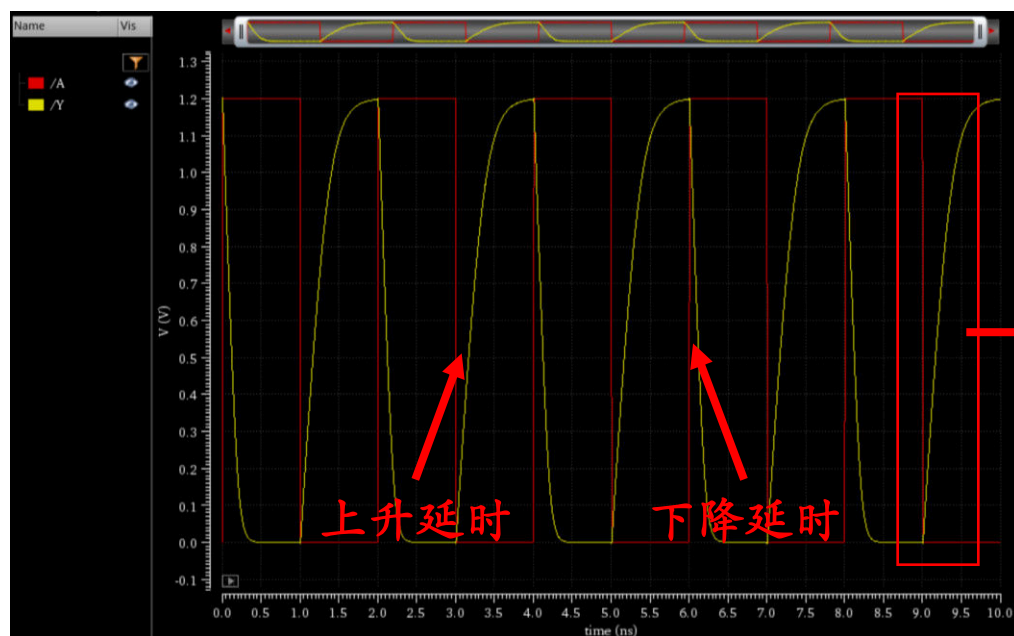
- 视图平移：方向键、滚轮、**Alt**+鼠标拖动
- 时间（水平）缩放：**Shift**+滚轮
- 电压（垂直）缩放：**Ctrl**+滚轮
- 局部放大：鼠标右键框选
- 恢复默认视图：**F**键
- 添加测量标记：鼠标指向波形后按**M**键
- 添加差分测量标记：添加一个测量标记后，鼠标指向波形另一处，然后按**D**键
- 清除测量标记：**Ctrl**+**E**
- 输入具体坐标：双击数字



例：利用差分测量标记测量反相器延时



- 分别放大至波形的上升沿和下降沿处
- 在输入信号与0.6V的交界处按M键，加入一个标记
- 在输出信号与0.6V的交界处按D键，加入一个差分标记
- 差分标记显示的dx为时间差，即反相器的延时





电路模块讲解说明

-以反相器为例

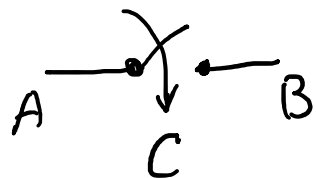


- Switch/sample and hold
- Inverter
- Logic gates (VML and CML)
- Amplifiers
- Latch/flip-flop/comparator
- Digital-to-analog converter (DAC)
- Voltage-control oscillator (VCO)

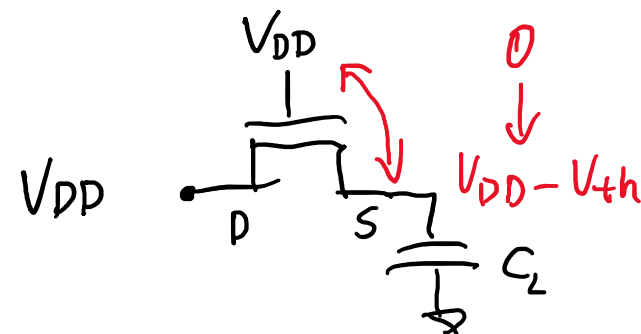
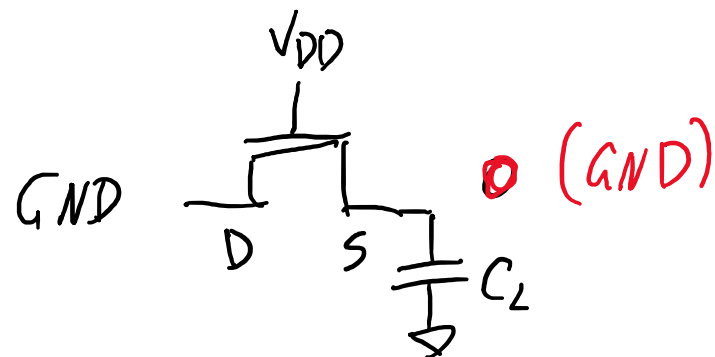
Switch



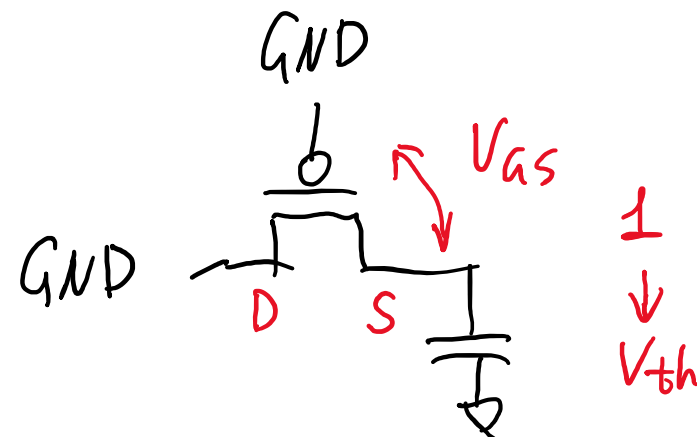
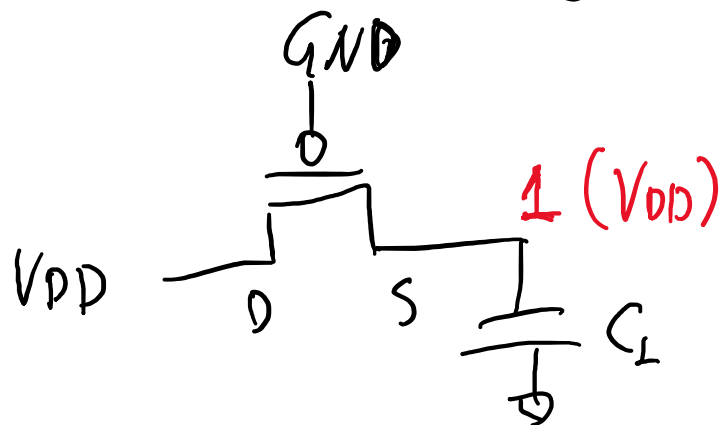
Symbol:



NMOS passes strong zero and a weak one.



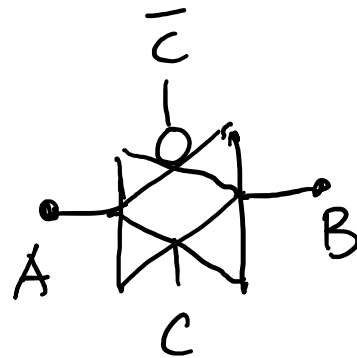
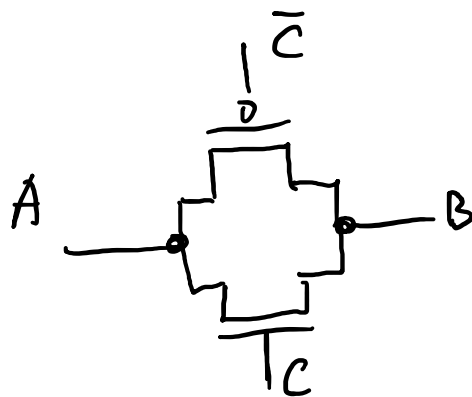
PMOS passes strong one and a weak zero.



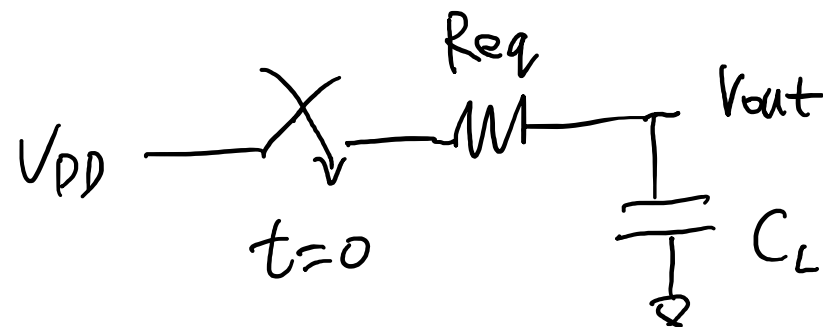
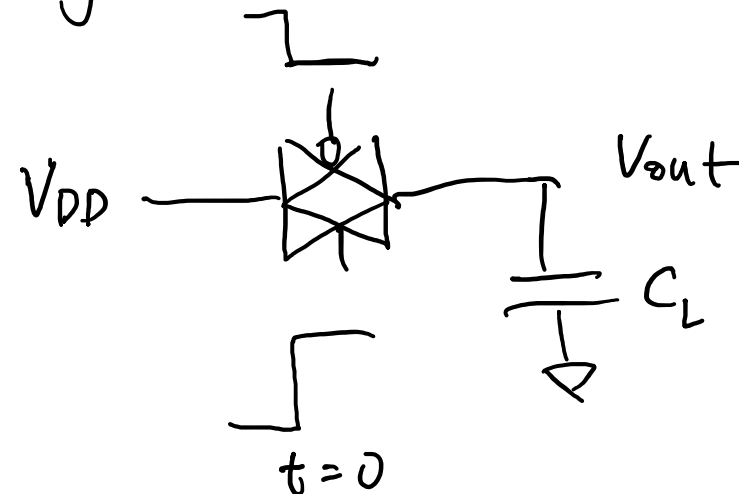
Switch



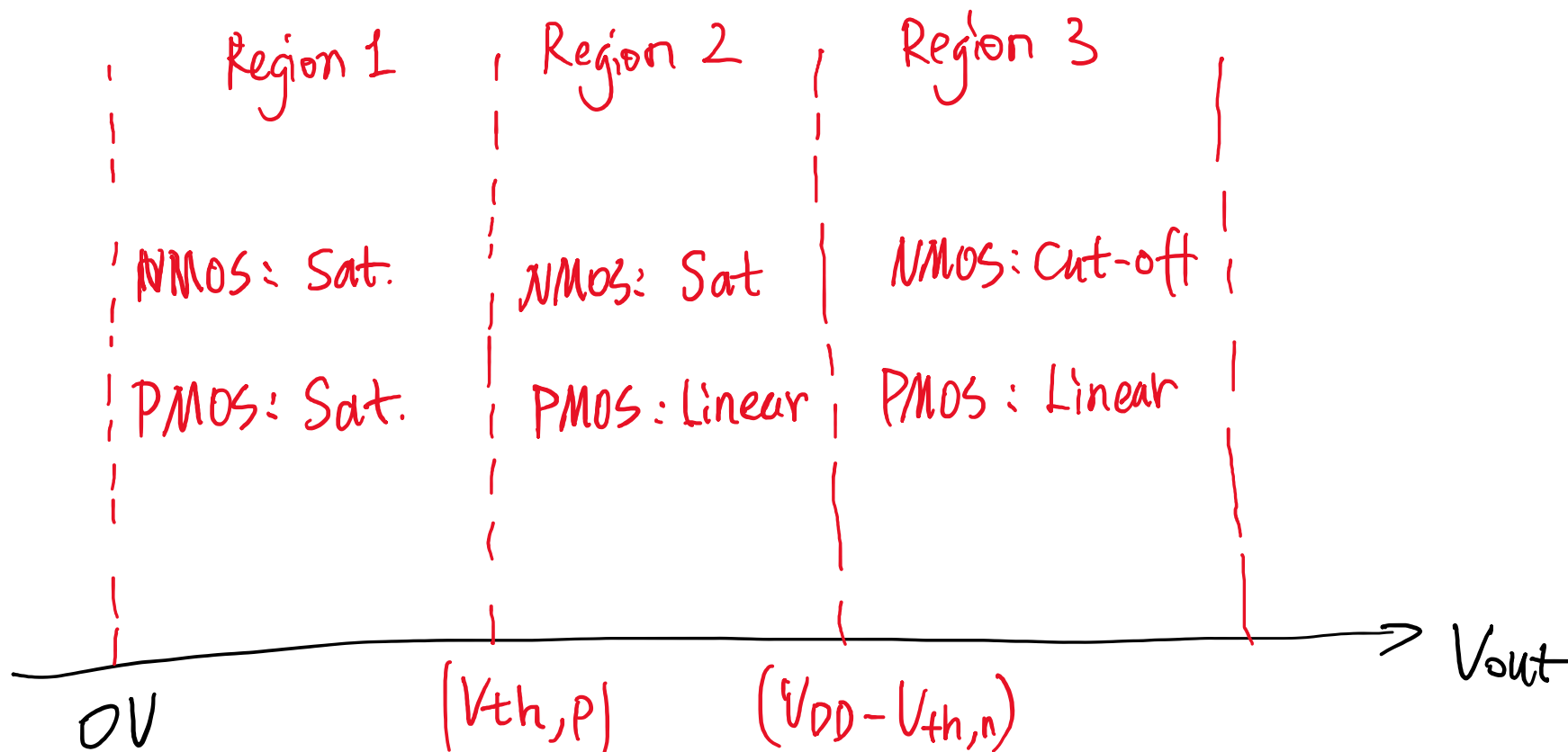
Transmission Gate



Dynamic model:



Switch



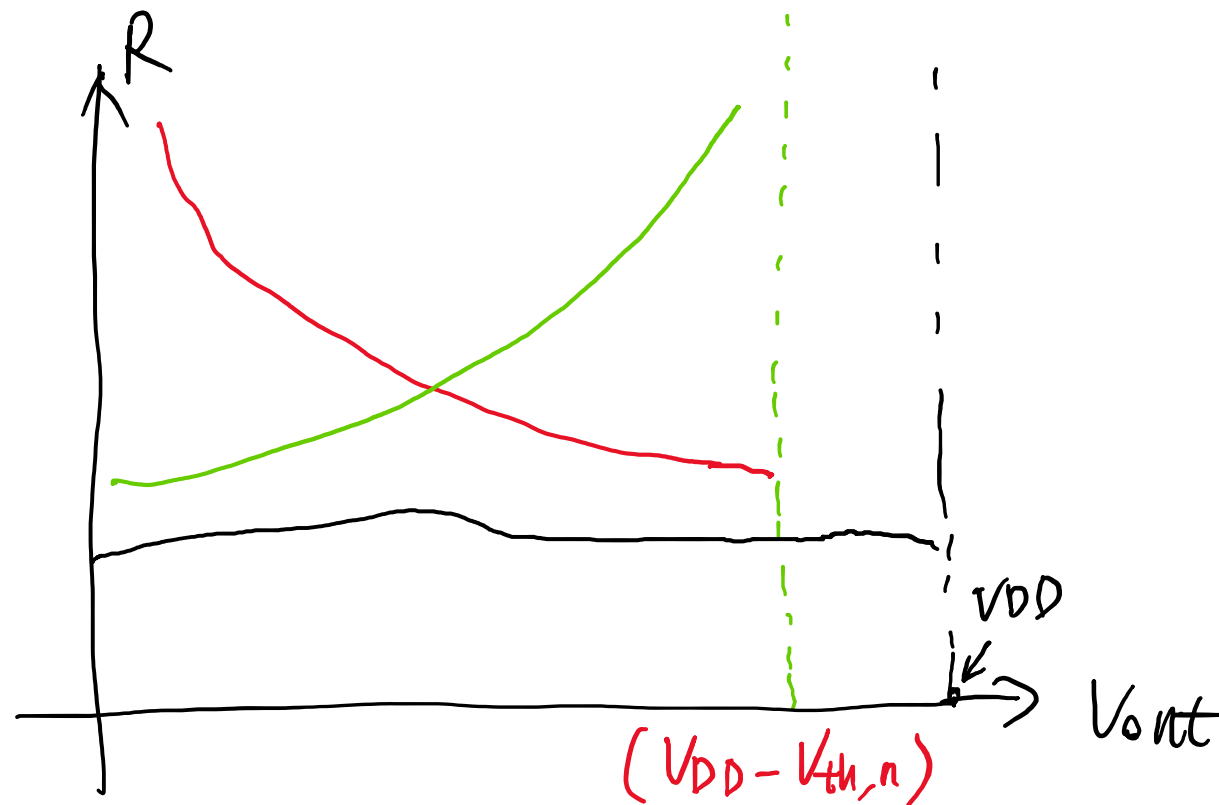
Switch



$$R_{eq,n} = \frac{V_{DD} - V_{out}}{I_{DS,n}}$$

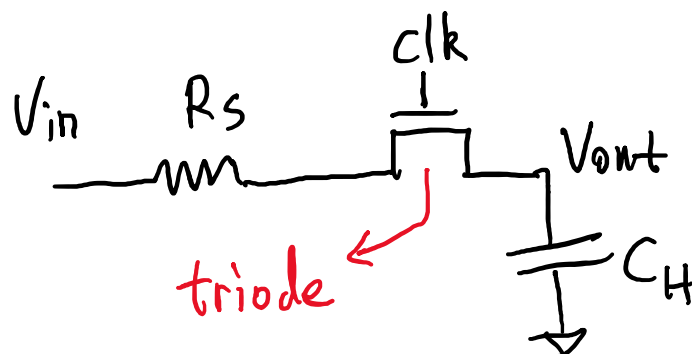
$$R_{eq,p} = \frac{V_{DD} - V_{out}}{I_{DS,p}}$$

$$R_{eq} = R_{eq,n} \parallel R_{eq,p}$$

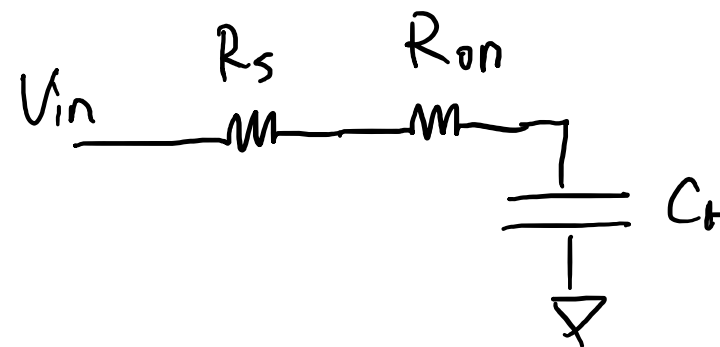


R_{eq} remains relatively constant.

Sample and Hold

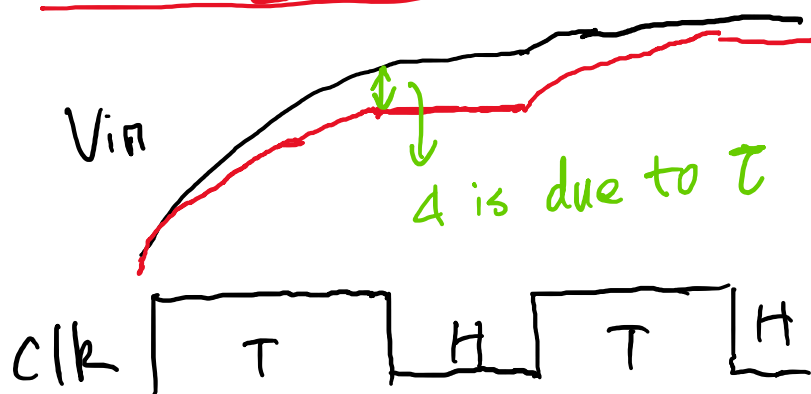


On
 $\Rightarrow$



$\Rightarrow$ RC filter with time constant $\tau = (R_s + R_{on}) C_H$

Tracking Bandwidth

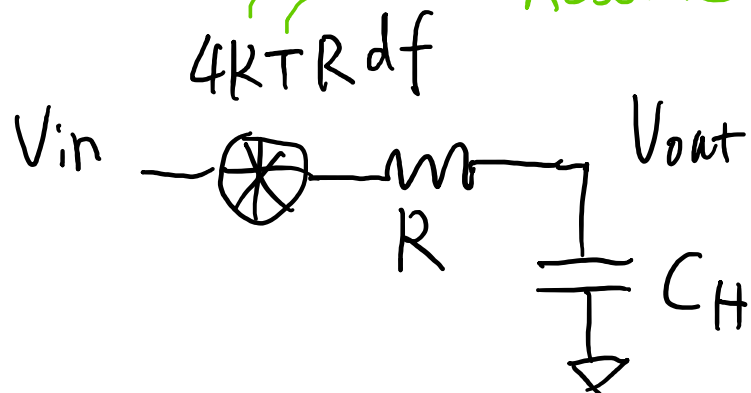


Accuracy	tracking time / τ
10%	2.3
1%	4.6
0.1%	6.9
0.01%	9.2
0.001%	11.5

Sample and Hold



Noise → Boltzmann's constant
→ Absolute temperature



$$V_N^2 = \int_0^\infty 4kTR \left| \frac{1}{1 + j2\pi f \cdot R C_H} \right|^2 df$$

@ output

$$= \frac{kT}{C_H}$$

Solution:
kT/c noise

cancellation

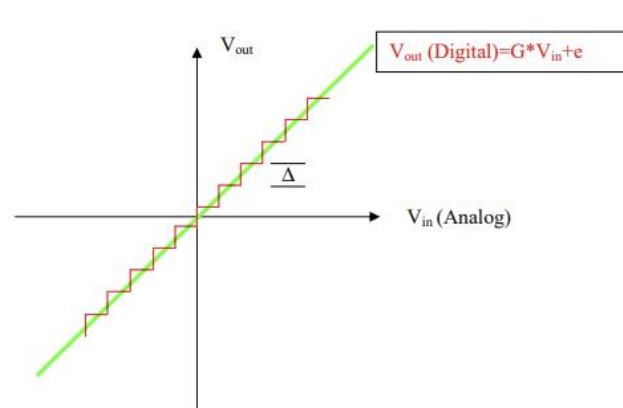
For a Nyquist rate ADC:
total quantization noise power $\approx \frac{\Delta^2}{12}$
where Δ is the LSB step.

$$\frac{kT}{C} \leq \frac{\Delta^2}{12}$$

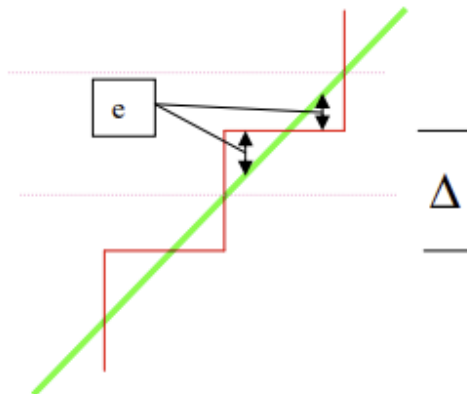
$$\Rightarrow C > 12kT \left(\frac{2^{\text{bit}} - 1}{V_{FS}} \right)^2$$

Resolution (bit)	Required $C_{min}(V_{FS}=1V)$
8	0.003 pF
12	0.8 pF
14	13 pF
16	206 pF

Quantization Noise



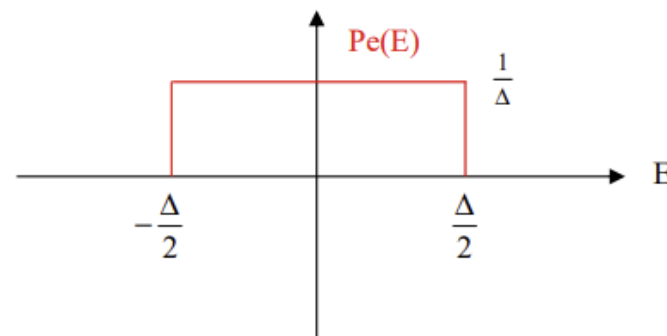
Staircase curve of a linear N Bit ADC Converter



Zoom in of Staircase

Staircase curve of a linear N Bit ADC Converter

- Assume that the input is “busy”, moderate signal level.
- Green curve is a scaled version of V_{in} without any quantization.
- Red curve is the ADC Output.
- Δ is the step size of the converter.



PDF of Quantization Error

- Quantization error is uniformly distributed.
- Integrates to 1

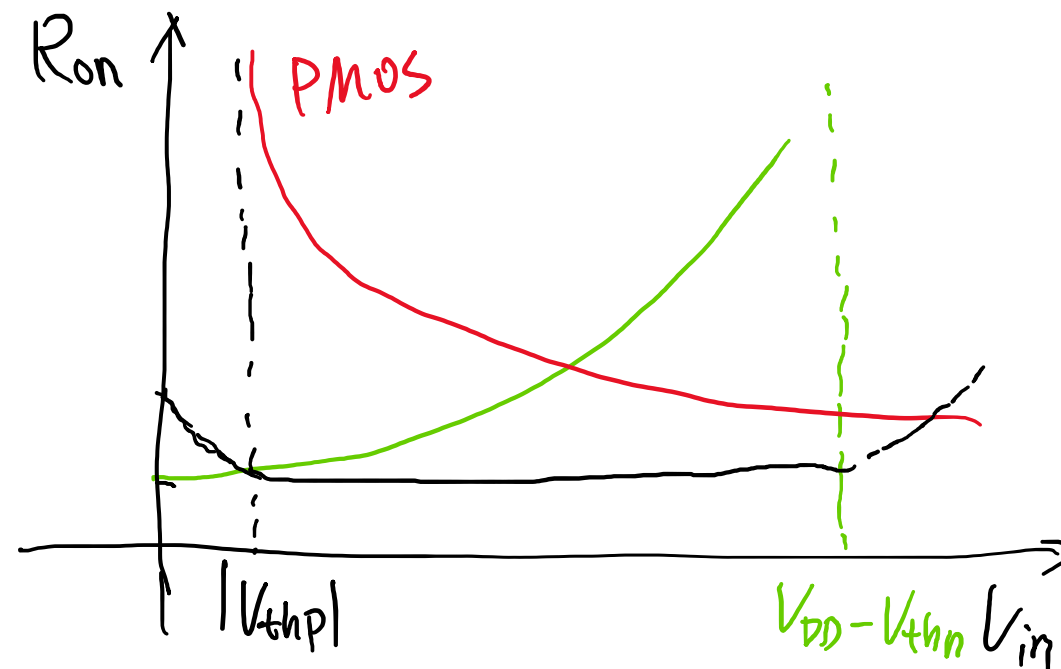
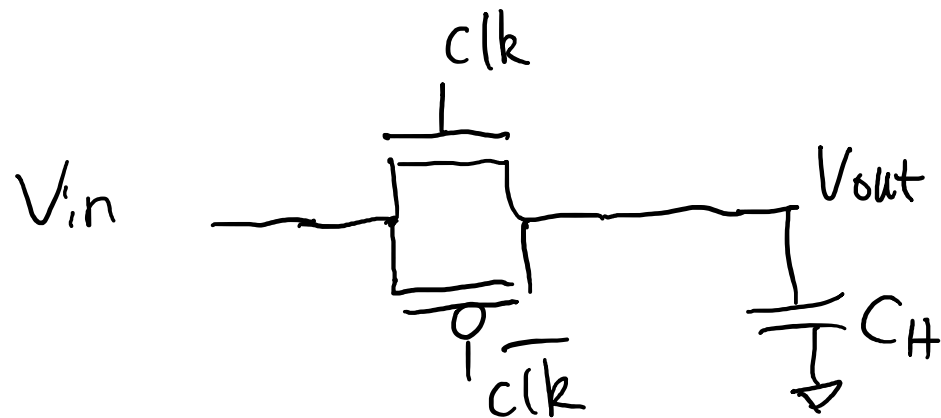
$$\sigma_{QNoise}^2 = E(e^2) = \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} \frac{1}{\Delta} e^2 de = \frac{1}{\Delta} \frac{e^3}{3} \Big|_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} = \frac{1}{\Delta} \left[\frac{\left(\frac{\Delta}{2}\right)^3}{3} - \frac{\left(-\frac{\Delta}{2}\right)^3}{3} \right] = \frac{1}{\Delta} \left[\frac{\Delta^3}{24} - \frac{-\Delta^3}{24} \right] = \frac{\Delta^2}{12}$$

- $\sigma_{QNoise}^2 = \text{Quantization Noise Power} = \frac{\Delta^2}{12}$
- $\sigma_{QNoise} = V_{QNoise_rms} = \frac{\Delta}{\sqrt{12}}$

Sample and Hold



Distortion (Input-dependent R_{on})



✓ constant R_{on} within $|V_{thp}| \sim (V_{DD} - V_{thn})$

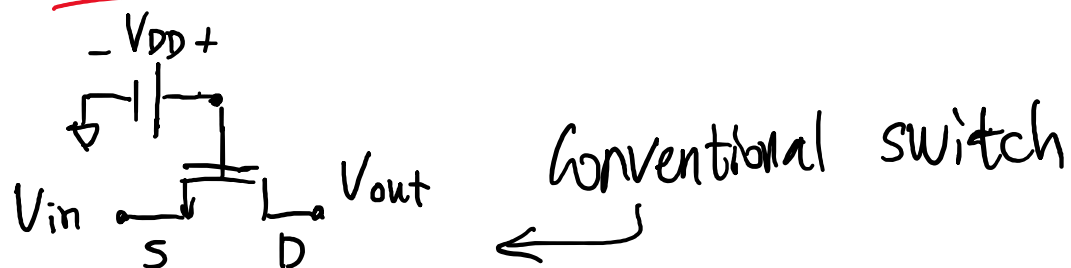
✓ reduce the overall $R_{on} \rightarrow$ smaller $\tau \rightarrow$ faster.

Sample and Hold



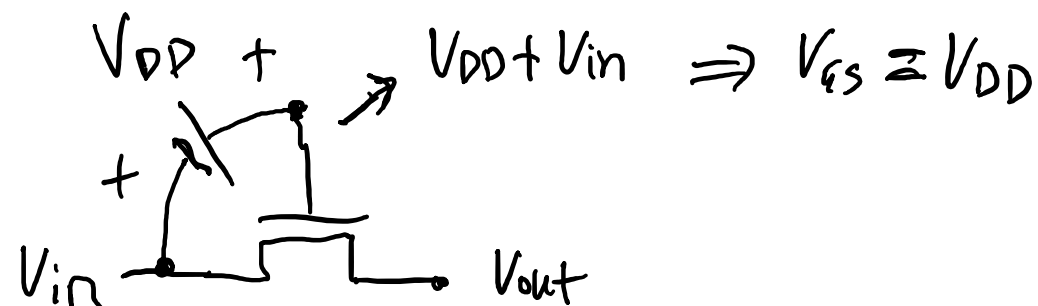
However, as the supply scales down, $|V_{thp}| \sim (V_{DD} - V_{thn})$ shrinks,
 $\Rightarrow$ Complementary switch becomes less efficient.

Bootstrap switch / Constant V_{as} Sampling



$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{as} - V_{th})} \text{ input-dependent}$$

$$\approx \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{DD} - V_{in} - V_{th})}$$



$$\Rightarrow R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{DD} - V_{th})}$$

ref: A. Abo, JSSC 1999.



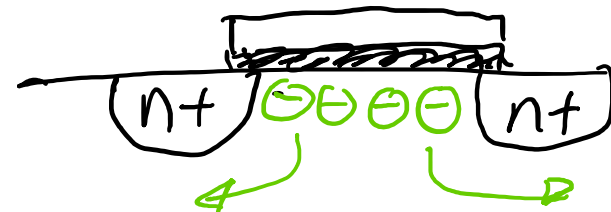
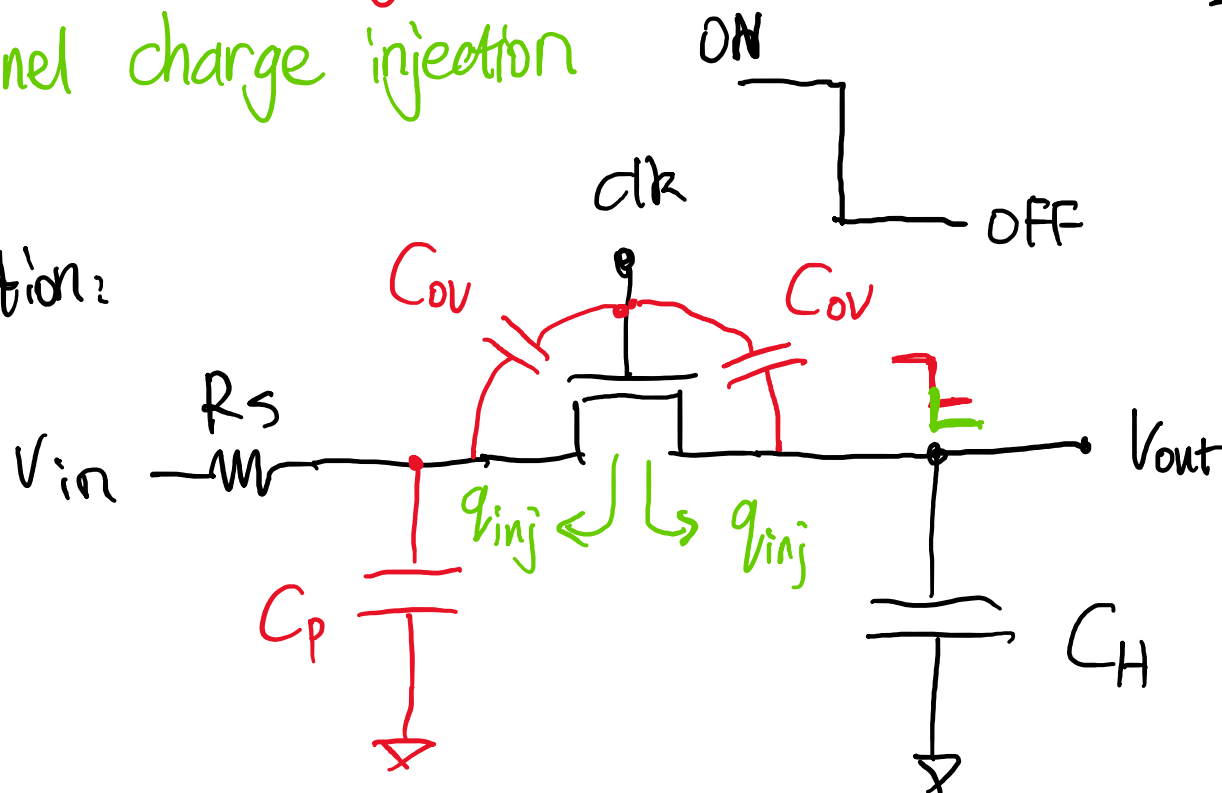
Sample and Hold



Other non-idealities of the switches

- ① clock feedthrough
- ② channel charge injection

Fast transition:

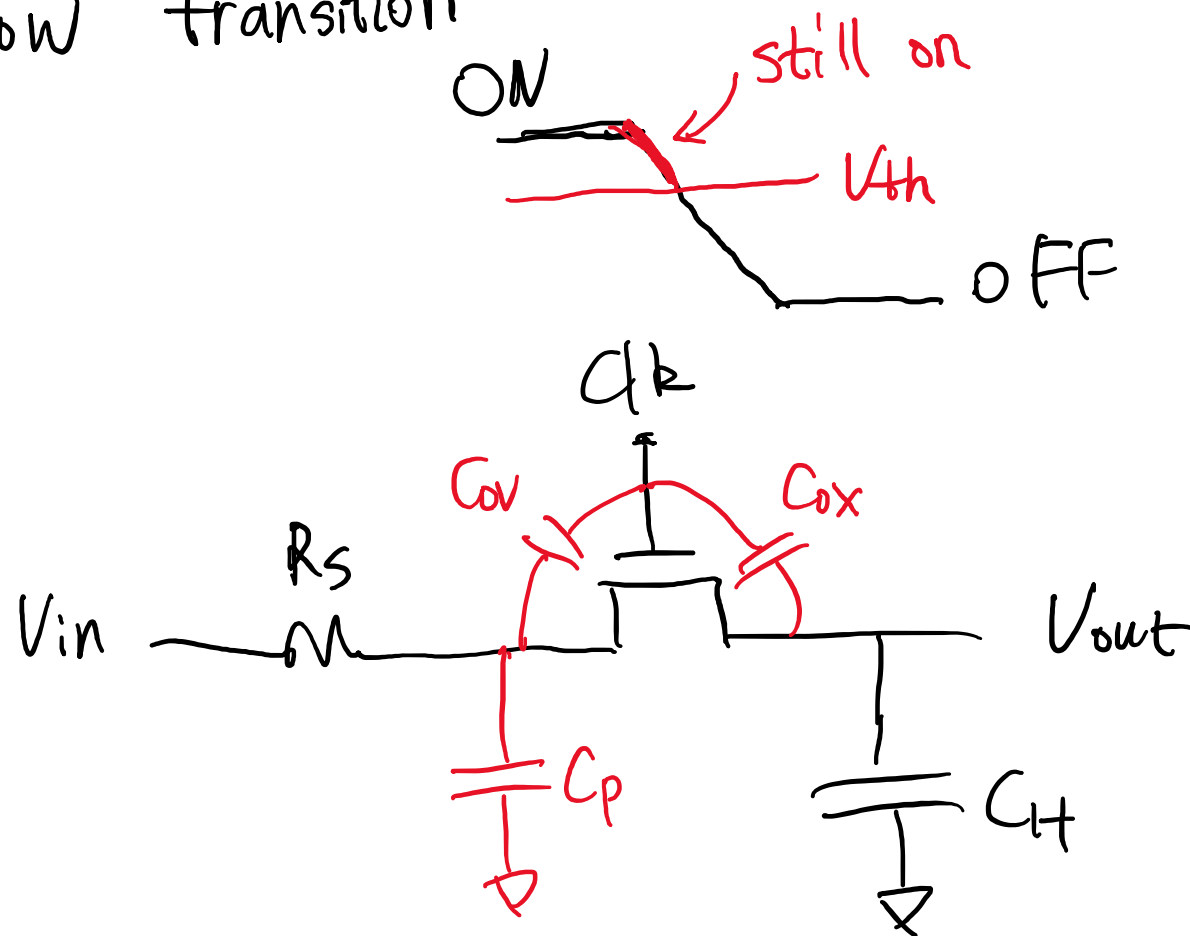


$$\Delta V_{out} = \frac{q_{inj}}{C_H} \quad (2)$$
$$+ \alpha \frac{C_{ov}}{C_{ov} + C_H} V_{swing} \quad (1)$$

Sample and Hold



slow transition



$$\Delta V_{out} = \frac{q_{inj}}{C_H} + \alpha \frac{C_{ov}}{C_{ov} + C_H} \cdot V_{swing}$$

Sample and Hold



charge partitioning depends on $\left\{ \begin{array}{l} \frac{C_H}{C_P} \\ \text{On resistance of the switch} \\ \text{slope of the Clk signal} \end{array} \right.$

Equal charge sharing if $\xrightarrow{\text{either}} C_H = C_P$ (independent of switching time)

$\xrightarrow{\text{or}}$ Switching time is much less than $R_{on} C$ time constant.

(independent of $\frac{C_H}{C_P}$)

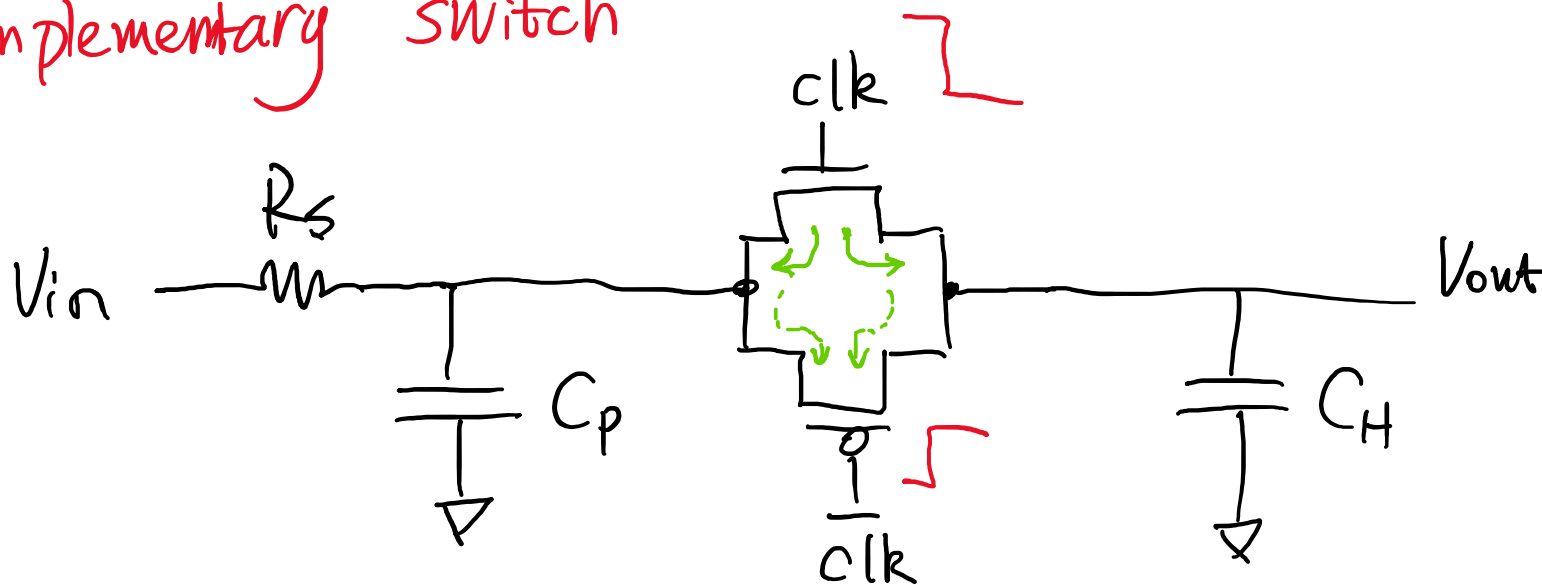
Assumption for
most of the
cases.

Sample and Hold



Mitigate the effect of charge injection & Clk feedthrough

- ① Make the clk transition as slow as possible. $\rightarrow$ limited speed.
- ② Complementary switch



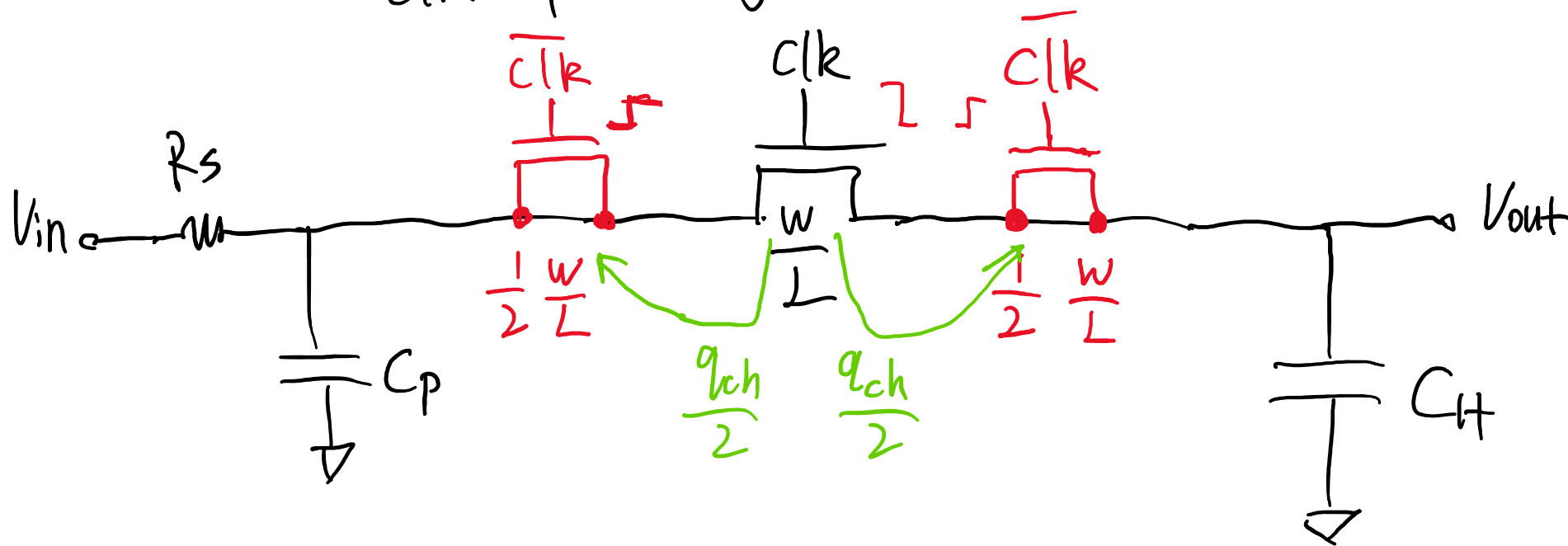
X hard to match NFET & PFET, as well as clk & $\overline{clk}$.

Sample and Hold



③ Dummy switch

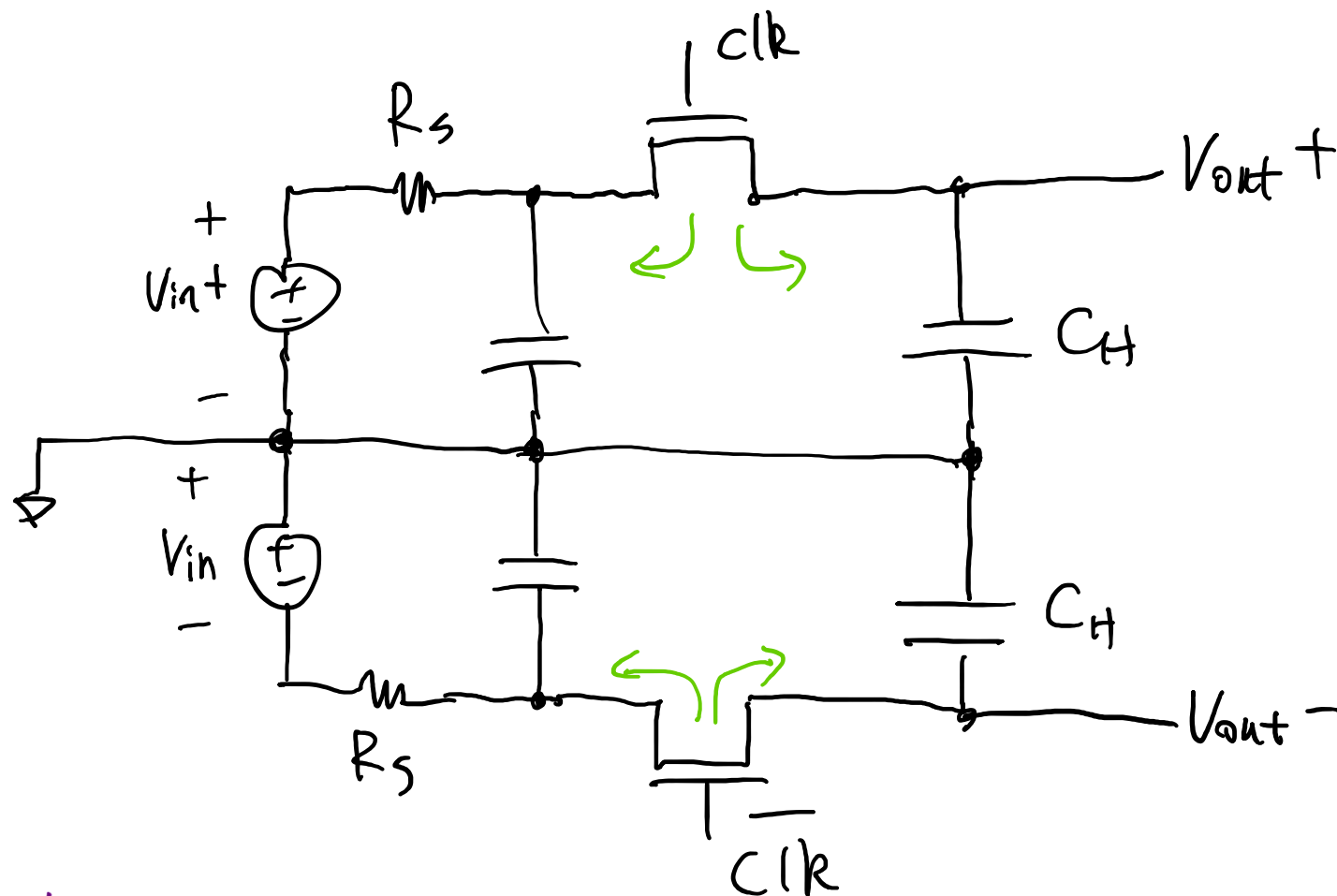
A half-size dummy switch is used to absorb the charge injection and cancel out clk feedthrough.



Sample and Hold



④ Fully-differential sampling



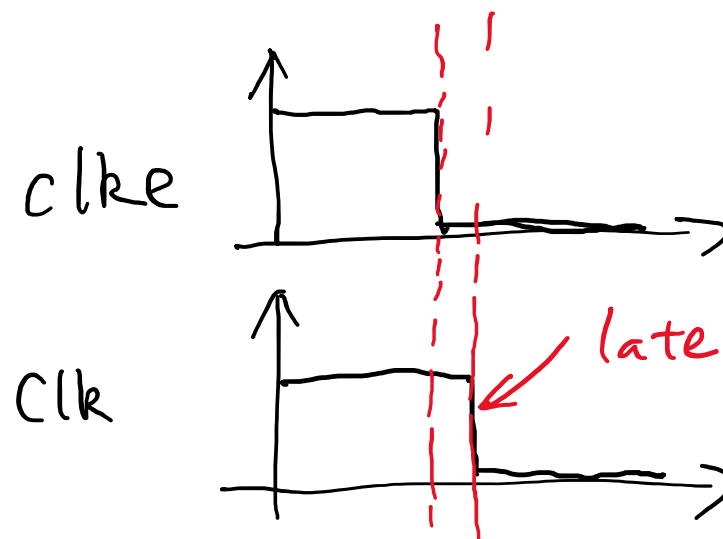
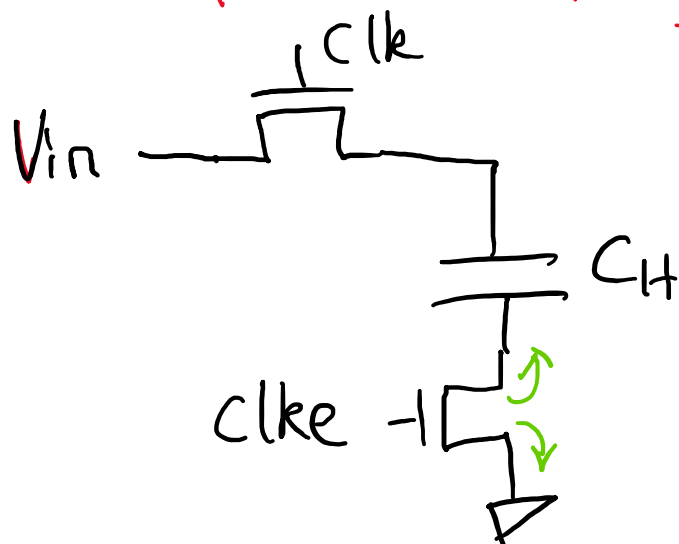
$$V_{out} = V_{out}^{+} - V_{out}^{-}$$

- charge injection
appears as
common-mode

Sample and Hold



⑤ Bottom-plate sampling

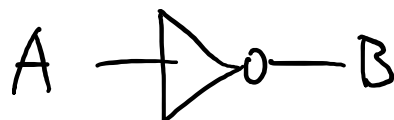


- Bottom switch injects charge on C_H , but it is constant since its V_{gs} is constant and eliminated in a differential implementation.
- Later on, no charge can be deposited on C_H since the bottom side is floating.

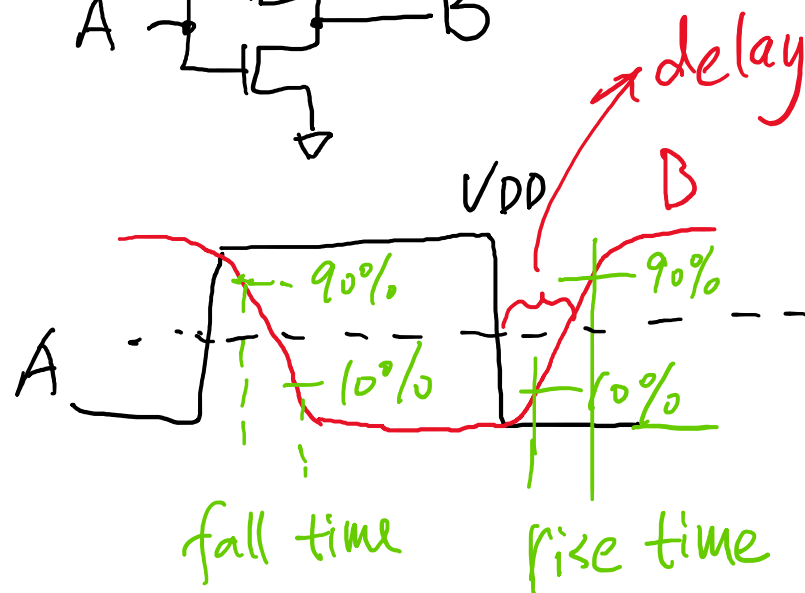
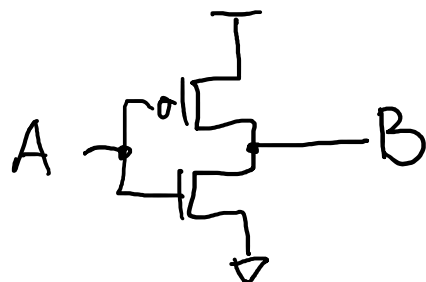
Inverter



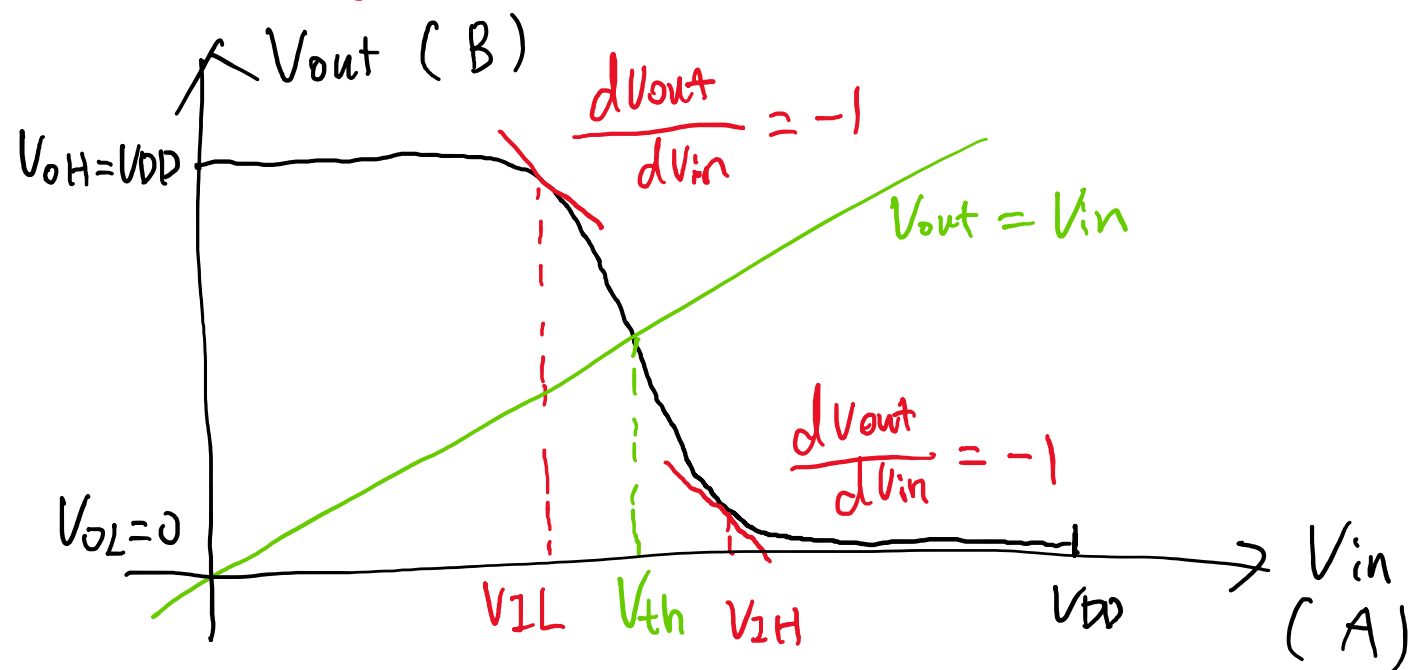
Symbol



A	B
0	1
1	0



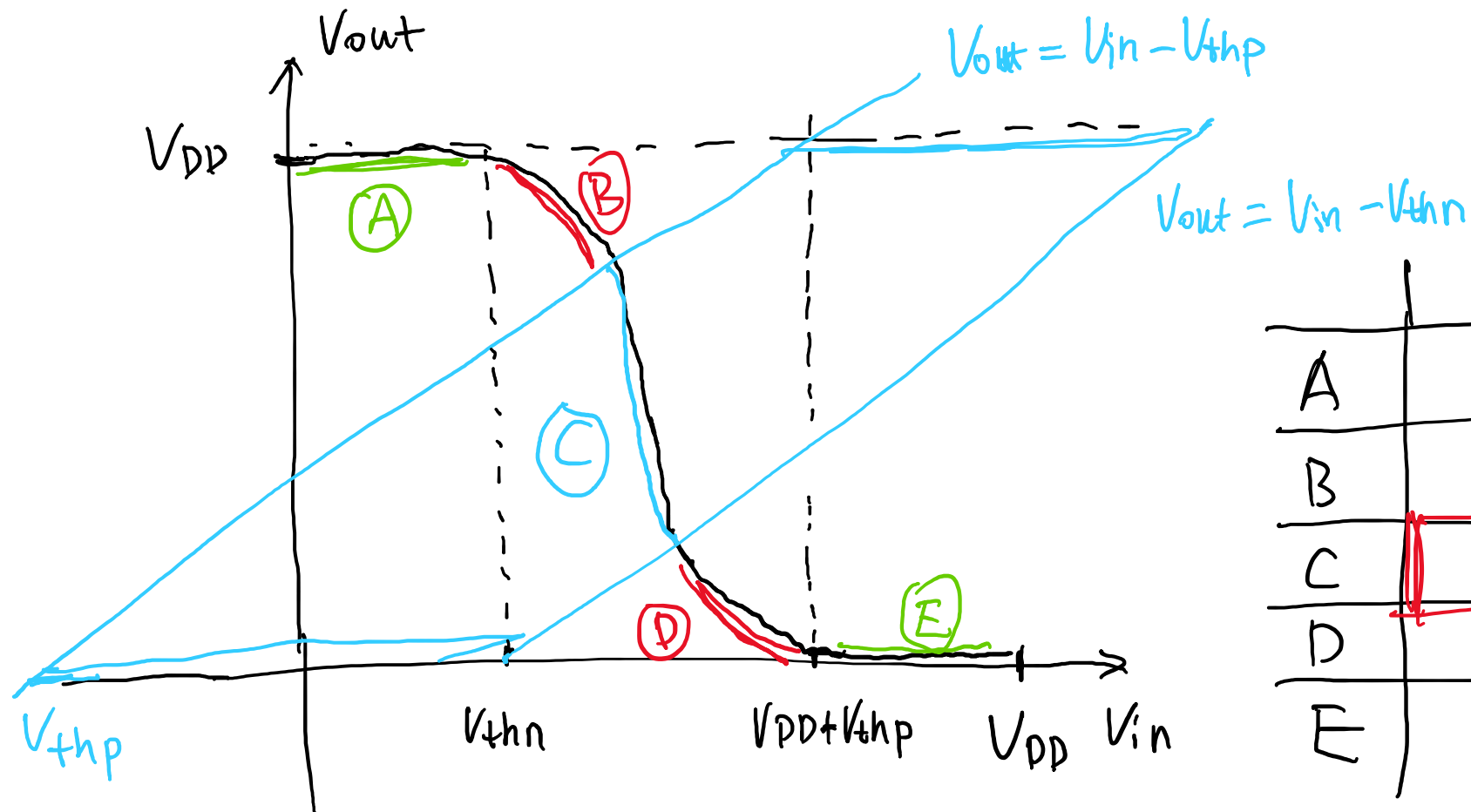
Voltage transfer function



V_{IL} : Maximum input voltage which can be interpreted as logic "1".

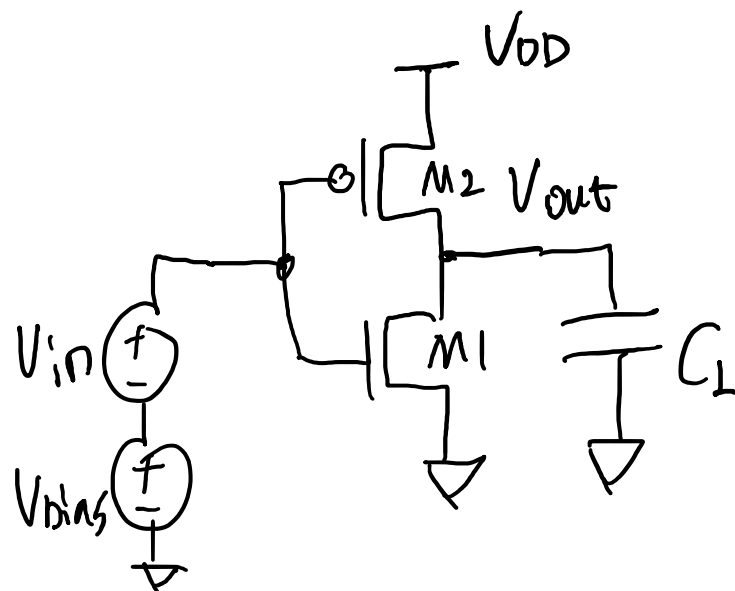
V_{IH} : Minimum input voltage logic "0".

Inverter



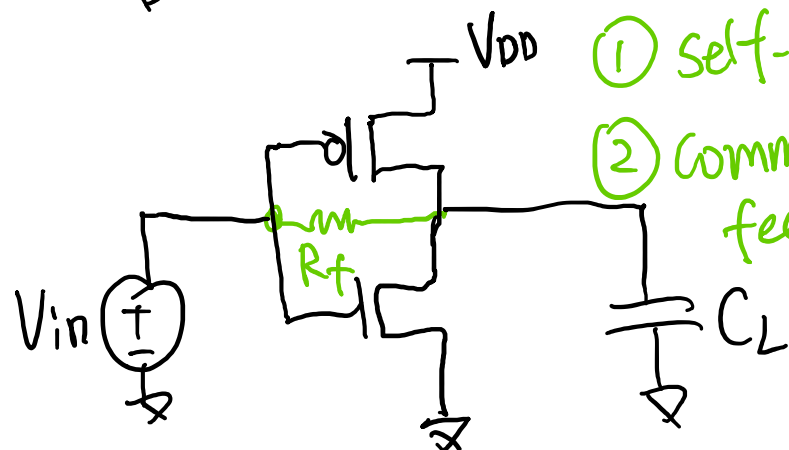
	NMOS	PMOS
A	Cut-off	Linear
B	Sat.	Linear
C	Sat.	Sat.
D	Linear	Sat.
E	Linear	Cut-off

Inverter-Based Amplifier



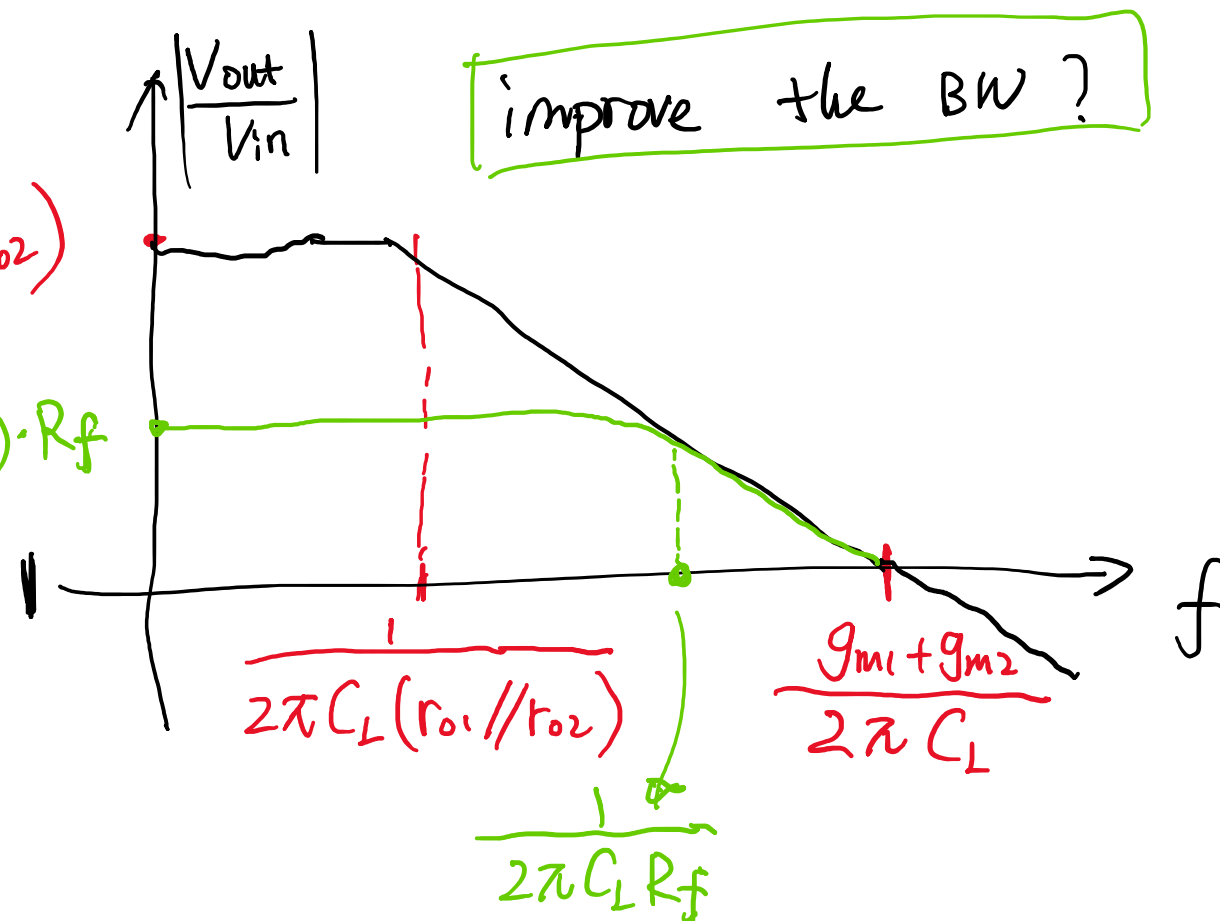
$$(g_{m1} + g_{m2})(r_{o1} // r_{o2})$$

$$\approx (g_{m1} + g_{m2}) \cdot R_f$$



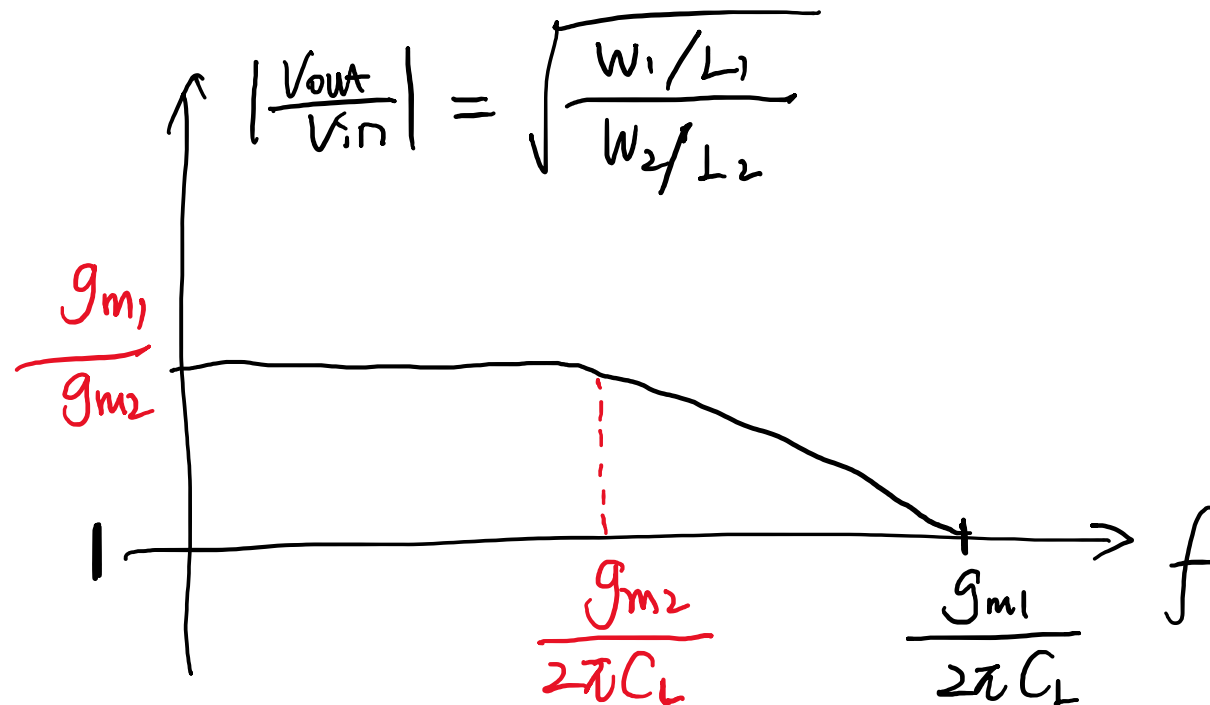
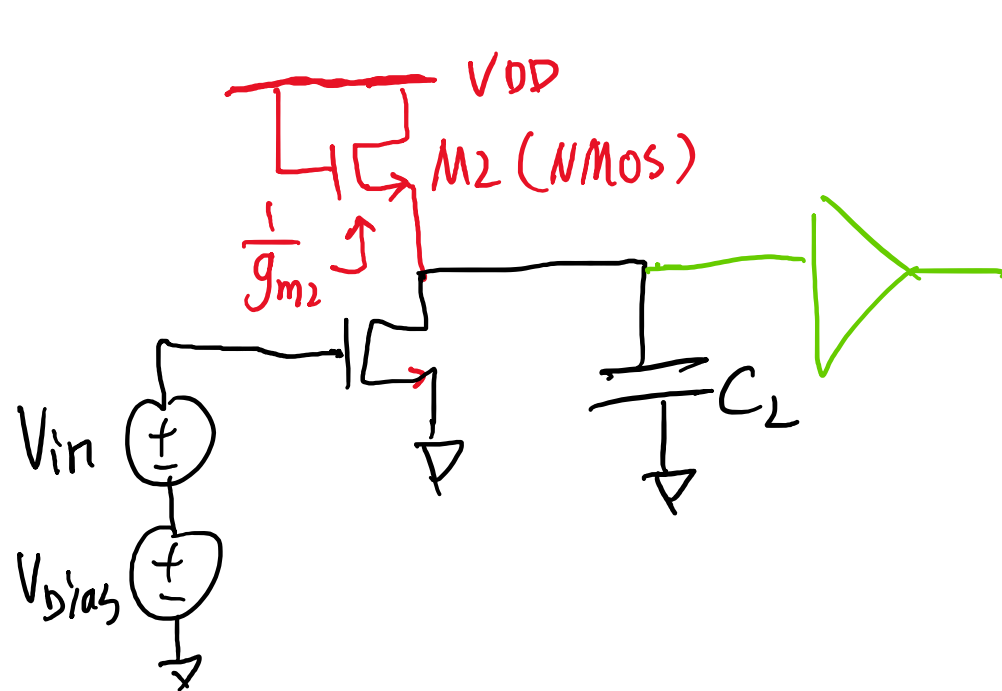
- ① self-biased
- ② common-mode feedback

from



AC simulation in Cadence.

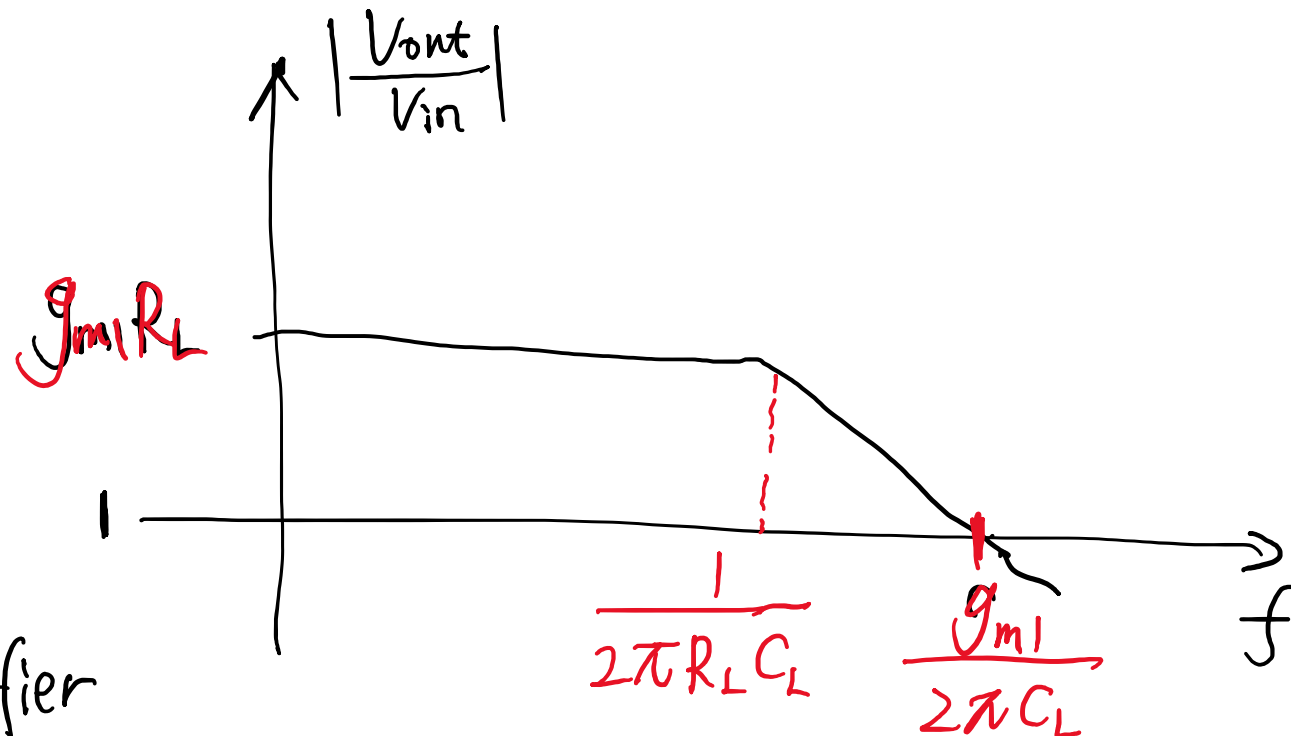
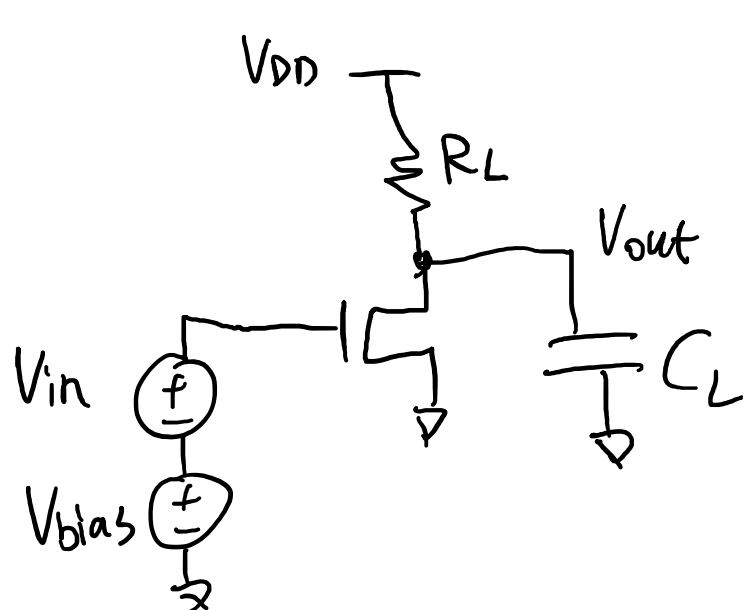
Inverter-Based Amplifier



★ there's no PMOS gate loading the input.

problem: V_{th} of M2 lowers the bias voltage of next stage.

Inverter-Based Amplifier

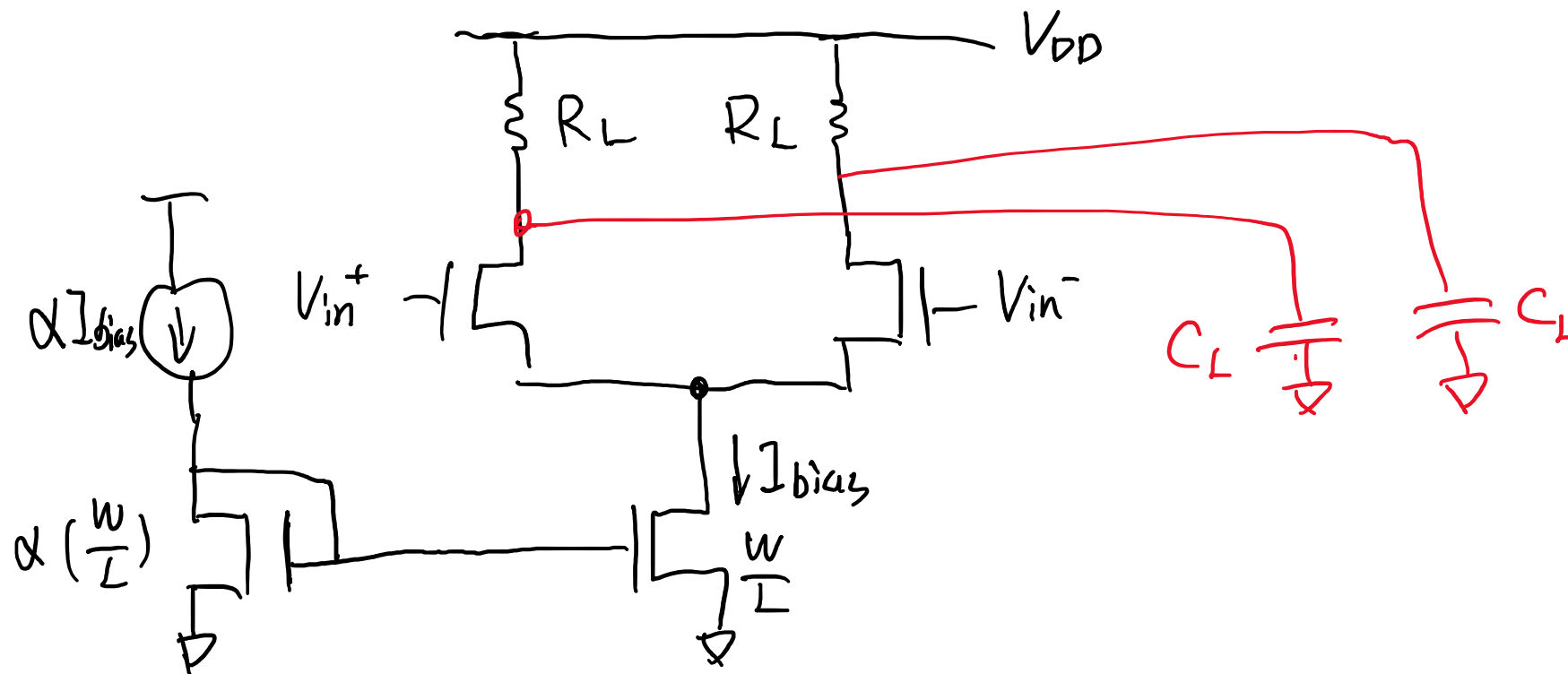


- ★ fastest non-enhanced amplifier
- ★ output can be up to V_{DD}
(allows following stage to achieve high f_t) 如果load为电感, 信号swing多大?
- ★ more linear settling compared to NMOS load.

Implement Using Differential Pairs



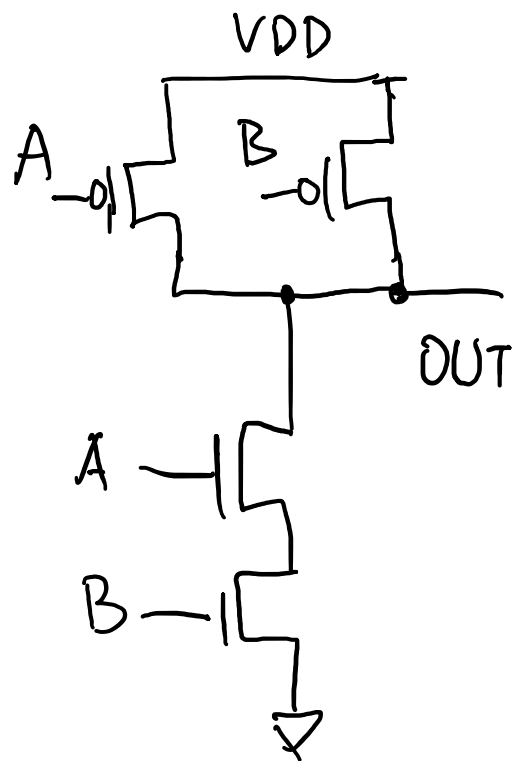
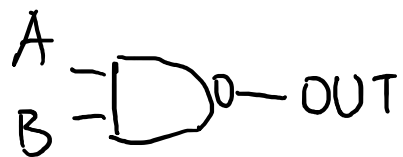
清华大学 集成电路学院
School of Integrated Circuits, Tsinghua University



- ✓ Better common-mode rejection
- ✗ Higher power consumption.

NAND/NOR

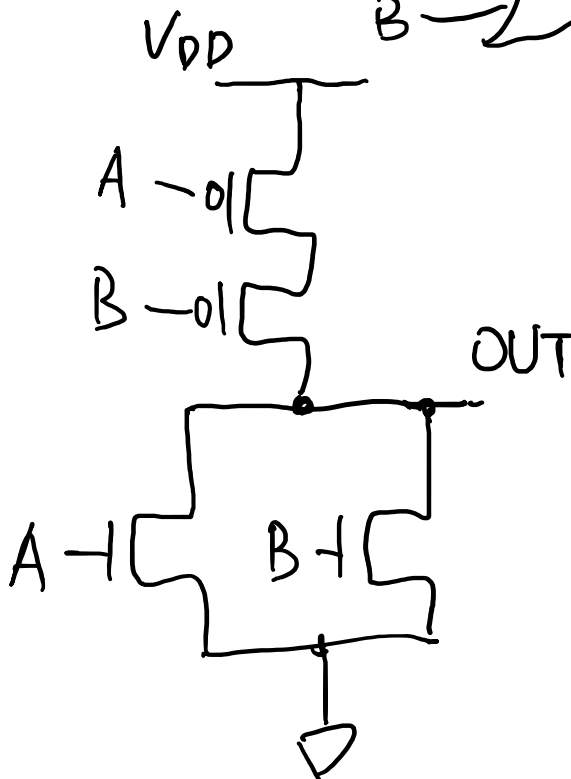
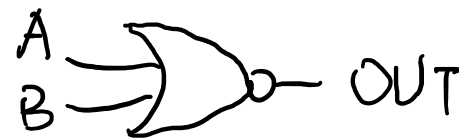
NAND



Truth table

A	B	OUT
0	0	1
0	1	1
1	0	1
1	1	0

NOR



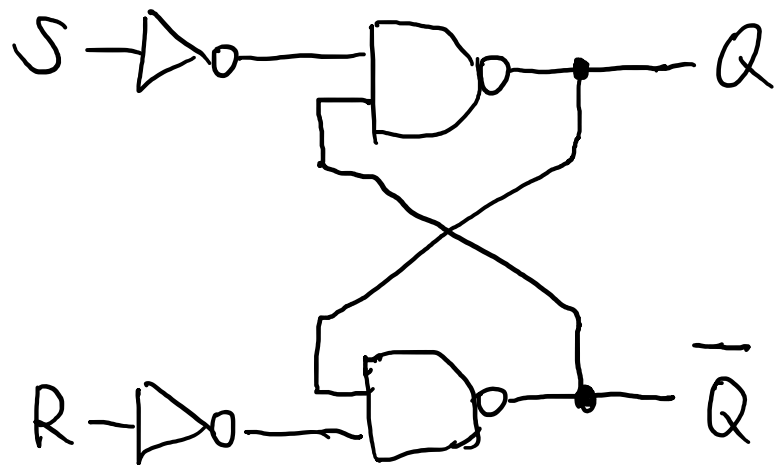
Truth table

A	B	OUT
0	0	1
0	1	0
1	0	0
1	1	0

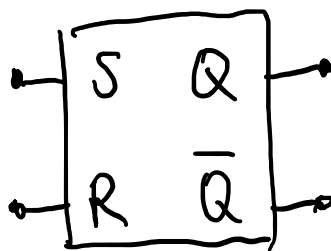
SR Latch



Set_Reset (SR) Latch



Symbol



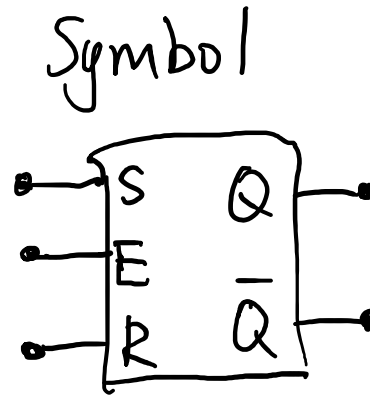
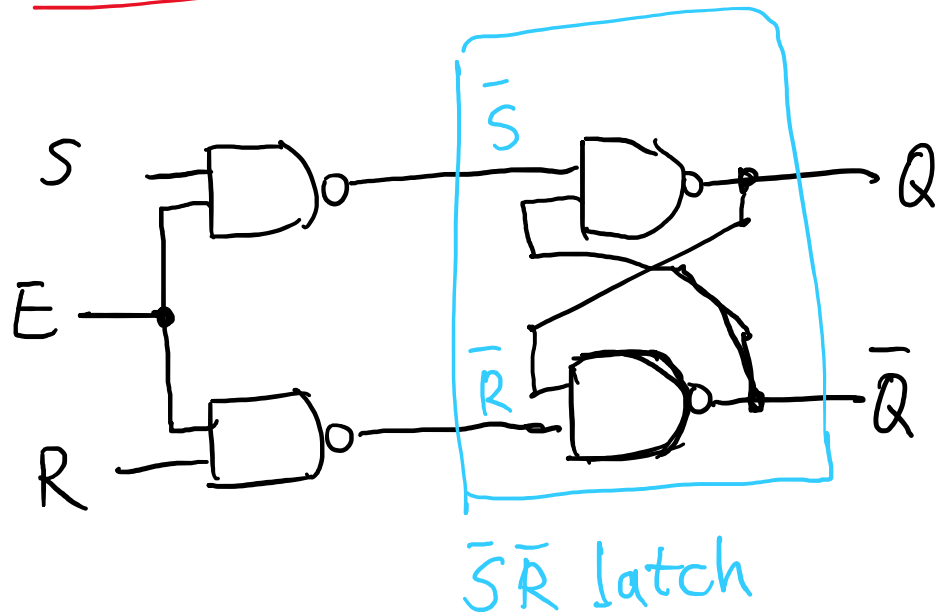
Truth table

S	R	Q	Action
1	1	X	Not Allowed
1	0	1	Set
0	1	0	Reset
0	0	Q	Hold

SR Latch



Synchronous (Clocked)(Gated) SR Latch



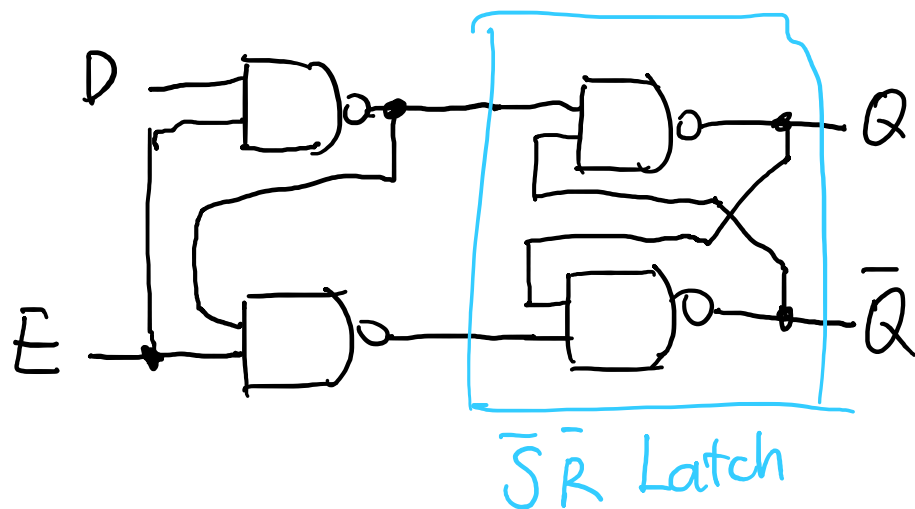
Truth table

E	Action
0	No Action - keep static
1	Same as SR Latch

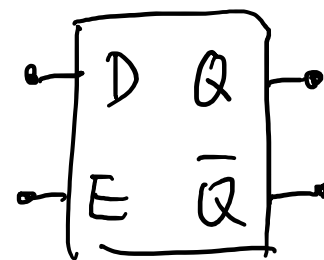
D Latch



Data (D) Latch (1)



Symbol



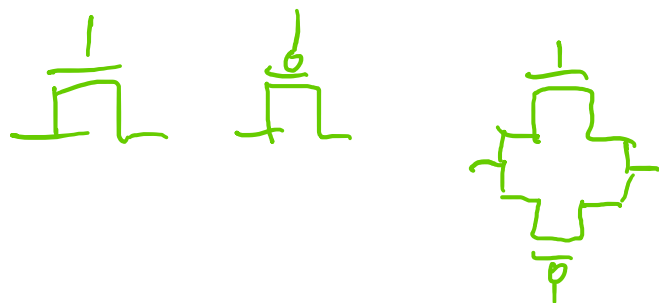
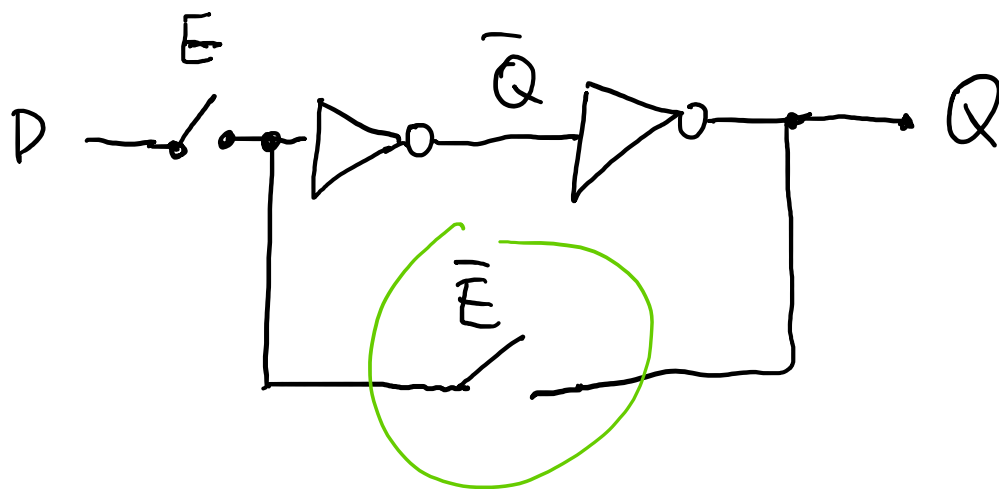
Truth table

$\bar{E}$	D	Q	$\bar{Q}$	Action
0	X	Q	$\bar{Q}$	Hold (No Action)
1	0	0	1	Reset
1	1	1	0	Set

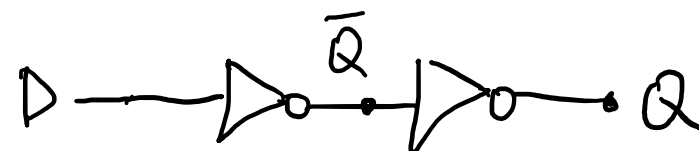
D Latch



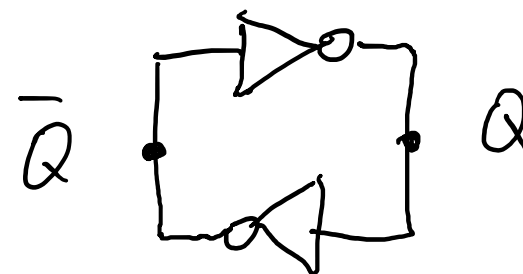
Data (D) Latch (2)



When $\bar{E} = 1$ (buffer)



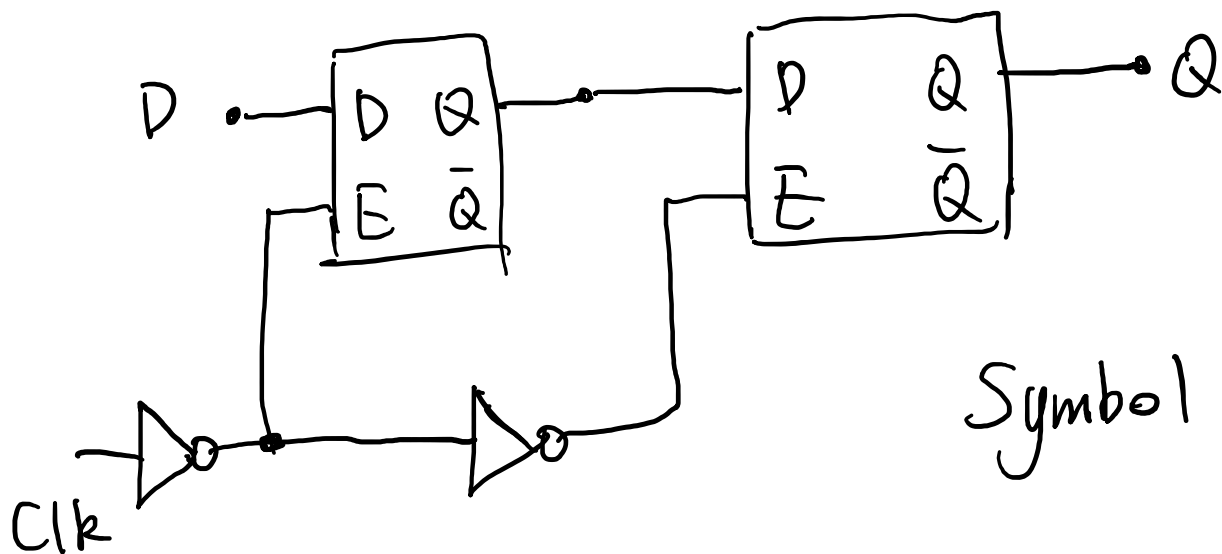
When $\bar{E} = 0$ (latch)



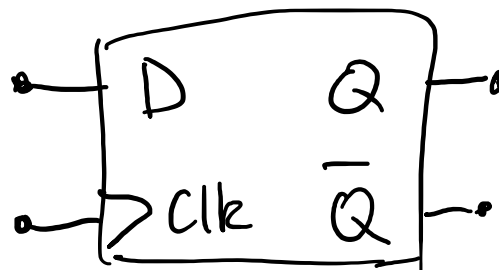
D Flip-Flop



Data (D) Flip-Flop (FF)



Symbol



clock	D	Q_{next}
rising edge	0	0
rising edge	1	1
Not rising	X	Q

D Flip-Flop



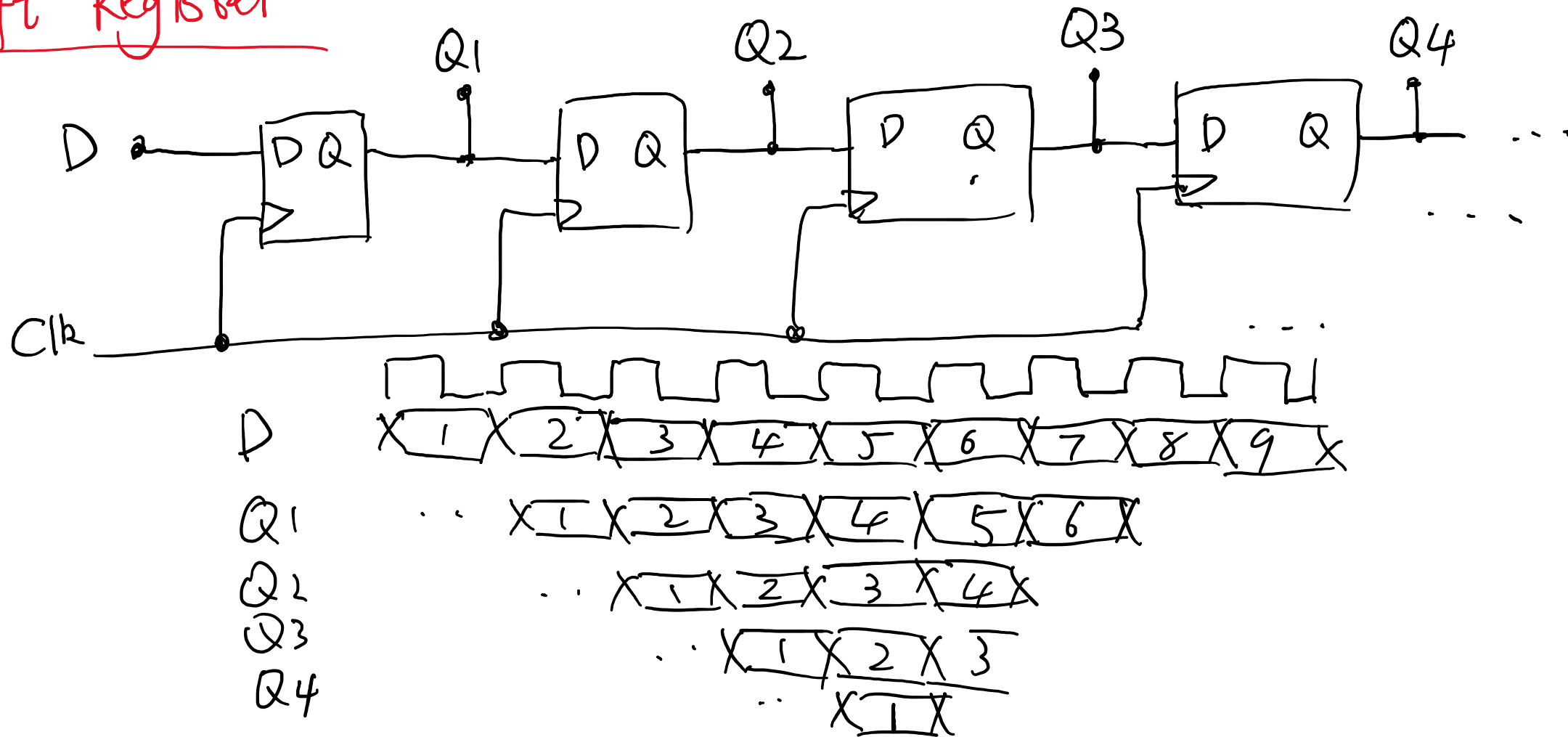
Different implementations of FF

1. transmission gate master-slave (MS) FF
2. C^2MOS MS FF
3. single-transistor-clocked MS FF
4. Hybrid latch FF (Pulsed-triggered latches)
5. semi-dynamic FF
6. Sense-amplifier-based FF
- ⋮

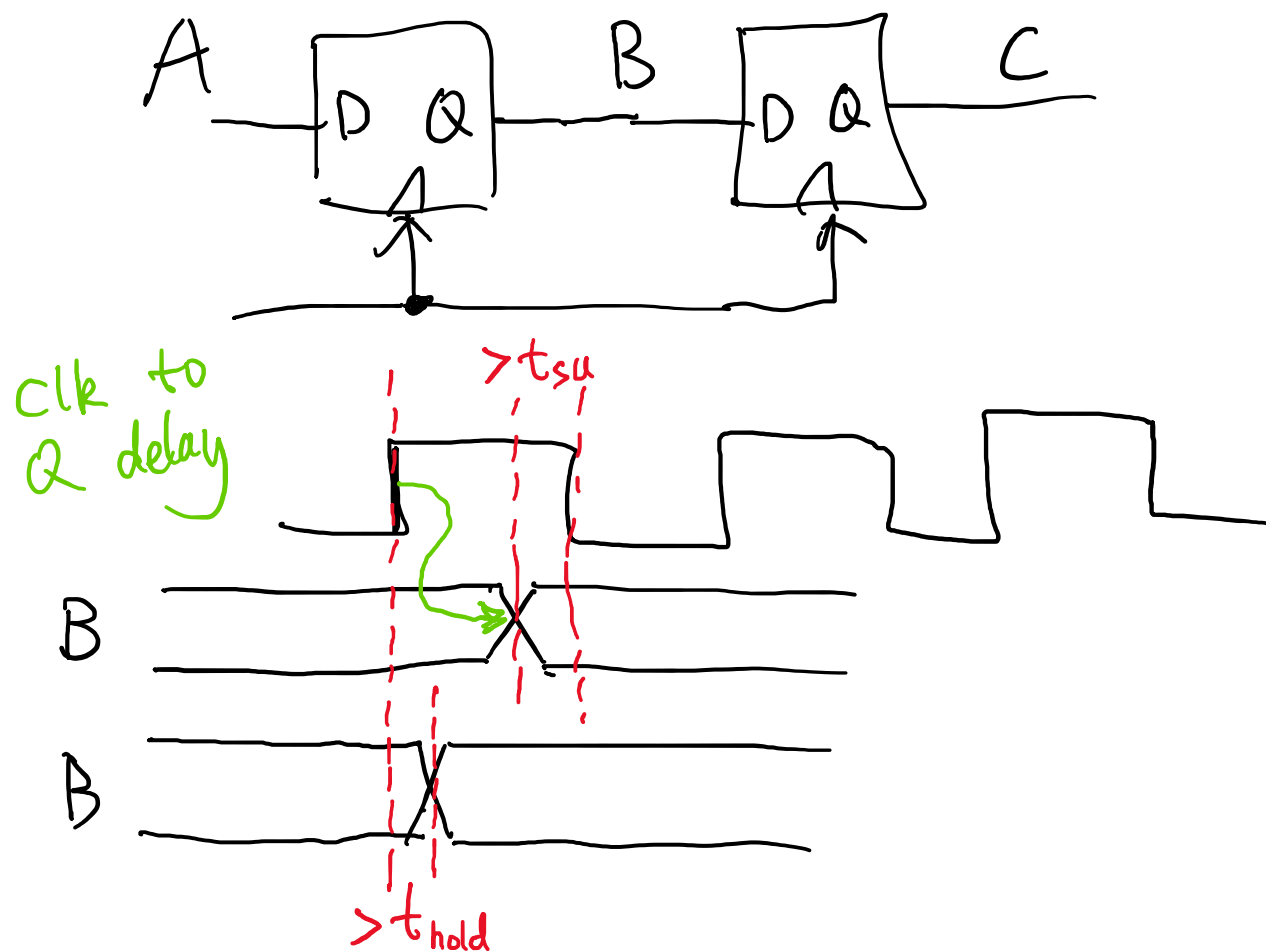
Shift Register



Shift Register



Setup/Hold Time



Solutions:

Setup time: ① $\uparrow$ size
violation

② $\uparrow$ supply

Hold time violation: ① insert buffers

② attach caps.

③ $\downarrow$ size

Current Mode Logic

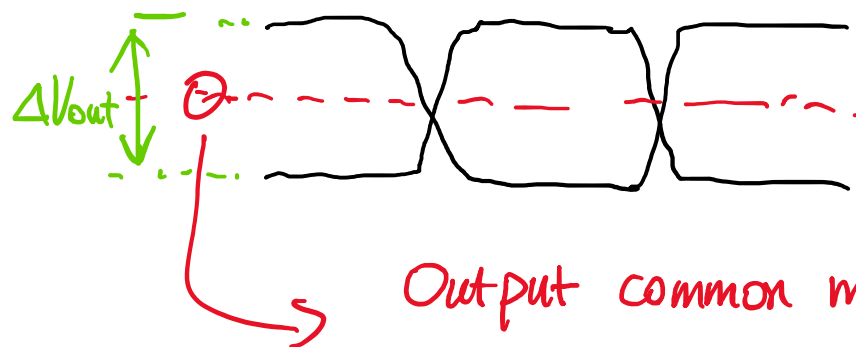
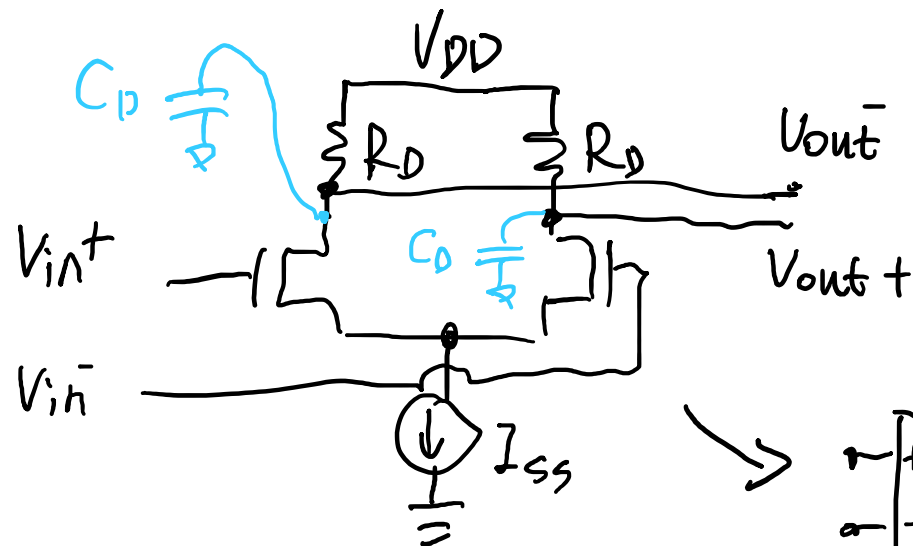


Current_Mode Logic (CML) $\Rightarrow$ Fast!

logic levels:

$$V_{OH} = V_{DD}$$

$$V_{OL} = V_{DD} - R_D I_{SS}$$



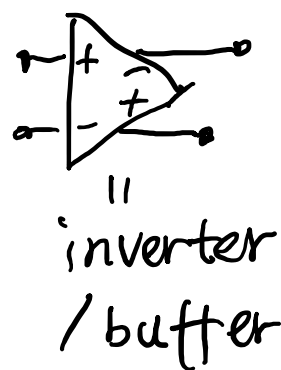
Output common mode

$$\text{Voltage} = V_{DD} - R_D \frac{I_{SS}}{2}$$

$$\Delta V_{out} = V_{OH} - V_{OL} = R_D I_{SS}$$

$$W_{-3dB} = \frac{1}{R_D C_D} = \frac{I_{SS}}{\Delta V_{out} C_D}$$

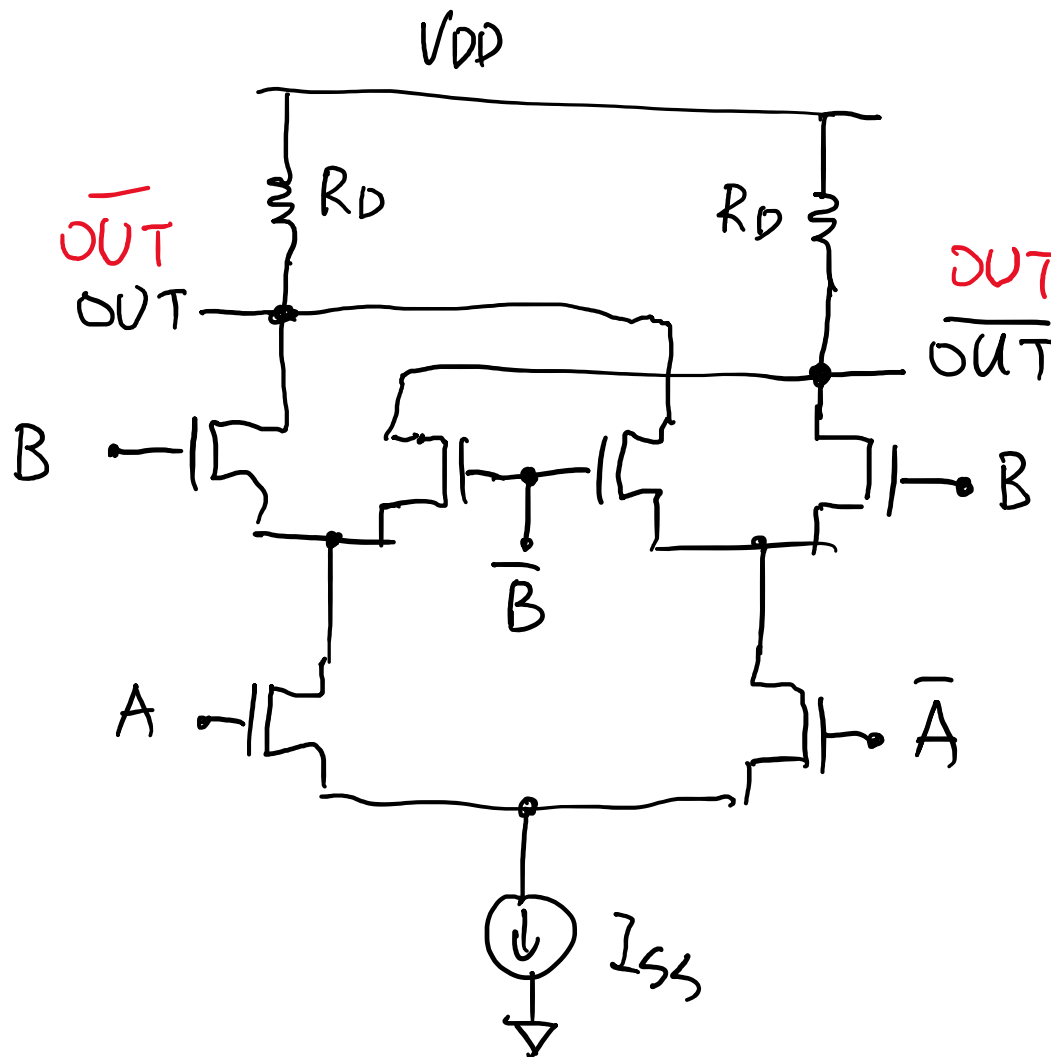
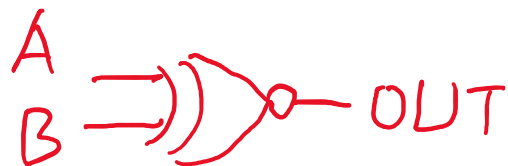
$$\Rightarrow I_{SS} = W_{-3dB} \times \Delta V_{out} \times C_D$$



$$P_{diss} = V_{DD} \cdot I_{SS}$$

CML AND/NAND

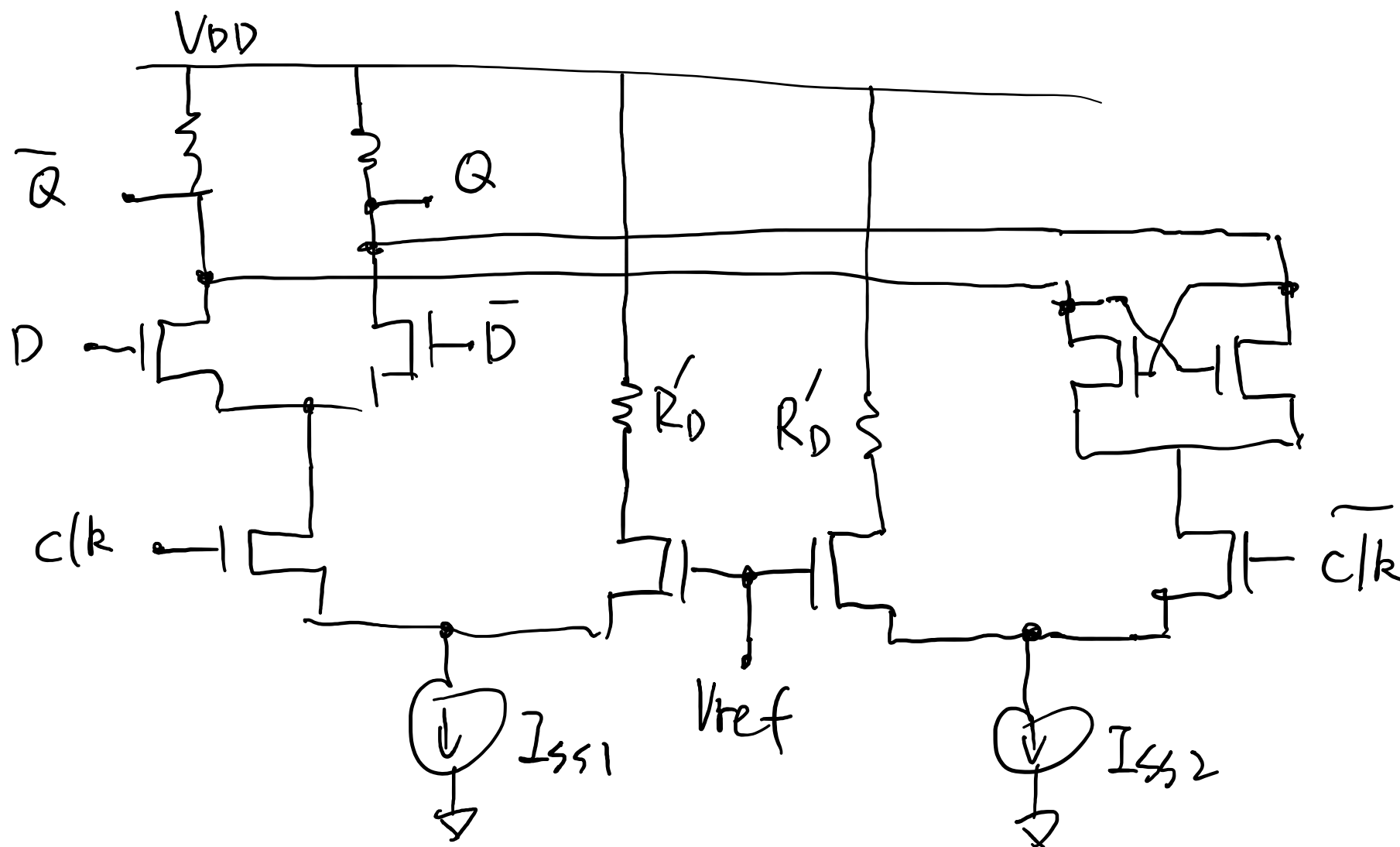




CML D Latch



CML D Latch



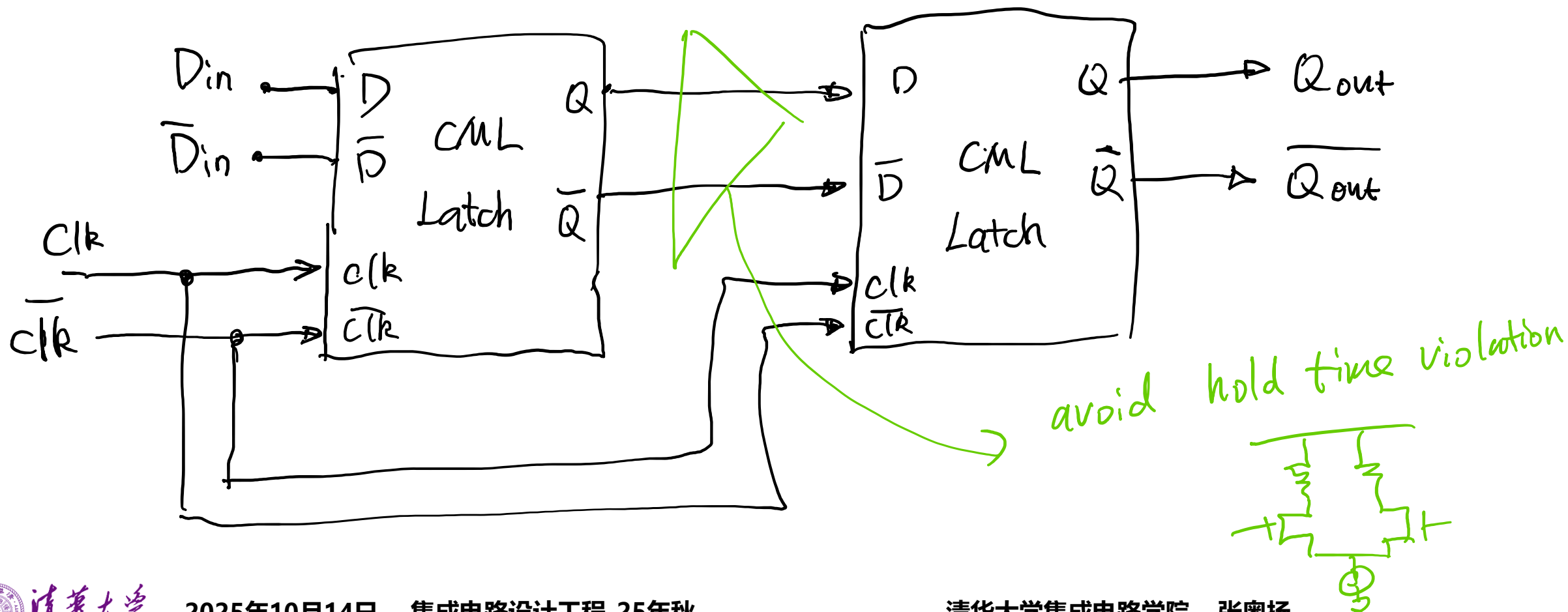
CML D Latch



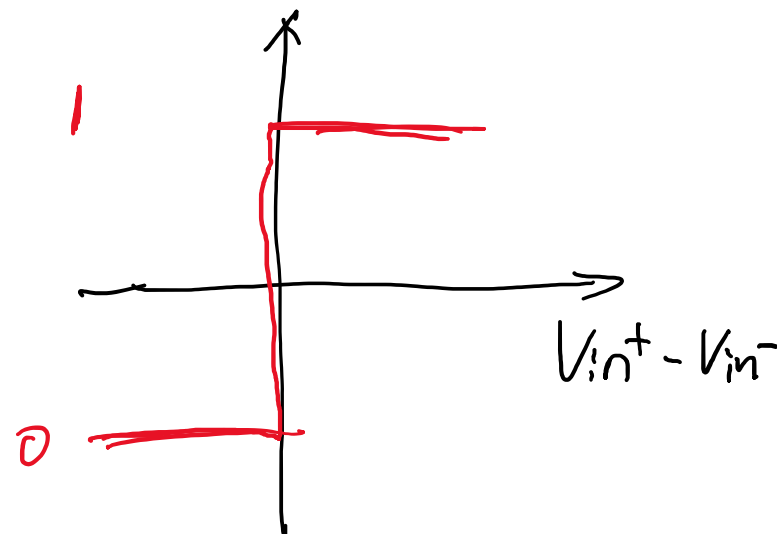
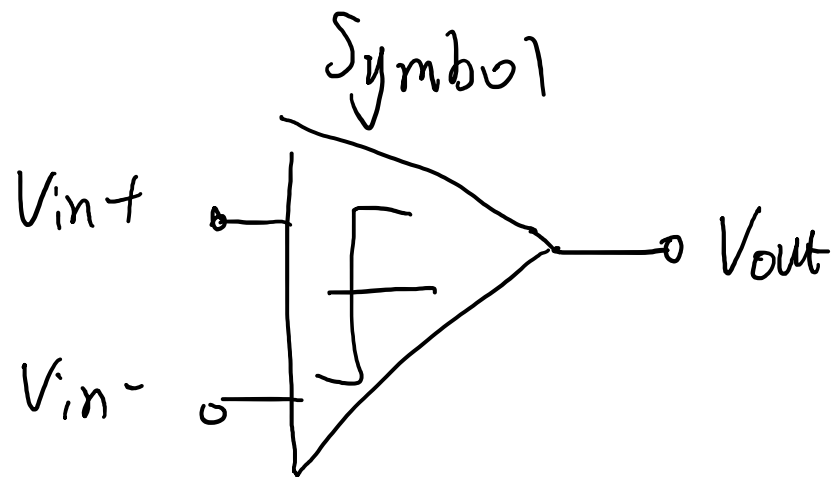
CML D Flip-Flop



Master-Slave CML D Flip-Flop



Comparator



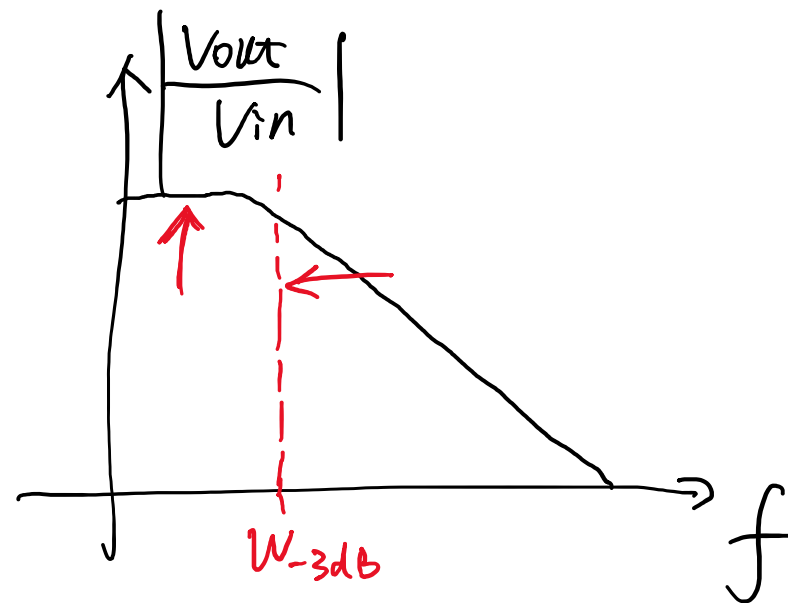
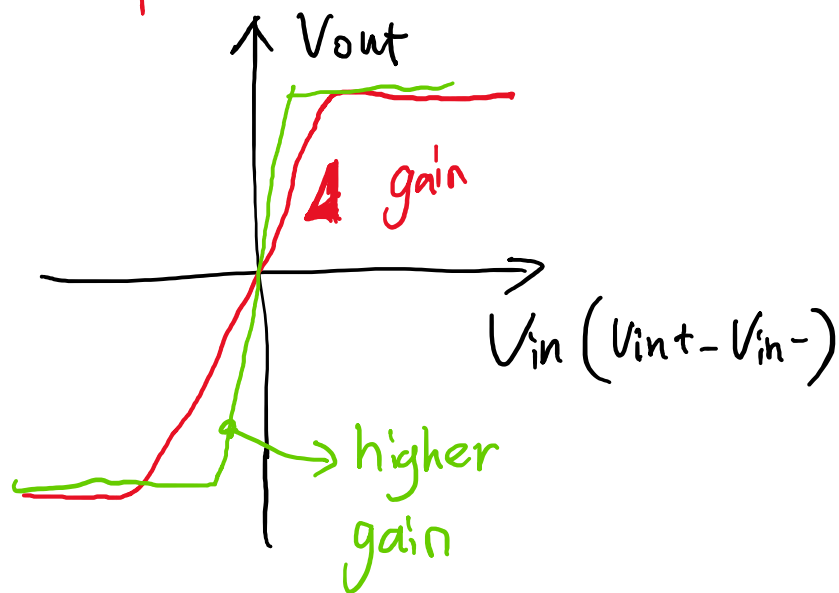
- Compare two input signals, and output the polarity.

$$\text{When } \begin{cases} V_{in+} > V_{in-} & \Rightarrow V_{out} = "1" \\ V_{in+} < V_{in-} & \Rightarrow V_{out} = "0" \end{cases}$$

Comparator

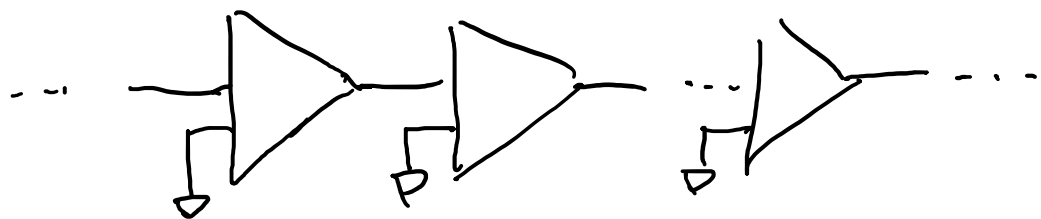


Comparator Build with High-Gain Amplifier



- High gain $\Rightarrow$ slow

Cascaded Lower-Gain Amplifiers

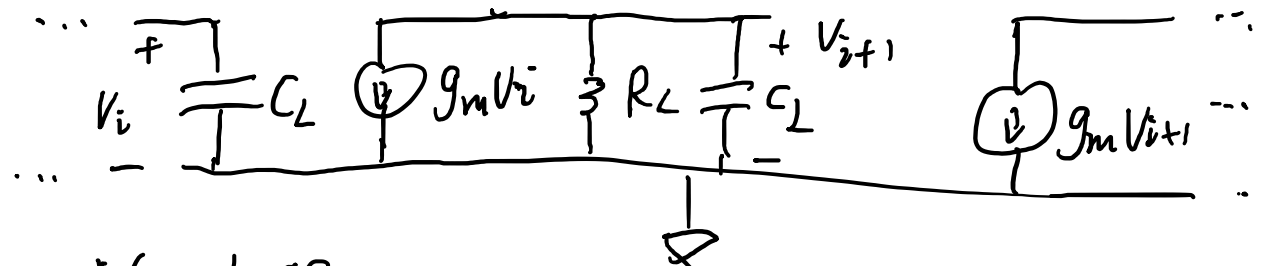


One stage:

$$A_1 = \frac{V_{out}}{V_{in}} = \frac{g_m R_L}{1 + j \frac{\omega}{\omega_0}}$$

$$\omega_0 = \frac{1}{R_L C_L}$$

3dB bandwidth



N stage:

$$A_N = \frac{(g_m R_L)^N}{1 + j \frac{\omega}{\alpha \cdot \omega_0}}$$

3dB bandwidth
of N-stage
amplifier

$$\alpha = (g_m R_L)^{N-1} \sqrt{2^{1/N} - 1}$$

Comparator



Fix total DC gain = 10000

N	α	$A_1(\text{DC})$
1	1	10000
2	64	100
3	236	21.5
4	435	10

$N \uparrow \Rightarrow \text{single-stage gain} \downarrow$

$N \uparrow \Rightarrow \text{bandwidth} \uparrow$

How to choose # of stage?

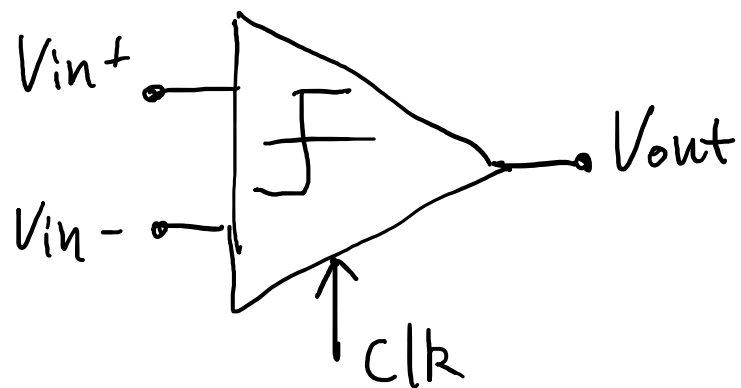
BW: $\frac{\text{gain}}{N} \downarrow$ } trade off

DC Offset: $\frac{\text{gain}}{N} \uparrow$

Comparator

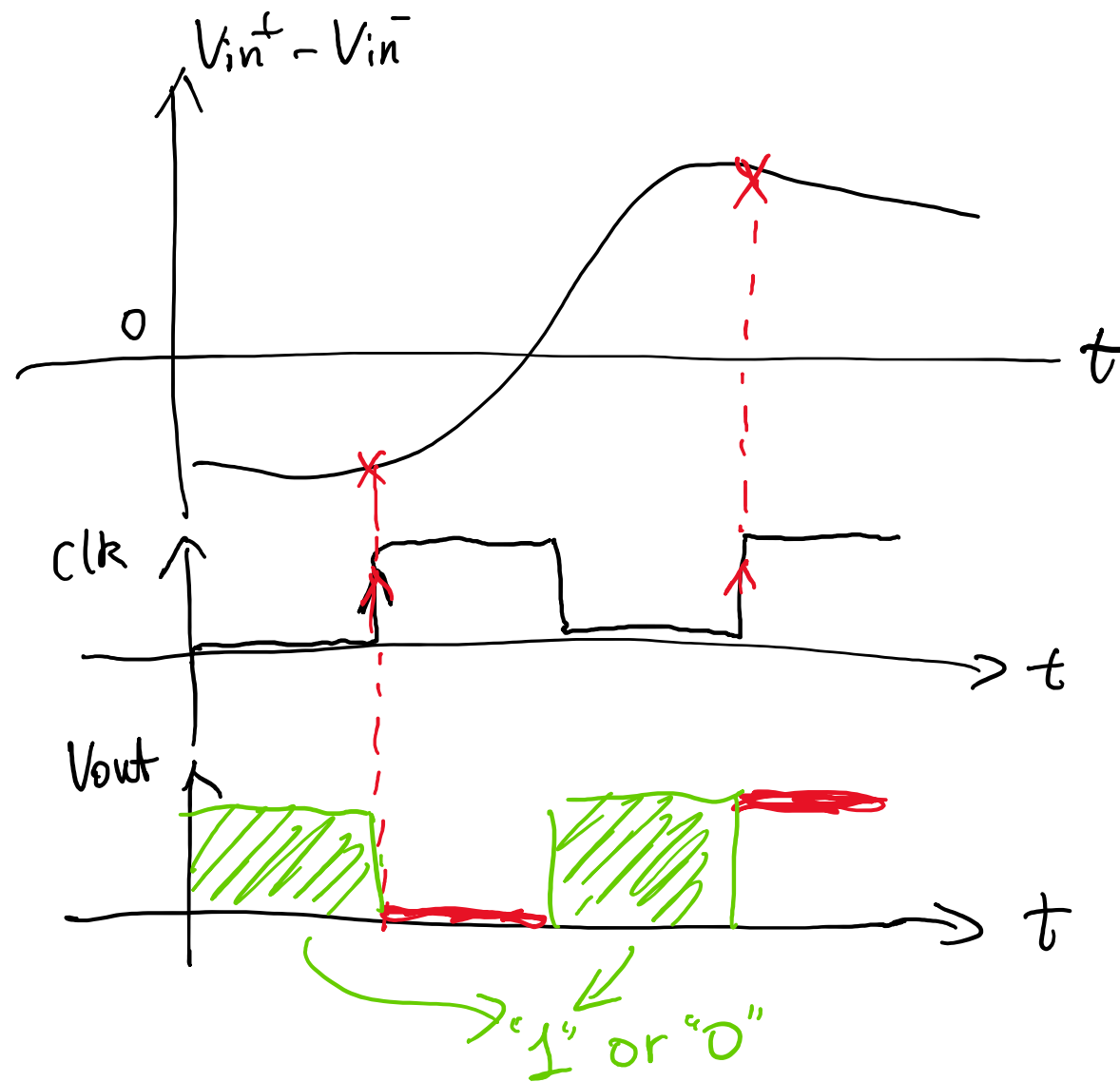


Latch-Based Comparator



When clk goes from "0" to "1",

$$\begin{cases} \text{If } V_{in+} - V_{in-} > 0, & V_{out} = "1" \\ \text{If } V_{in+} - V_{in-} < 0, & V_{out} = "0" \end{cases}$$

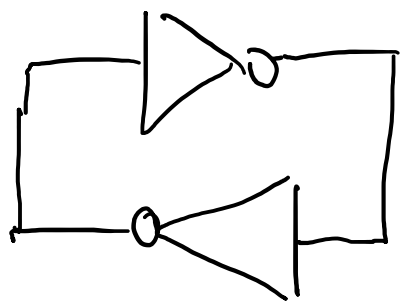


Comparator

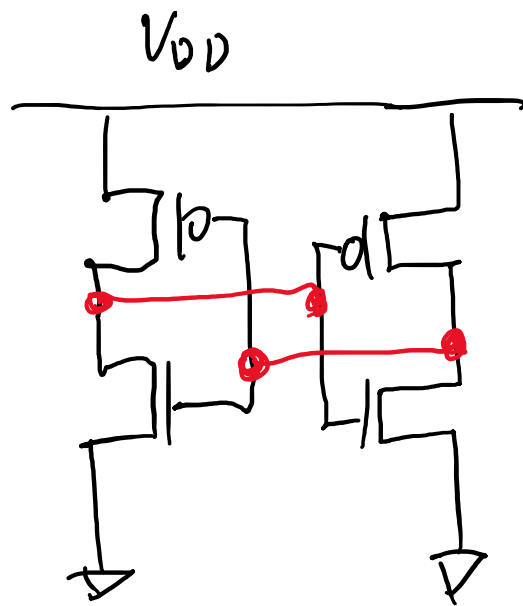


The amplification of a comparator need not be linear.
 $\Rightarrow$ leverage the regeneration of a latch

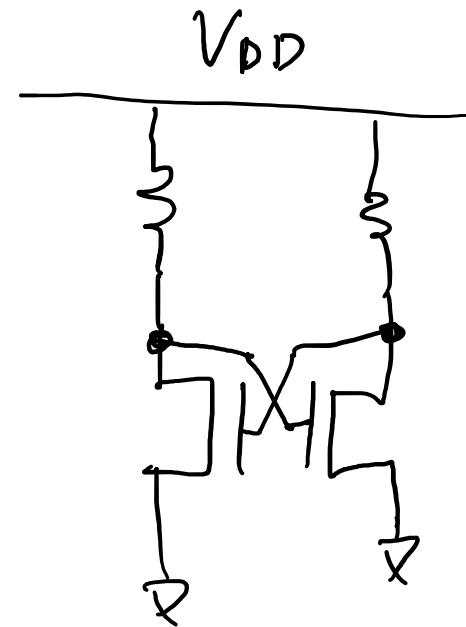
A simple latch



=



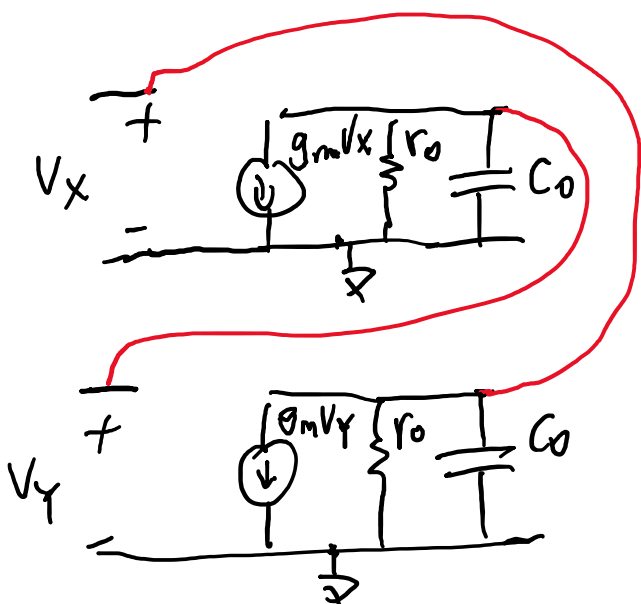
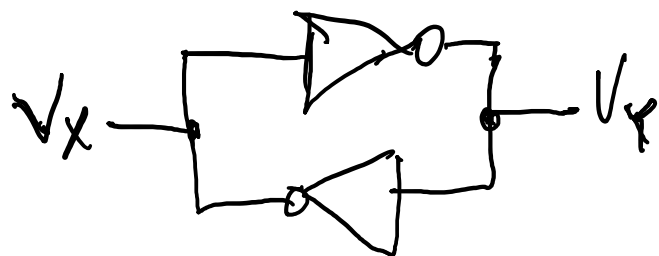
or



Comparator



Latch $\approx$ single-pole amplifier + positive feedback



$$g_m = g_{m,n} + g_{m,p}$$

$$r_o = r_{o,n} // r_{o,p}$$

C_o : input + output caps

$$\text{KCL: } \Rightarrow \begin{cases} -g_m V_x = \frac{V_y}{r_o} + C_o \frac{dV_y}{dt} & (1) \\ -g_m V_y = \frac{V_x}{r_o} + C_o \frac{dV_x}{dt} & (2) \end{cases}$$

$$(2) - (1) \Rightarrow g_m (V_x - V_y) = \frac{V_x - V_y}{r_o} + C_o \frac{d(V_x - V_y)}{dt}$$

$$\Rightarrow \frac{g_m}{C_o} \left(1 - \frac{1}{g_m r_o} \right) dt = \frac{d(V_x - V_y)}{V_x - V_y}$$

Comparator



Integrate both sides

$$\frac{g_m}{C_o} \left(1 - \frac{1}{g_m r_o}\right) \int_{t_0}^t dt = \int_{V_0}^V \frac{d(V_X - V_Y)}{V_X - V_Y}$$

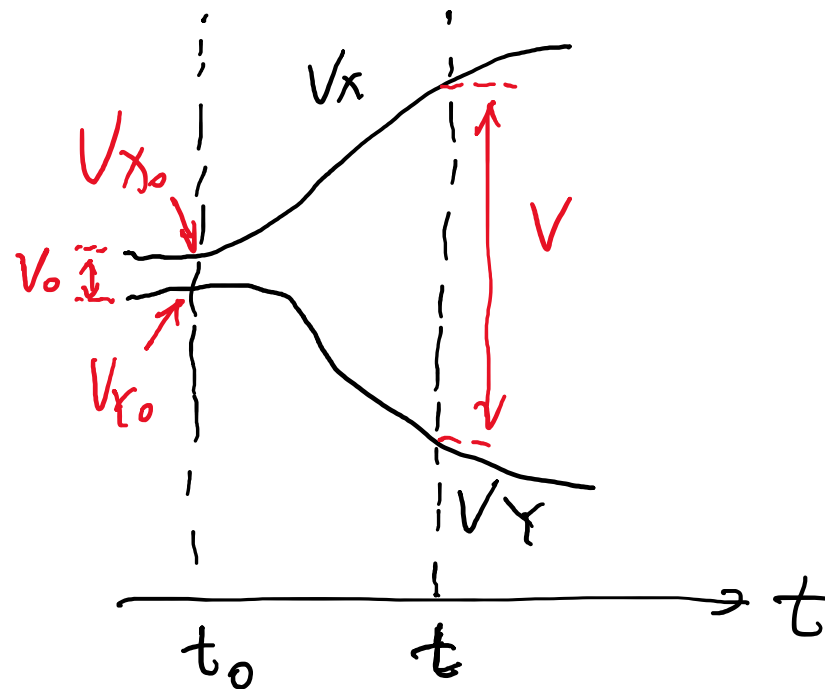
$$\Rightarrow V = V_0 e^{t \left(\frac{r_o C_o}{g_m r_o - 1} \right)}$$

When $g_m r_o \gg 1$

$$\Rightarrow V = V_0 e^{t (r_o C_o)}$$

$$t = \frac{C_o}{g_m} \ln \left(\frac{V}{V_0} \right)$$

gain of comparator



Comparator



Summary

✓ Latch-based comparator is faster compared to cascade of open-loop amplifier.

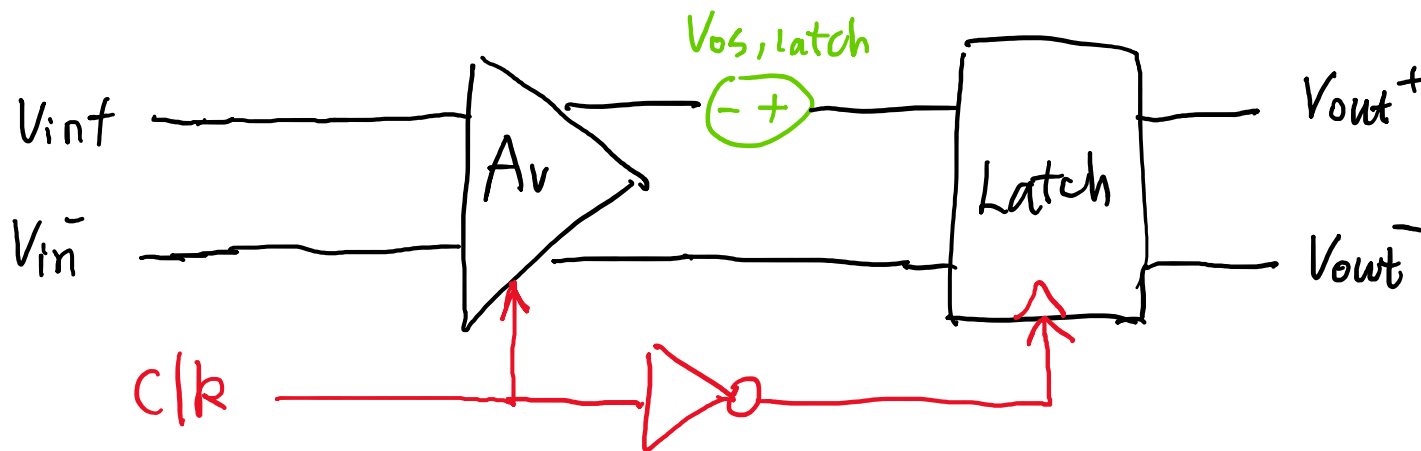
x Main issue: high input referred offset voltage.

Solution: 1. use pre-amplifier prior to the comparator
2. offset cancellation (discuss later).

Comparator



Reduce Offset $\Rightarrow$ Pre-Amplifier + Latch



★ Pre-amp improves the comparator delay.

Comparator delay:

$$t \approx \frac{C}{g_m} \ln \left(\frac{V_{out}}{A_v V_{in}} \right)$$

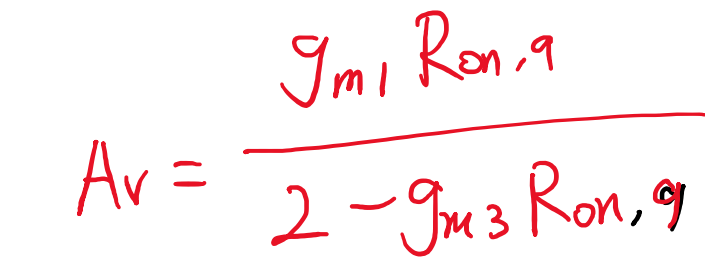
Ignore pre-amp offset,

$$V_{os, input} \downarrow = V_{os, latch} \cdot \frac{1}{A_v} \uparrow$$

★ leverage the unused half clock cycle.

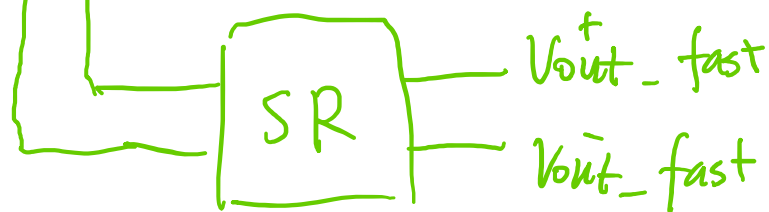
$A_v \uparrow$ $t \downarrow$

Hand-drawn schematic of a differential pair pre-amplifier. The circuit consists of two NMOS transistors, M_1 and M_2 , whose sources are connected to a common ground. The gates of M_1 and M_2 are connected to a common input node, which is also connected to a current source I_{SS} . The input signals are labeled V_{in}^+ and V_{in}^- . The output of the circuit is labeled CLK . The entire circuit is enclosed in a red oval. Below the circuit, there is a red triangle containing the text A_v , followed by the text $pre-amp$ and GND .



choose $g_{m3} R_{on} = 1$

$$\Rightarrow A_v = g_{m1} R_{on,q}$$



Use SR latch to speed up.

Comparator



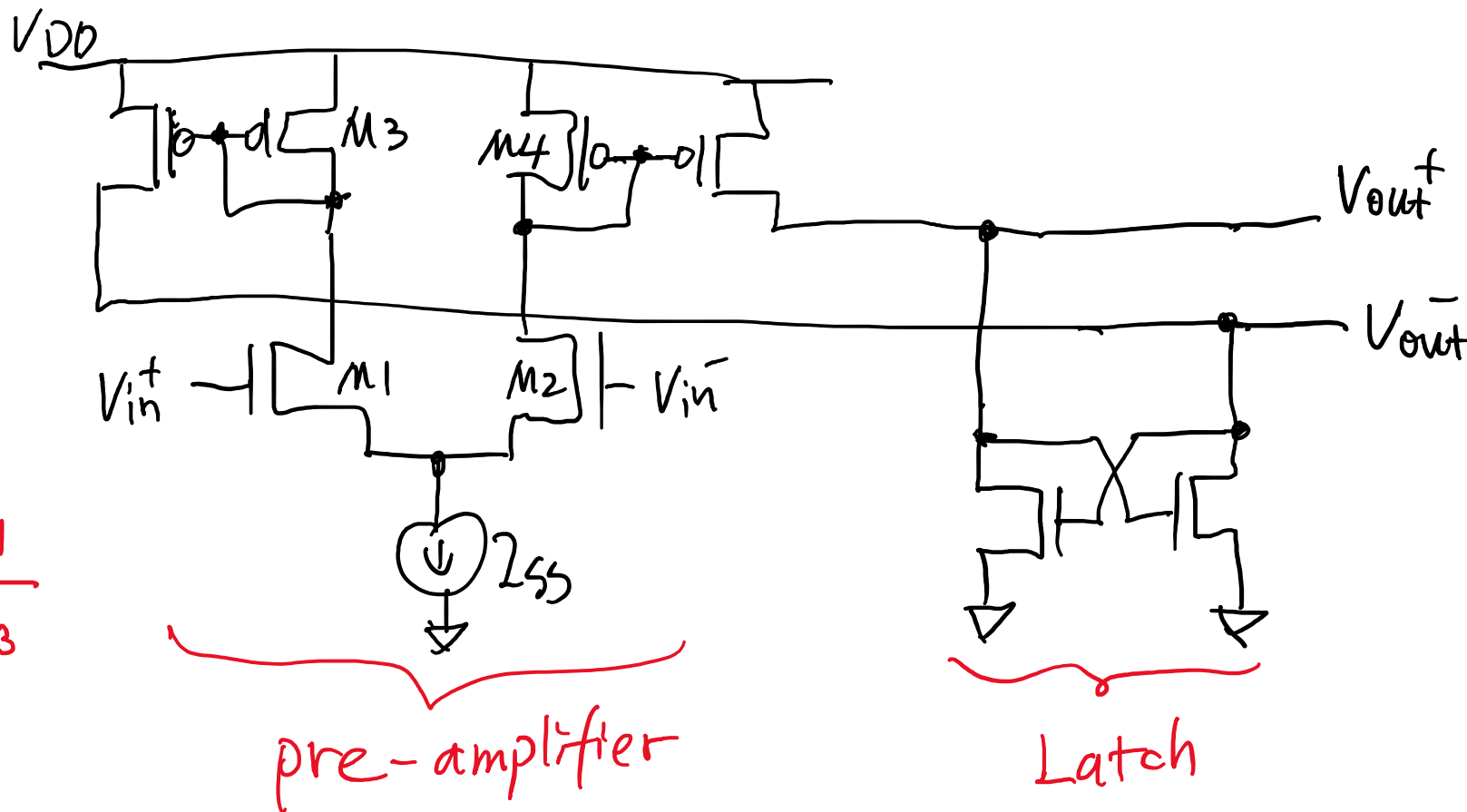
Example:



Comparator



Example: Reduced kickback



$$A_v = \frac{g_{m1}}{g_{m3}}$$

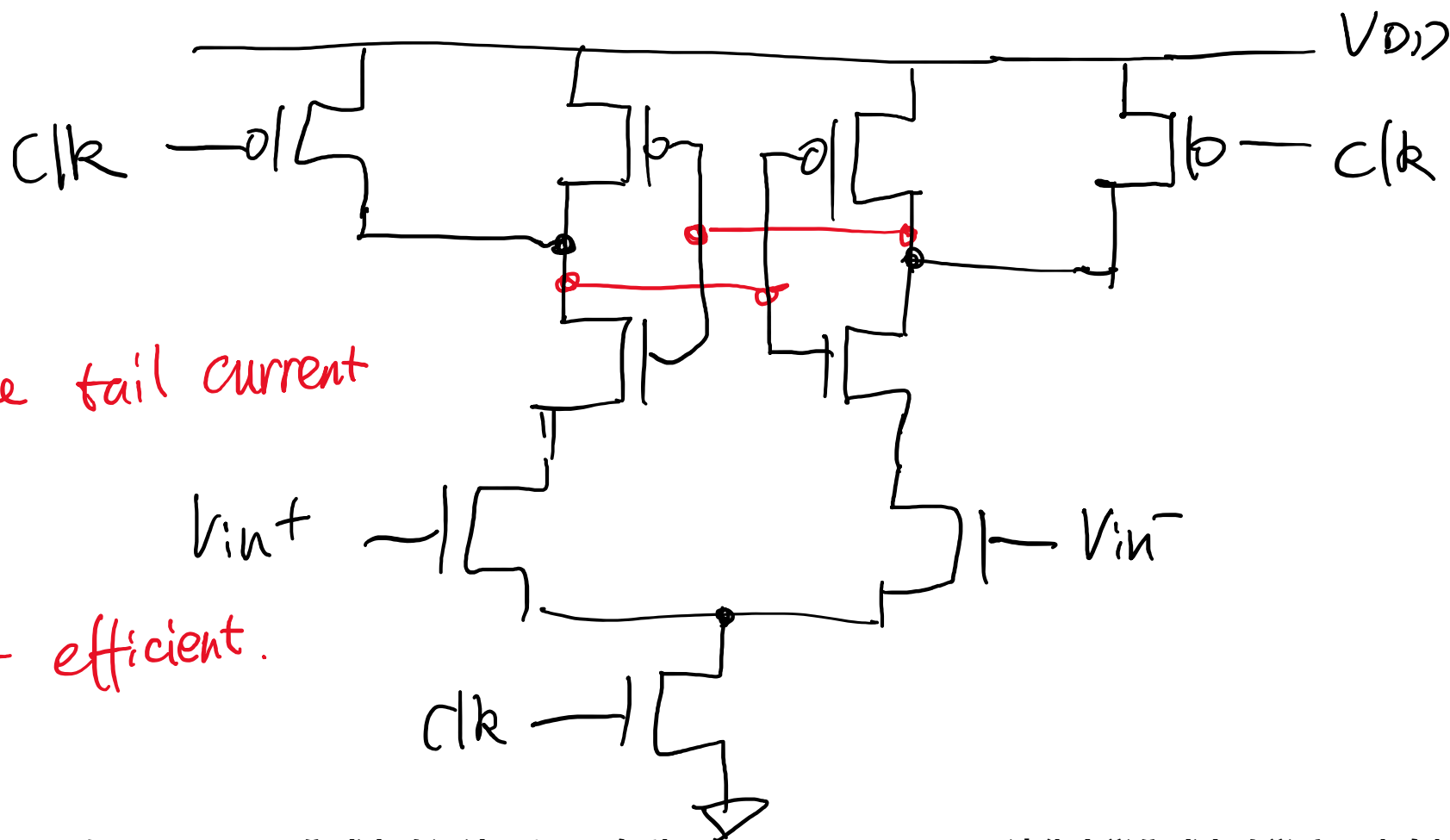
pre-amplifier

Latch

Comparator



Dynamic Comparator



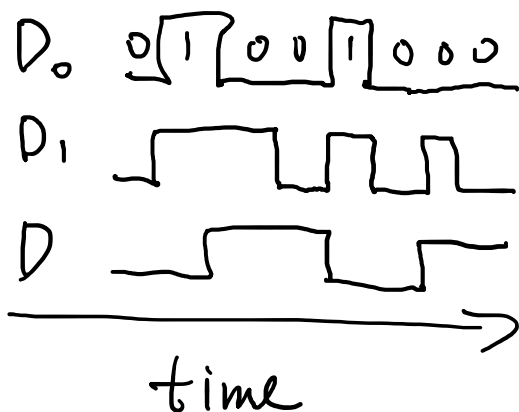
Remove the tail current
Source.



More power efficient.

Binary DAC

$\{D_0, D_1, D_2, \dots\}$
Digital



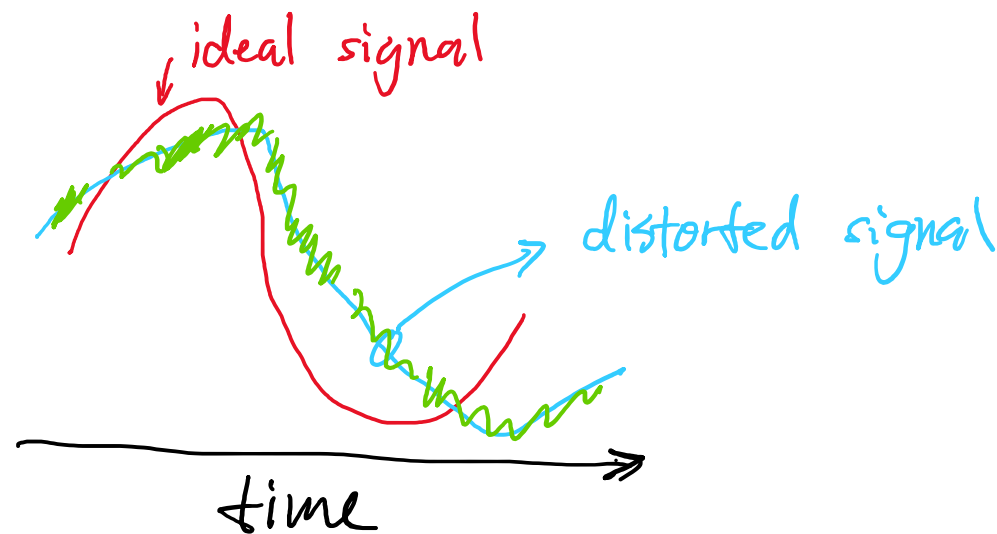
of bits

$N-1$

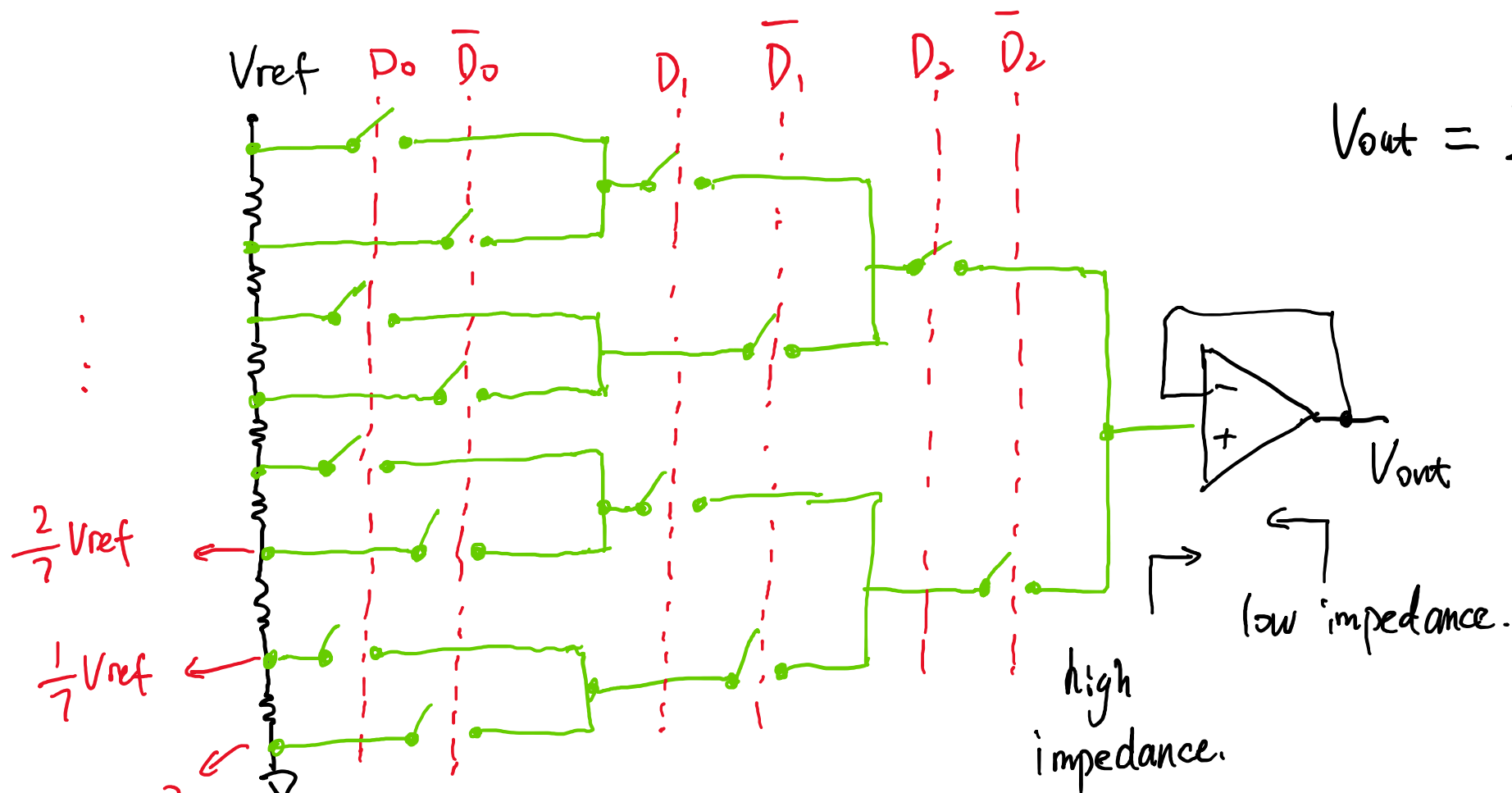
$$\sum_{i=0}^{N-1} \alpha D_i \cdot 2^i$$

scaling factor

Analog (Voltage/Current)



Resistor (R) String DAC



$$V_{out} = \sum_{i=0}^2 \frac{V_{ref}}{7} D_i \cdot 2^i$$

|
scaling factor.

The relative mismatch between two resistors is

$$\frac{\Delta R}{R} = \frac{\Delta \rho}{\rho} + \frac{\Delta L}{L} - \frac{\Delta W}{W} - \frac{\Delta t}{t} + \frac{\Delta R_c}{R}$$

↓
resistivity
↓
length
↓
width
↓
thickness

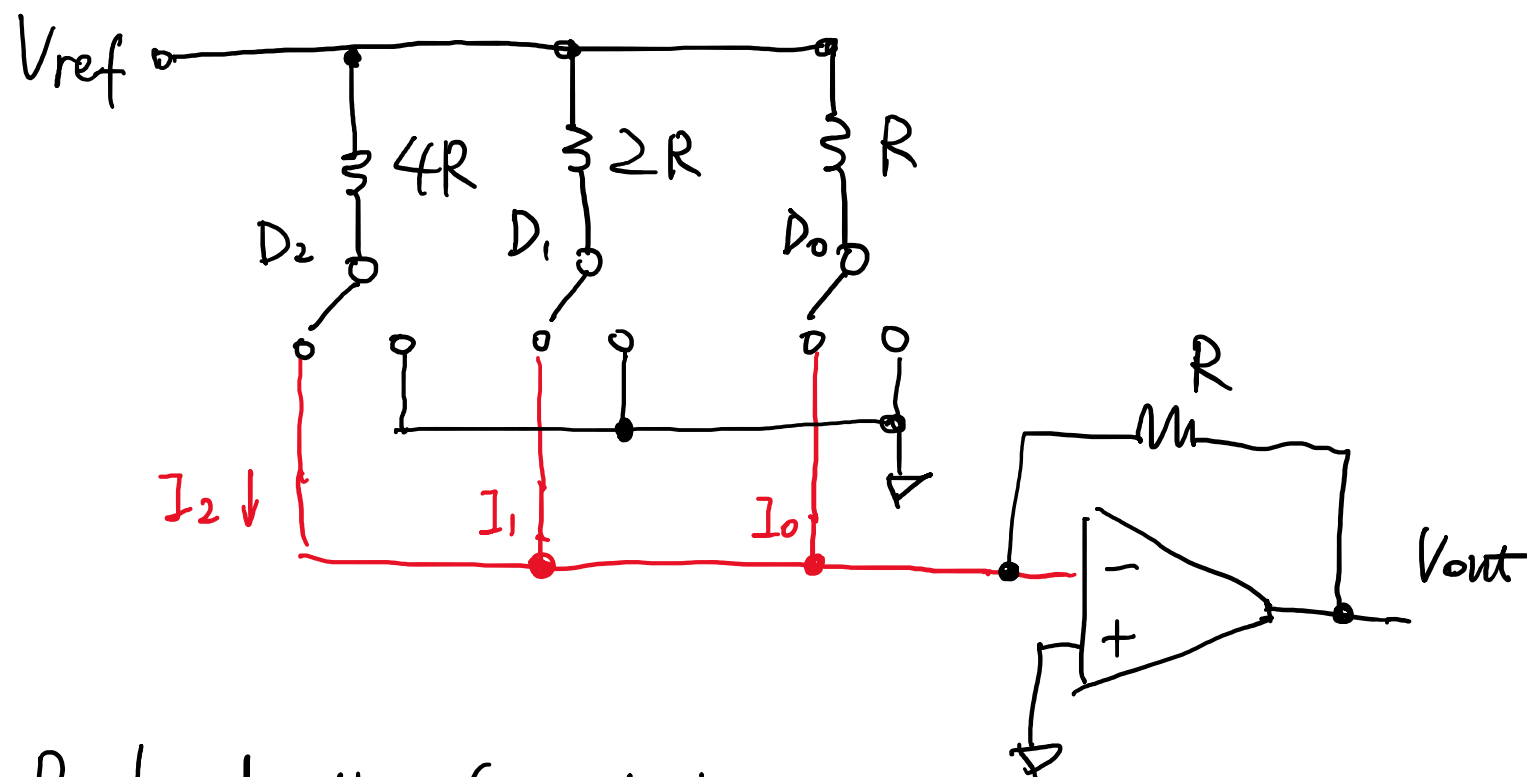
→ contact
resistance

↓

$$\text{Integral non-linearity (INL)} = \frac{V_{\text{ref}}}{\sqrt{4N}} \cdot \frac{\Delta R}{R} \quad (\text{from [Razavi]})$$

— RDAC is not preferred by high-speed applications.

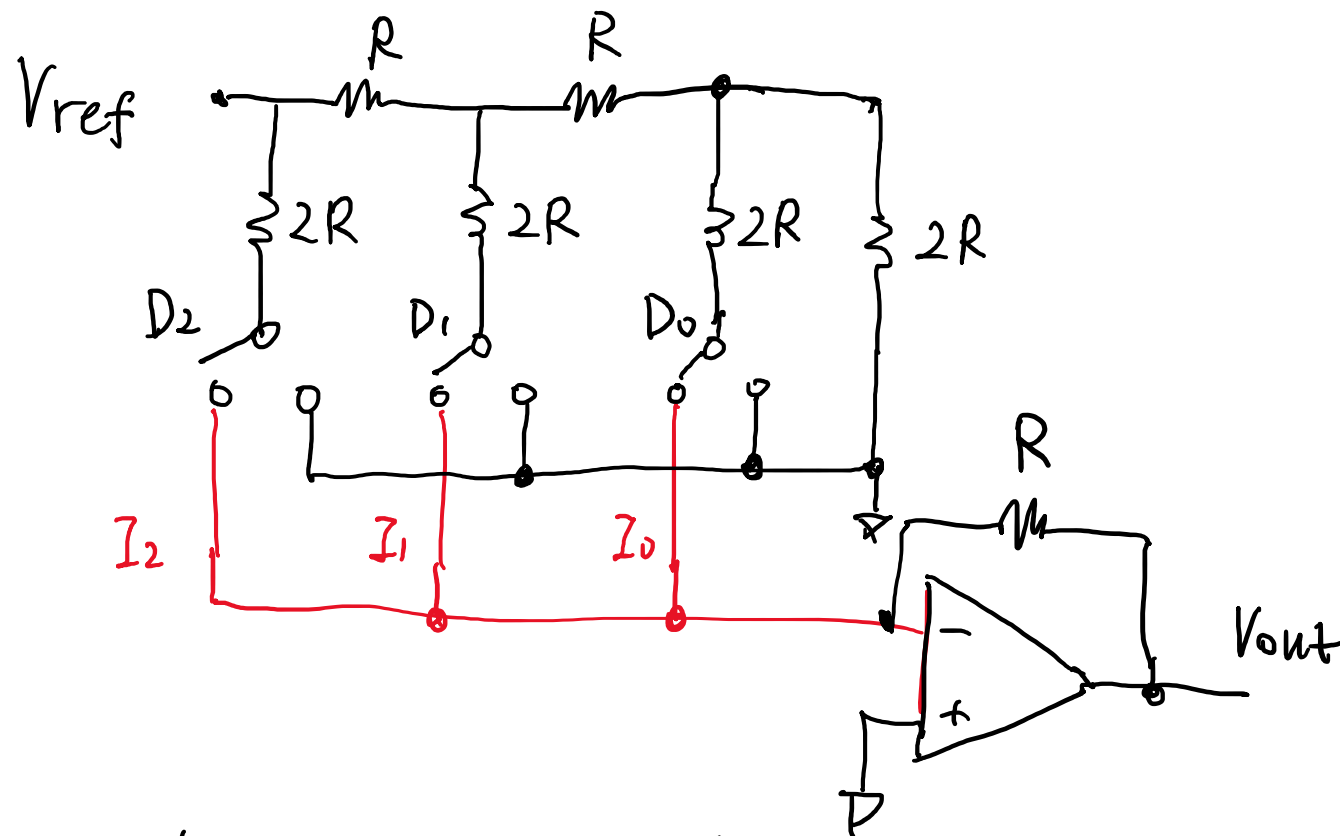
Binary weighted resistor DAC



$$V_{out} = \sum_{i=0}^2 (I_i R) D_i 2^i$$

- Reduced # of switch : one switch per bit.

R-2R DAC

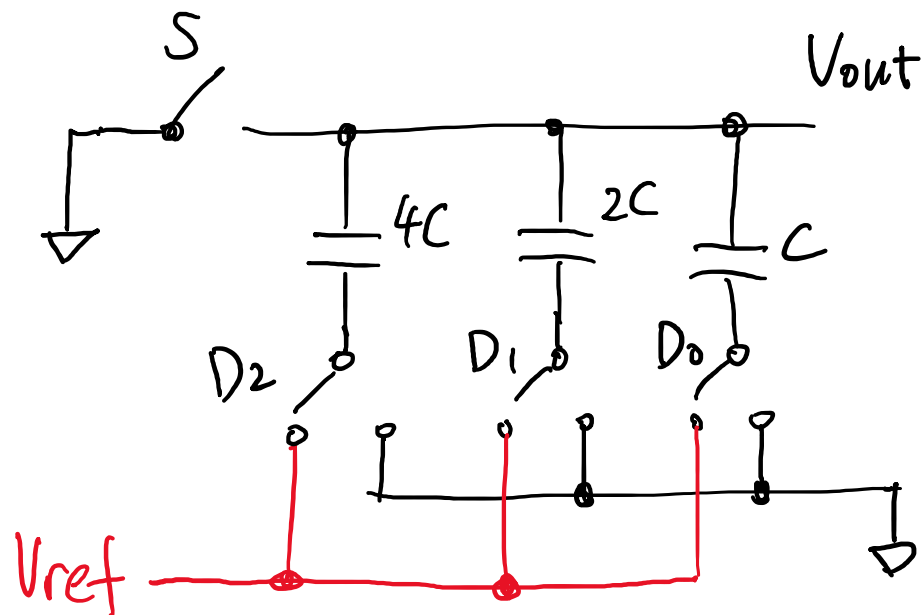


$$V_{out} = \sum_{i=0}^2 (I_i R) D_i 2^i$$

R DAC)

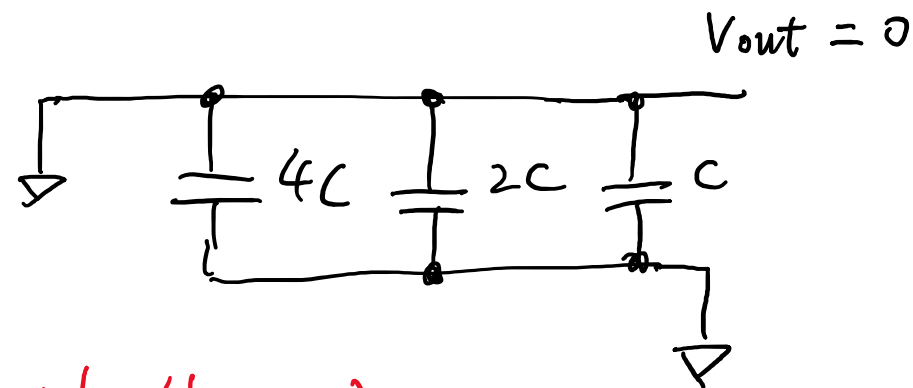
- The large resistor values can be reduced (compared to binary weighted

Capacitor (C) DAC

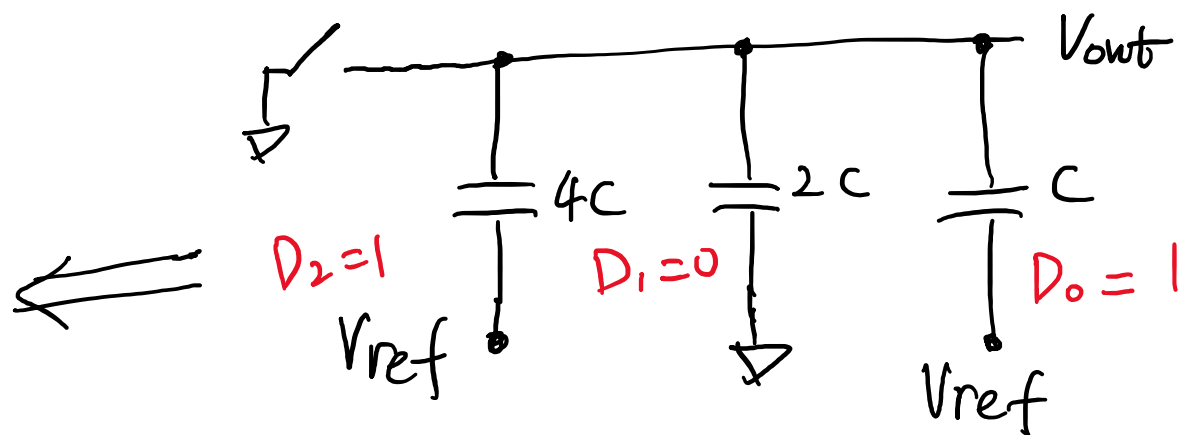


$$V_{out} = \sum_{i=0}^2 \frac{V_{ref}}{7} D_i 2^i$$

① reset phase



② evaluation phase



The capacitor mismatch is

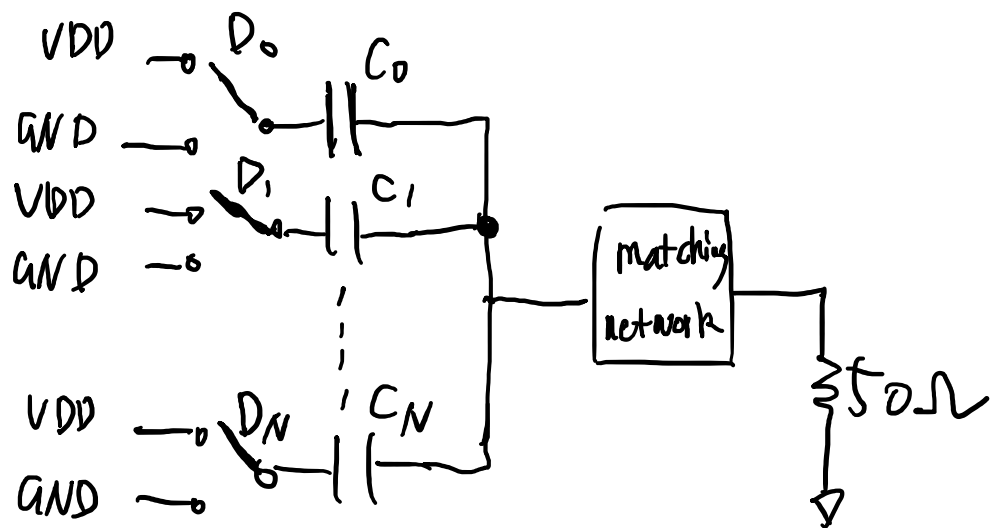
$$\frac{\Delta C}{C} = \frac{\Delta W}{W} + \frac{\Delta L}{L} - \frac{\Delta t_{ox}}{t_{ox}}$$

The capacitor non-linearity is due to the voltage dependent capacitance, which can be modelled as,

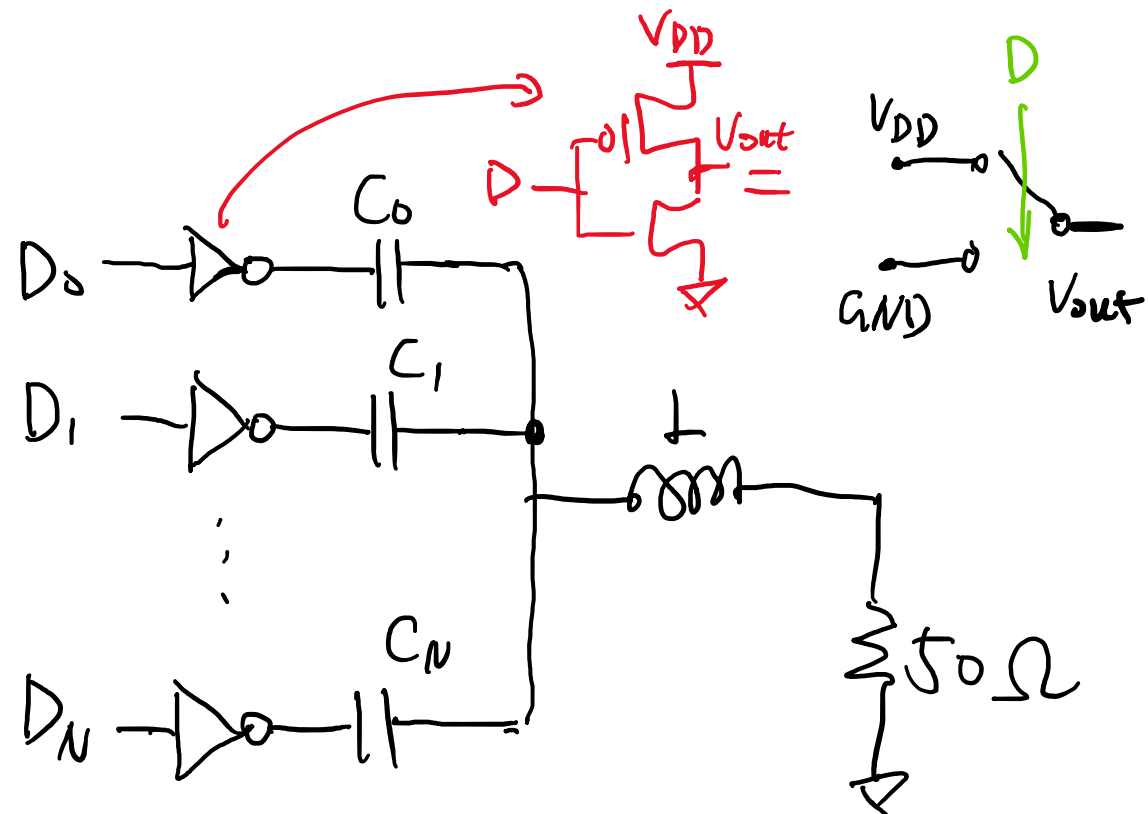
$$C = C_0 + C_0 \alpha_1 V + C_0 \alpha_2 V^2 + \dots$$

Switched Capacitor Power Amplifier (SCPA)

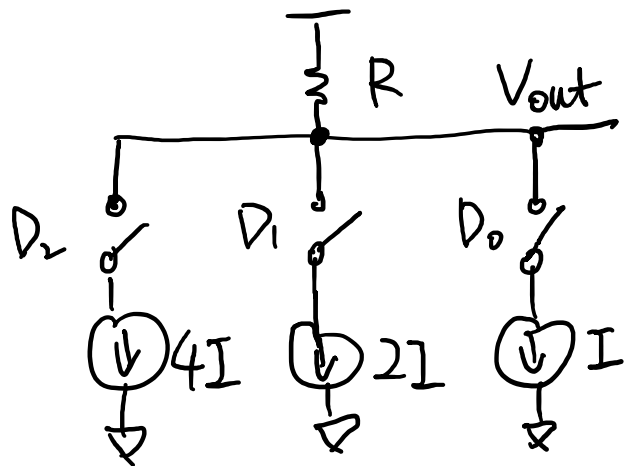
or SC Power DAC



$\Rightarrow$



Current-steering DAC



$$V_{out} = \sum_{i=0}^2 RI \cdot D_i 2^i$$

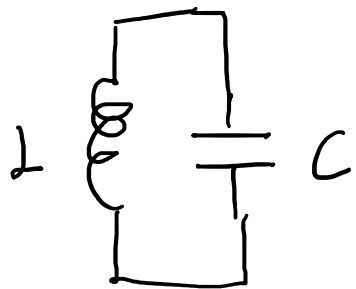
Scaling

Optimal design $\Rightarrow$ Segmented DAC

LC Oscillator



LC tank



Oscillating frequency

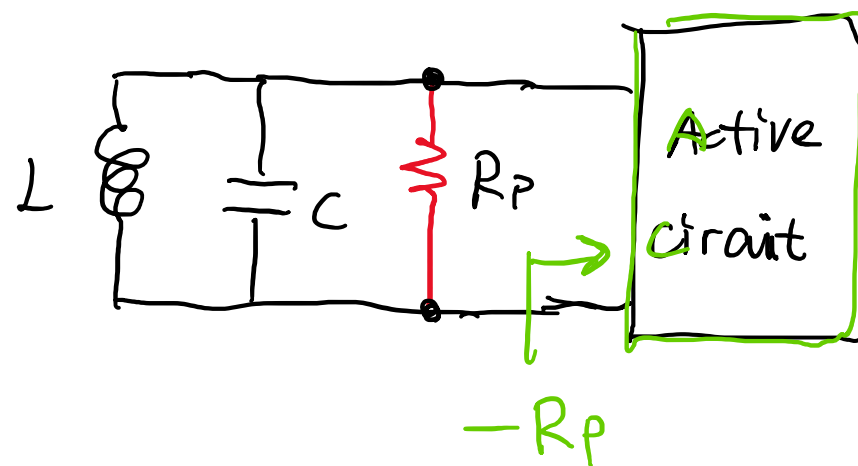
$$\omega_{osc} = \frac{1}{\sqrt{LC}}$$

$$Z_L + Z_C = 0 \quad @ \quad \omega_{osc}$$

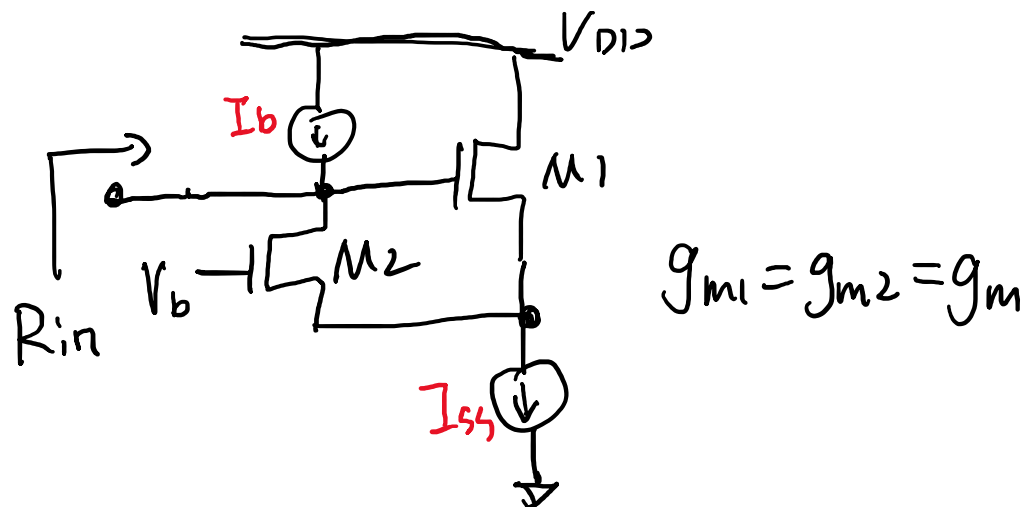
In practical:



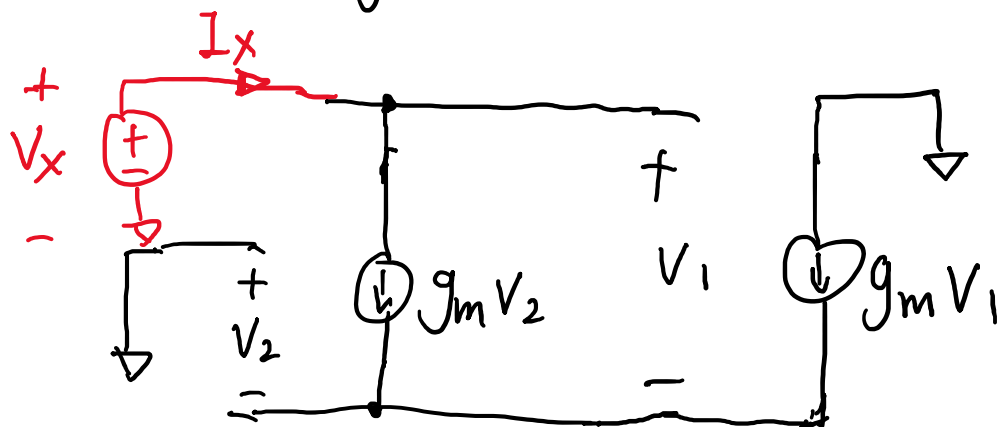
Real
Oscillator



LC Oscillator



Small-Signal Model



$$I_x = g_m V_2 = -g_m V_1$$

$$V_x = V_1 - V_2$$

$$= -\frac{I_x}{g_m} - \frac{I_x}{g_m}$$

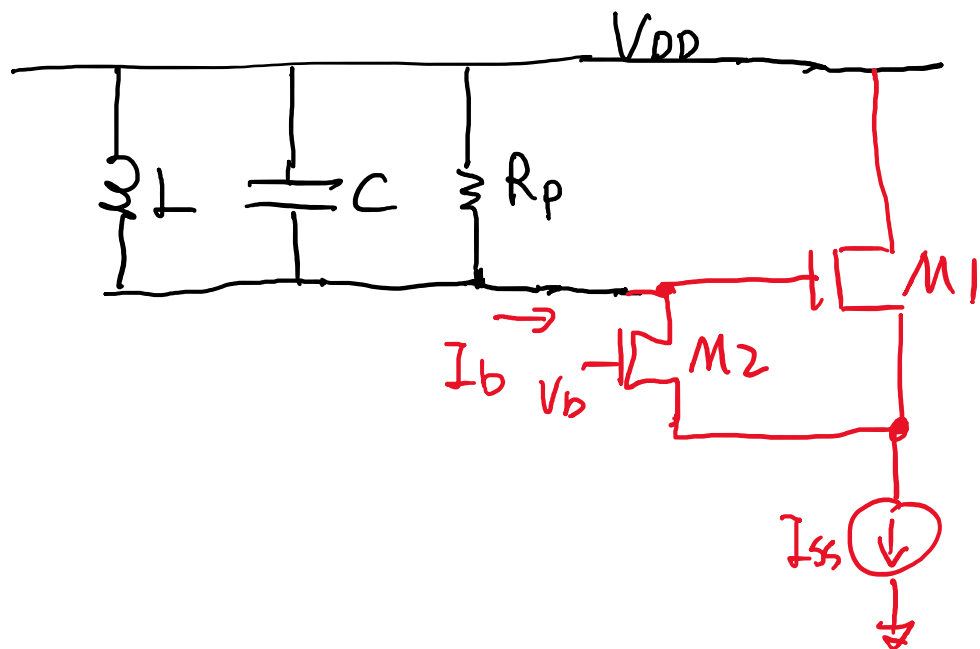
$$\Rightarrow \boxed{\frac{V_x}{I_x} = -\frac{2}{g_m}}$$



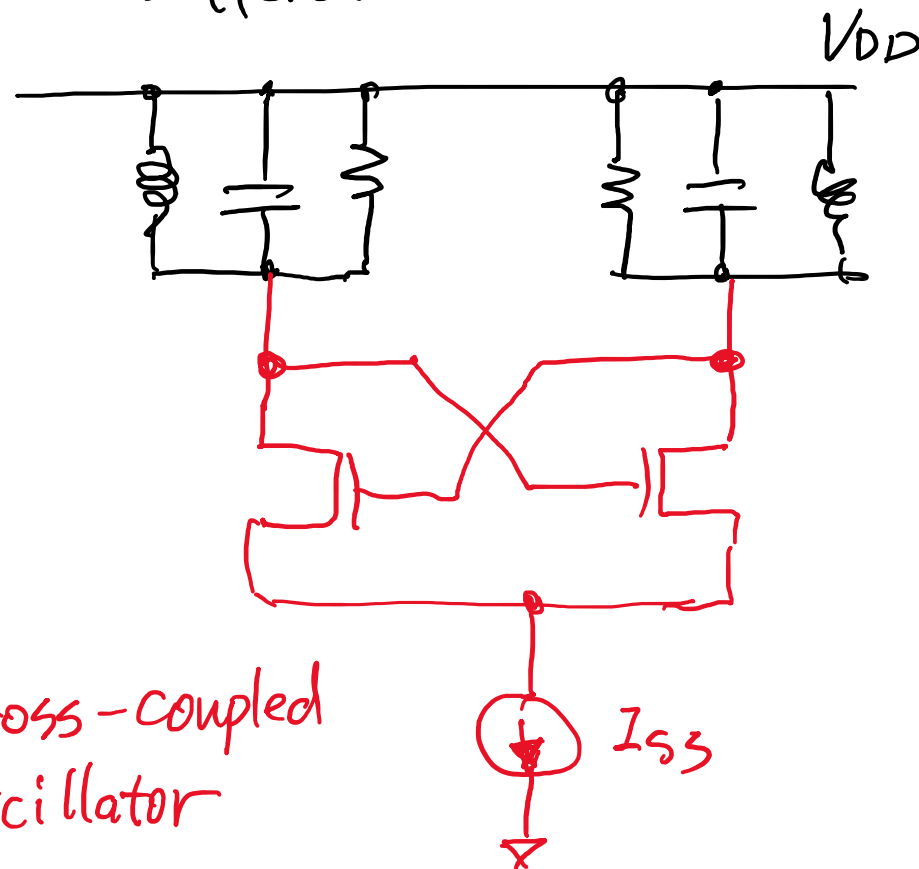
LC Oscillator



LC tank + active circuit



Differential version

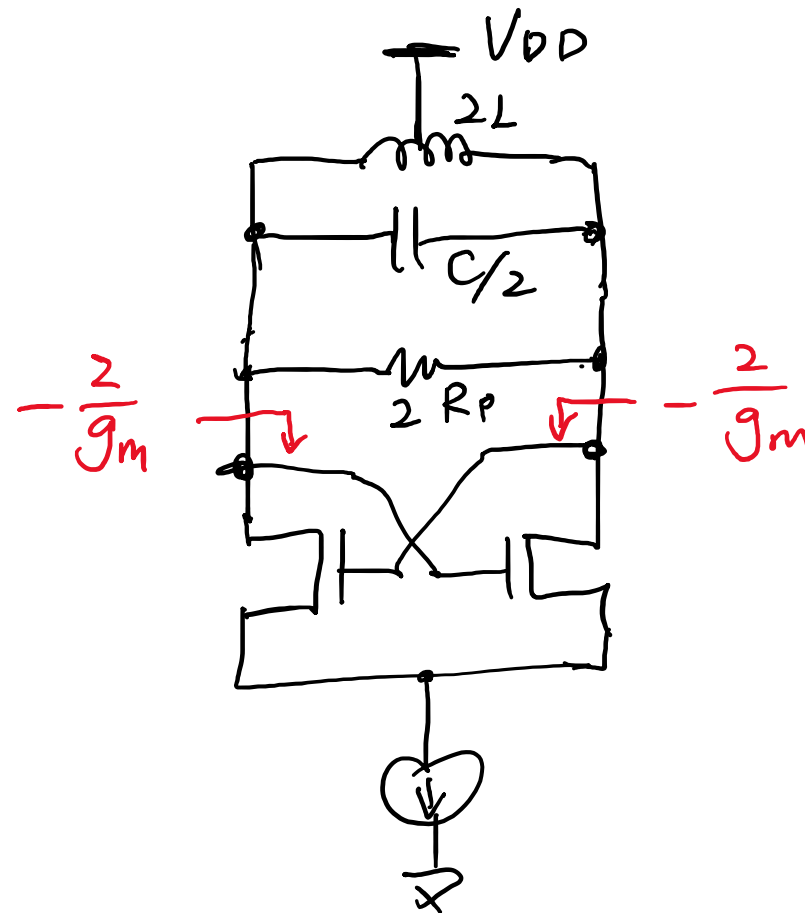
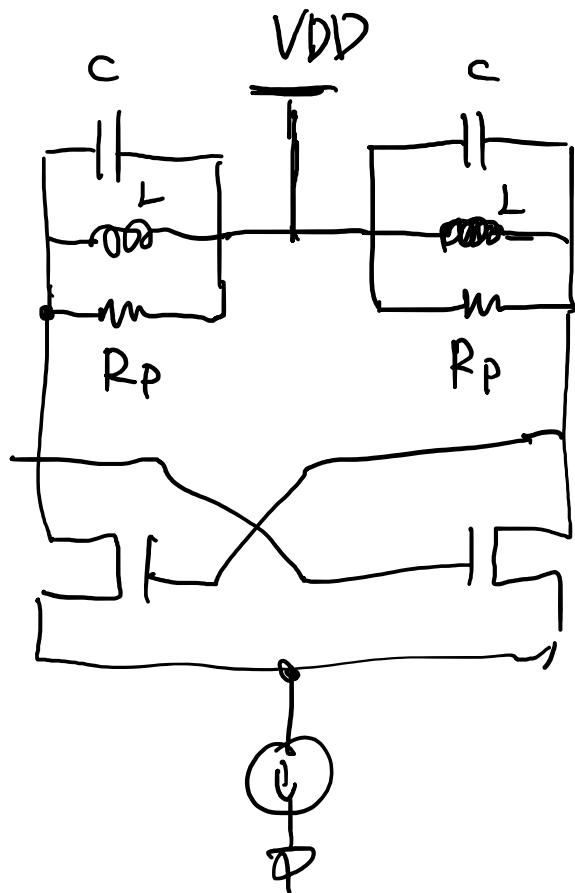


Cross-coupled
Oscillator

LC Oscillator



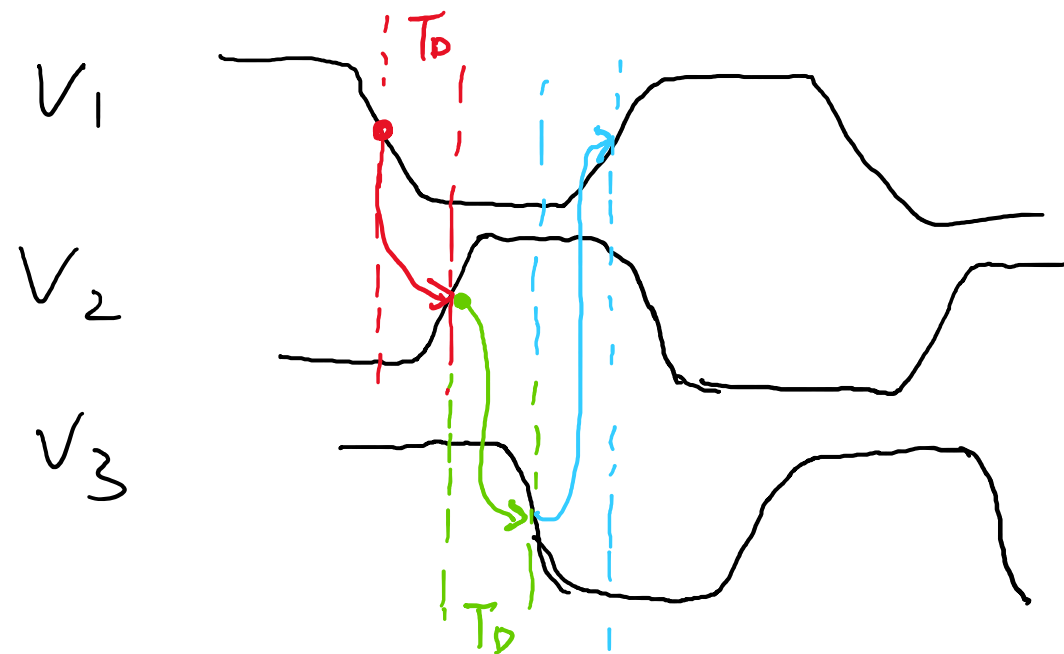
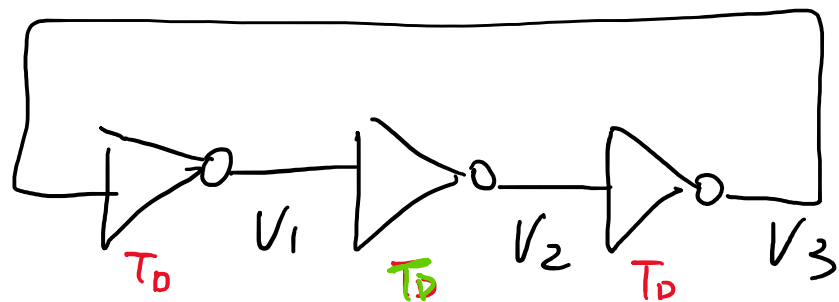
Equivalent Circuit of Cross-Coupled Oscillator



Ring Oscillator



Ring Oscillator (RO)

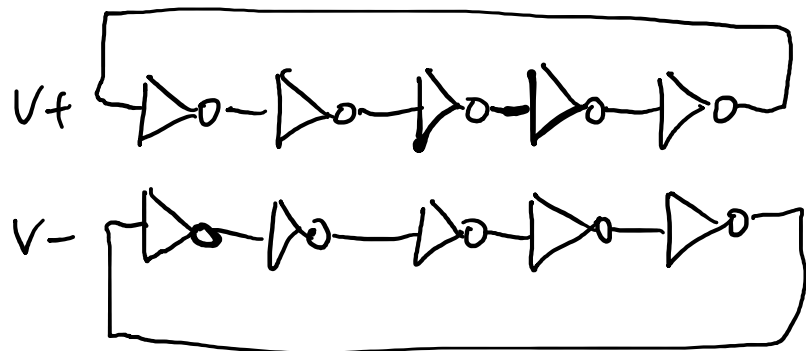


Ring Oscillator

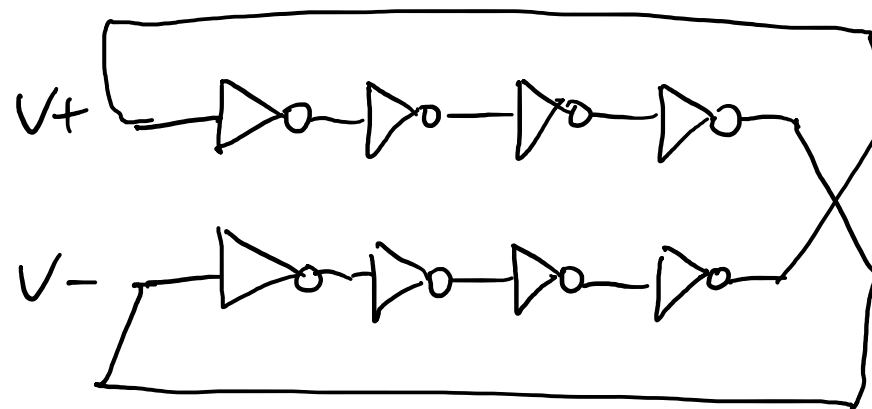


Differential RO

5-stage RO



4-stage RO



Ring Oscillator



More "differential" RO

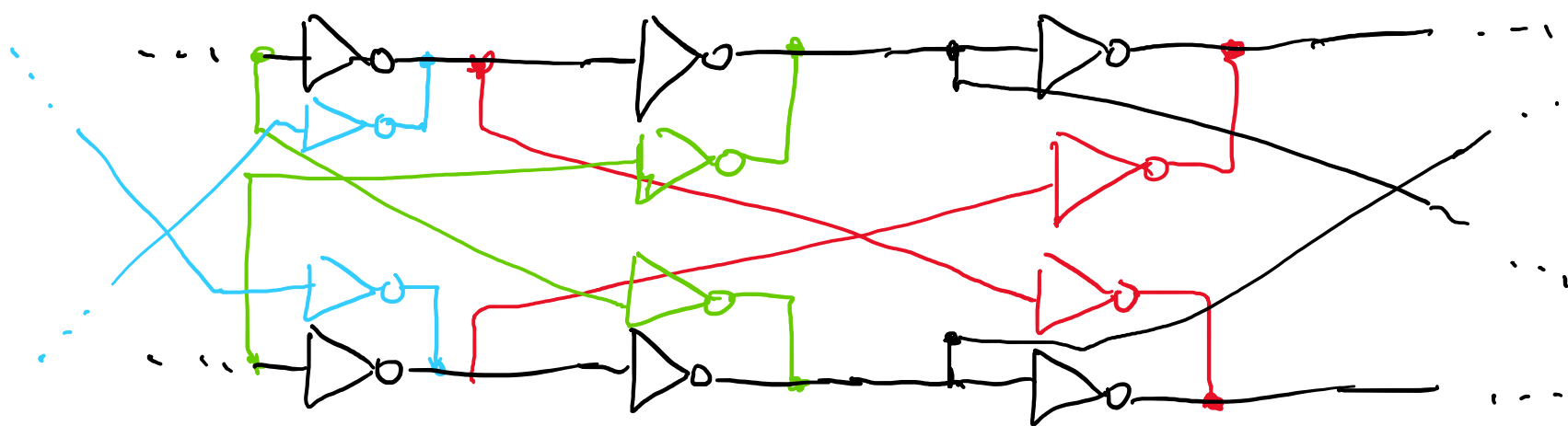


⇒ cross-coupling

Ring Oscillator



Faster design $\Rightarrow$ Feedforward cross-coupling

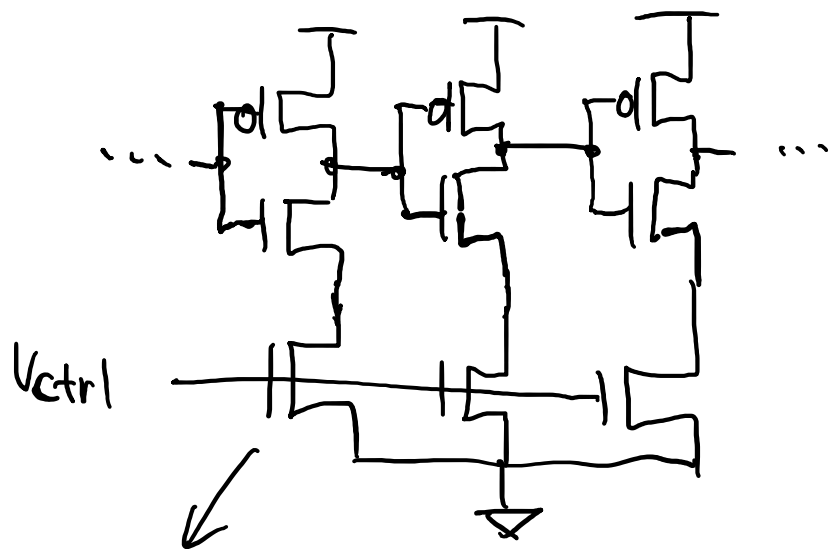


Voltage-Controlled Oscillator (VCO)

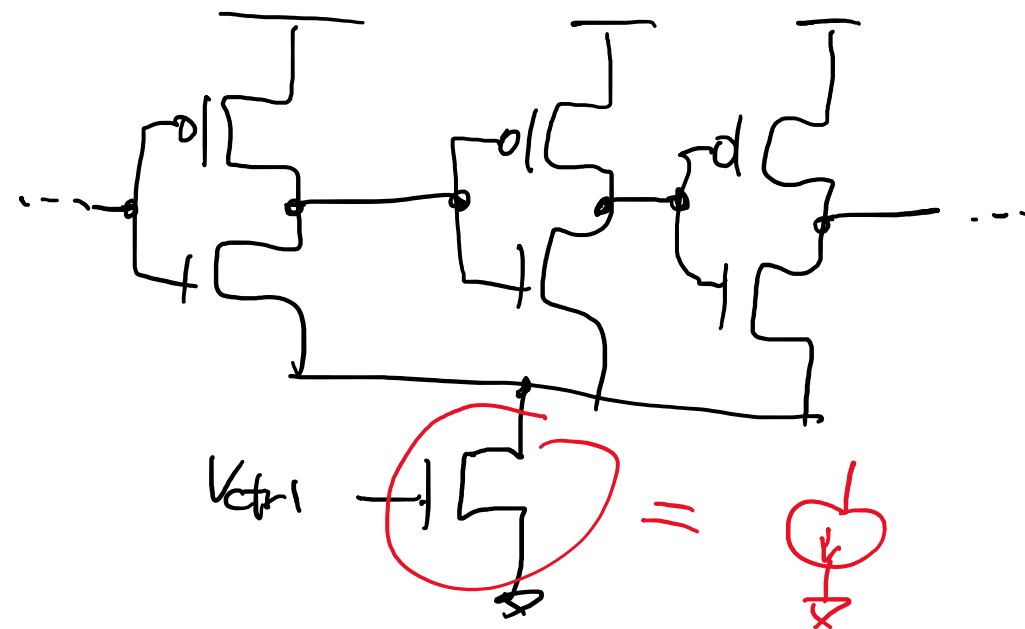


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Current starving VCO



individual current-starved
transistor



Shared tail transistor $\Rightarrow$ in saturation
for most of V_{ctrl} range $\Rightarrow$ less prone to
supply noise